Master's thesis

Master of Science (M.Sc.)
Department of Business
Major: Data Science

Topic:

Improving the robustness of binary classifiers in cryptocurrency exchange rate forecasts

Author: Can Atalay

Matriculation Number: 27698638
First supervisor: Prof. Dr. Talha Ali Khan
Second supervisor: Prof. Dr. Raja Hashim Ali
Submitted on: 24.07.2025

University of Applied Sciences Europe
Iserlohn · Berlin · Hamburg

EIGENSTÄNDIGKEITSERKLÄRUNG / STATEMENT OF AUTHORSHIP

Atalay
Name | Family Name

Can
Vorname | First Name

27698638
Matrikelnummer | Student ID Number

Improving the Robustness of Binary Classifiers in Cryptocurrency Exchange Rate Forecasts
Titel der Examsarbeit | Title of Thesis

Ich versichere durch meine Unterschrift, dass ich die hier vorgelegte Arbeit selbstständig verfasst habe. Ich habe mich dazu keiner anderen als der im Anhang verzeichneten Quellen und Hilfsmittel, insbesondere keiner nicht genannten Onlinequellen, bedient. Alles aus den benutzten Quellen wörtlich oder sinngemäß übernommen Teile (gleich ob Textstellen, bildliche Darstellungen usw.) sind als solche einzeln kenntlich gemacht.

Die vorliegende Arbeit ist bislang keiner anderen Prüfungsbeh.rde vorgelegt worden. Sie war weder in gleicher noch in ähnlicher Weise Bestandteil einer Prüfungsleistung im bisherigen Studienverlauf und ist auch noch nicht publiziert. Die als Druckschrift eingereichte Fassung der Arbeit ist in allen Teilen identisch mit der zeitgleich auf einem elektronischen Speichermedium eingereichten Fassung.

With my signature, I confirm to be the sole author of the thesis presented. Where the work of others has been consulted, this is duly acknowledged in the thesis' bibliography. All verbatim or referential use of the sources named in the bibliography has been specifically indicated in the text.

The thesis at hand has not been presented to another examination board. It has not been part of an assignment over my course of studies and has not been published. The paper version of this thesis is identical to the digital version handed in.

23/07/2025 Potsdam
Datum, Ort | Date, Place

Unterschrift | Signature

University of Europe for Applied Sciences

Declaration on the use of generative Artificial Intelligence (AI) systems

Atalay	Can
Name | Family Name	Vorname | First Name
27698638	Improving the robustness of binary classifiers in cryptocurrency exchange rate forecasts
Matrikelnummer | Student ID Number	Titel Prüfungsarbeit | Title of the exam

I have used the following artificial intelligence (AI)-based tools in the creation of my work:

1. OpenAI GPT-4.1
2. OpenAI o3-mini
3. Gemini 2.5 Pro
4. Google Translate

I further declare that

☑ I have actively informed myself about the performance and limitations of the above-mentioned AI tools,
☑ I have marked the passages taken from the above-mentioned AI tools,
☑ I have checked that the content generated with the help of the above-mentioned AI tools and adopted by me is actually accurate,
☑ I am aware that, as the author of this work, I am responsible for the information and statements made in it.

I have used the AI-based tools mentioned above as shown in the table below.

AI-based support tool	**Usage**	**Parts of the work affected**	**Remarks**
GPT-4.1 (OpenAI)	• Converting handwritten equations to latex equations • Code generation (python) for visualization tasks	• Chapter 3 (Mathematical Foundations) • Chapters 2, 5 (Literature Review, Results and Discussion)	The mistakes were manually corrected
o3-mini (OpenAI)	• Code generation (python) for data processing and ML workflows	all chapters	The mistakes were manually corrected
Gemini 2.5 Pro (Google)	• Conversion of online resource URLs used into BibTeX citations	• Bibliography	All citations are valid, can confirm w/ URLs
Google Translate (Google)	• Translation of the abstract into German	• Zusammenfassung	Language support

Place, date, signature of the student

Potsdam, 23.07.2025,

Abstract

Binary classification based exchange rate forecasting research is gaining popularity, but is not yet nearly as prevalent as regression based efforts in financial forecasting research. The cryptocurrency market, being a newer addition to financial markets, especially suffers from this condition with very few articles written on classification based exchange rate forecasting. In addition, measures of randomness borrowed from chaos theory that are frequently used in exchange rate forecasting, such as the Hurst exponent, are not common in this intersection of research. Moreover, there is no robustness-centric discussion of classification based exchange rate forecasting in the crypto market or mention of cointegration based techniques such as the use of synthetic pairs, which raises concerns regarding the generalization properties of the models discussed.

This work uses synthetic cryptocurrency pairs to conduct binary classification using an ensemble ML model. The LGBM classifier used is supported via an additional custom layer that predicts directional market movement based on trend continuation. This custom layer is supported by Hurst filtering, allowing for market interaction during windows of low mean reversion. With hyperparameter tuning of the LGBM classifier during the testing/-validation phases and a separate testing phase, a robustness-first performance evaluation framework is built, quantifying the generalization gap.

This novel approach successfully integrates a mean-reversion filter with a trend-continuation cue. The generalization gap is successfully closed to a very significant degree for an array of synthetic cryptocurrency pairs. The improved robustness is also observed in multiple timeframes at selected thresholds. In addition, the Hurst filtering is able to simultaneously improve testing performance while closing the generalization gap, displaying a very promising profile for directly improving ML performance in the cryptocurrency market. The obtained performance is also demonstrated to outperform buy-and-hold strategies within the crypto market.

This thesis is (one of) the first work(s) in the literature that focuses on classification based exchange rate forecasting using synthetic pairs in the cryptocurrency market. In addition, it presents a procedural framework for building robustness-first approaches to exchange rate forecasting.

Zusammenfassung

Die Forschung zur binären Klassifikation von Wechselkursen gewinnt an Popularität, ist aber noch nicht annähernd so weit verbreitet wie regressionsbasierte Ansätze in der Finanzprognoseforschung. Der Kryptowährungsmarkt, ein neuerer Bereich der Finanzmärkte, leidet besonders darunter, da nur wenige Artikel zu klassifikationsbasierten Wechselkursprognosen veröffentlicht wurden. Zudem sind Zufallsmaße aus der Chaostheorie, die häufig in der Wechselkursprognose verwendet werden, wie der Hurst-Exponent, in diesem Forschungsbereich nicht üblich. Darüber hinaus fehlt eine robustheitsorientierte Diskussion der klassifikationsbasierten Wechselkursprognose im Kryptomarkt oder die Erwähnung kointegrationsbasierter Techniken wie der Verwendung synthetischer Paare, was Bedenken hinsichtlich der Generalisierungseigenschaften der diskutierten Modelle aufwirft.

Diese Arbeit verwendet synthetische Kryptowährungspaare zur binären Klassifizierung mithilfe eines Ensemble-ML-Modells. Der verwendete LGBM-Klassifikator wird durch eine zusätzliche benutzerdefinierte Ebene unterstützt, die die Richtung der Marktbewegung basierend auf der Trendfortsetzung vorhersagt. Diese benutzerdefinierte Ebene wird durch Hurst-Filterung unterstützt, was Marktinteraktionen in Zeiträumen mit geringer Mittelwertumkehr ermöglicht. Durch Hyperparameter-Tuning des LGBM-Klassifikators während der Test-/Validierungsphasen und einer separaten Testphase wird ein Rahmen für die Robustheitsbewertung der Performance entwickelt, der die Generalisierungslücke quantifiziert.

Dieser neuartige Ansatz integriert erfolgreich einen Mean-Reversion-Filter mit einem Trendfortsetzungssignal. Die Generalisierungslücke wird für eine Reihe synthetischer Kryptowährungspaare erfolgreich und signifikant geschlossen. Die verbesserte Robustheit ist auch in mehreren Zeiträumen bei ausgewählten Schwellenwerten zu beobachten. Darüber hinaus verbessert der Hurst-Filter die Testleistung und schließt gleichzeitig die Generalisierungslücke. Dies zeigt ein vielversprechendes Profil für die direkte Verbesserung der ML-Performance im Kryptowährungsmarkt. Die erzielte Performance übertrifft nachweislich Buy-and-Hold-Strategien im Kryptomarkt.

Diese Arbeit ist eine der ersten Arbeiten in der Literatur, die sich mit klassifizierungsbasierter Wechselkursprognose unter Verwendung synthetischer Paare im Kryptowährungsmarkt befasst. Darüber hinaus präsentiert sie einen prozeduralen Rahmen für die Entwicklung robustheitsorientierter Ansätze zur Wechselkursprognose.

Contents

List of Figures ix

List of Tables xi

1 Introduction 1
- 1.1 Background and Motivation . . . 1
- 1.2 Problem Statement . . . 3
- 1.3 Research Objectives . . . 4
- 1.4 Scope . . . 4
- 1.5 Contributions . . . 4
- 1.6 Thesis Outline . . . 5

2 Literature Review 7
- 2.1 Introduction to ER Forecasting in Financial Markets . . . 7
- 2.2 ML Approaches in ER Forecasting . . . 9
  - 2.2.1 Supervised Learning . . . 9
  - 2.2.2 Unsupervised Learning . . . 17
  - 2.2.3 Reinforcement Learning . . . 18
  - 2.2.4 Hybrid and Ensemble Models . . . 20
  - 2.2.5 Feature Engineering and Data Preprocessing . . . 21
  - 2.2.6 Challenges and Considerations . . . 22
- 2.3 ER Forecasting with Binary Classifiers . . . 23
- 2.4 The Problem of Generalization and Robustness in Financial ML . . . 25
- 2.5 Applications of Chaos Theory in Financial Forecasts . . . 26
- 2.6 Summary . . . 27

3 Mathematical Foundations 29
- 3.1 Market Interaction Outcomes in BinClass-based ER Forecasting . . . 29
  - 3.1.1 Long Strategy, BinClass-based Outcomes . . . 29
  - 3.1.2 Short Strategy, BinClass-based Outcomes . . . 37
- 3.2 Strategy Success based on Overall Model Performance . . . 44

4 Methodology 49
- 4.1 Data Sourcing . . . 49
  - 4.1.1 Pair Selection . . . 49
- 4.2 Data Preprocessing and Feature Engineering . . . 50
  - 4.2.1 Generation of the Synthetic Pairs . . . 50
  - 4.2.2 Data Grouping . . . 52
  - 4.2.3 Feature Engineering . . . 54
- 4.3 Implementation of Chaos-Theory Inspired Techniques . . . 55
- 4.4 Machine Learning Workflow . . . 56
  - 4.4.1 LGBM Classifier . . . 57
  - 4.4.2 Custom ML Layer . . . 58
  - 4.4.3 Ensemble Model and Hyperparameter Tuning . . . 58

4.4.4 Training, Validation, Testing Process 59
4.5 Cloud Infrastructure and Experiment Setup 59
4.6 Evaluation 60

5 Results and Discussion **63**
5.1 Results Listed by Grouping 63
5.1.1 1000-Tick Candlestick Group 63
5.1.2 50-Tick Candlestick Group 67
5.1.3 20-Tick Candlestick Group 68
5.2 Pairwise Results 70
5.3 Best Hyperparameters for the LGBM Classifier 71
5.3.1 1000-Tick Candlestick Group 71
5.3.2 50-Tick Candlestick Group 71
5.3.3 20-Tick Candlestick Group 71
5.3.4 Summary of Hyperparameter Tuning 71
5.4 Portfolio Impact of the Results 72

6 Conclusions and Future Work **75**
6.1 Conclusions 75
6.2 Future Work 76

Bibliography **79**

List of Figures

1.1 The price of 1 BTC vs. 1 oz. of Gold . . . 1
1.2 Binance API Ecosystem . . . 2
1.3 Binary Classification in ML . . . 2
1.4 Generalization Gap in ML . . . 3

2.1 The Evolution of ER Forecasting . . . 8
2.2 Machine Learning in ER Forecasting . . . 9
2.3 Linear Regression Fit (SLR / MLR) . . . 10
2.4 Support Vector Regression (SVR) . . . 11
2.5 Random Forest Regression . . . 11
2.6 XGBoost vs. LGBM . . . 12
2.7 FFW Neural Network . . . 12
2.8 Simple CNN Diagram . . . 13
2.9 CNN-LSTM Hybrid Model for ER Forecasting of Stocks . . . 13
2.10 Recurrent Neural Networks . . . 14
2.11 LSTM Layer Example / LSTM Operation . . . 14
2.12 Gated Recurrent Unit (GRU) Operation . . . 15
2.13 Example Transformer Architecture . . . 16
2.14 Binary Classification Model . . . 17
2.15 Clustering (Unsupervised) for Market Regime Identification . . . 18
2.16 Reinforcement Learning for ER Forecasting . . . 19
2.17 Deep Reinforcement Learning Using a Deep-Q-Network . . . 20
2.18 Hybrid Models vs. Ensemble Models . . . 20
2.19 Weekend Gaps in Traditional Markets . . . 21
2.20 Overfitting in Forecasting Applications . . . 22
2.21 BinClass-based ER Forecasting . . . 24
2.22 Hurst exponent . . . 27

3.1 Profitability Condition Visualized . . . 46
3.2 Non-robust forecasting failure . . . 46

4.1 Selected Pairs Visualized: MLNUSDT . . . 50
4.2 Selected Pairs Visualized: PLUMEUSDT and SIRENUSDT . . . 51
4.3 Synthetic Pair MLNUSDT/PLUMEUSDT . . . 52
4.4 Synthetic Pair MLNUSDT/SIRENUSDT . . . 53
4.5 Synthetic Pair SIRENUSDT/PLUMEUSDT . . . 54
4.6 Hurst Exponent for MLNUSDT/PLUMEUSDT . . . 55
4.7 Hurst Exponent for MLNUSDT/SIRENUSDT . . . 56
4.8 Hurst Exponent for SIRENUSDT/PLUMEUSDT . . . 57
4.9 Trend Continuation Cue with Custom Layer . . . 58
4.10 Experiment Workflow on the Cloud . . . 60

5.1 MLNUSDT/SIRENUSDT pair, 1000, unfiltered . . . 63
5.2 MLNUSDT/SIRENUSDT pair, 1000, filtered . . . 64

5.3 ROC Curve for MLNUSDT/PLUMEUSDT 64
5.4 MLNUSDT/SIRENUSDT pair, 1000, improvement 65
5.5 MLNUSDT/PLUMEUSDT pair, 1000, improvement 65
5.6 SIRENUSDT/PLUMEUSDT pair, 1000, improvement 66
5.7 Robustness Improvements, All Pairs, 1000 66
5.8 MLNUSDT/SIRENUSDT pair, 50, improvement 67
5.9 MLNUSDT/PLUMEUSDT pair, 50, improvement 67
5.10 SIRENUSDT/PLUMEUSDT pair, 50, improvement 68
5.11 Robustness Improvements, All Pairs, 50 68
5.12 MLNUSDT/PLUMEUSDT pair, 20, improvement 69
5.13 Improvements in the Generalization Gap, Overall 69
5.14 Improvements in the Generalization Gap, Pairwise 70
5.15 LGBM Classifier Hyperparameters, Tuned, Pairwise 72
5.16 Account Growth with Robust Predictions 73
5.17 Non-robust case, account balance loss 74

List of Tables

4.1 Selected Trading Pairs and Trading Period 50
4.2 Synthetic Trading Pairs and Trading Period 52
4.3 Candlestick group tick-size for each pair 53
4.4 Input Features and Their Lag Context Window 54
4.5 Technical Indicators and Their Period Settings 55
4.6 Hyperparameter Tuning Setup for `LGBMClassifier` 59
4.7 Time Series Dataset Splits 59
4.8 Compute and Memory Specifications of the VM 60

5.1 Model Evaluation Metrics, 1000 63
5.2 Model Evaluation Metrics, 50 68
5.3 Model Evaluation Metrics, 20 69
5.4 Generalization Gap Improvement Across Tick Group Sizes 70
5.5 Best Hyperparameters, 1000 71
5.6 Best Hyperparameters, 50 71
5.7 Best Hyperparameters, 20 71

1 Introduction

This chapter introduces the master's thesis; summarizing the background, research objectives, and the scope of the study, as well as providing an overview of the following chapters.

1.1 Background and Motivation

Cryptocurrencies have remarkably outperformed, in terms of market returns, traditional asset classes in the past decade, attracting significant global demand and research interest [1][2]. As seen in Figure 1.1, the price of Bitcoin has increased more than 100-fold, even after 2015. In this context, forecasting market movements in the cryptocurrency market has been a significant area of research in the past decade, for the purposes of risk assessment, cost optimization and portfolio management. In addition, automated trading systems that aim to predict market movements have been researched and developed in private and academic circles.

Figure 1.1: The price of 1 BTC vs. 1 oz. of Gold
[1]

These systems interact with centralized [3] and decentralized exchanges (CEX/DEX) with automation capabilities that facilitate trading in the crypto market (see Figure 1.2).

When forecasting is conducted, depending on the build, these systems may consider the market as a whole, treat subsets of the market as separate dynamical systems, or combine various subsets of the space to generate and interact with synthetic processes that are not directly available on these EXs. Based on the performance of the forecasting agents deployed and the underlying software architecture, significant performance improvements can be expected compared to manual methods in terms of risk/portfolio management, trading fee optimization and effective response latency.

Exchange rate (ER) forecasting research is often carried out using regression techniques, for instance, to predict the ER of a specific instrument at a certain time in the future. Various machine learning (ML) techniques are employed, with complexities ranging from basic linear regression (LR) to deep learning and reinforcement learning. Classification techniques have also been used for ER forecasting in financial markets, via a remodelling of the process under investigation.

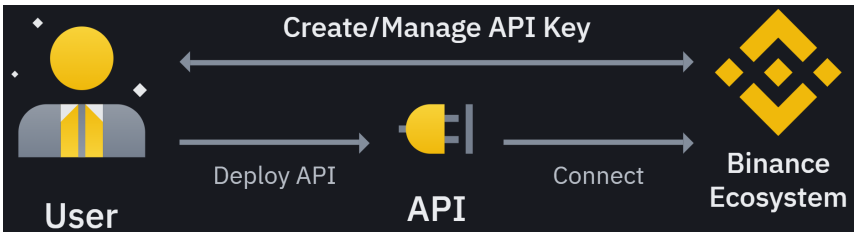

Figure 1.2: Binance API Ecosystem
[3]

In classification-based ER forecasting, market movement outcomes that cover a predetermined subset of the probability space can be defined. The outcomes (see Figure 1.3) defined can then be classified using ML techniques. In this context, binary or multi-class classifiers can be utilized, in addition to ensemble models that combine multiple classifier outcomes using mathematical functions such as descriptive statistics with decision thresholds or logic gates.

Various measures of error are consulted for drawing conclusions regarding the performance of the forecasting agents deployed in classification-based ML, such as accuracy and precision, based on the research objectives. These measures of error are often used as the end-goals of ER forecasting research, aiming to improve forecasting performance.

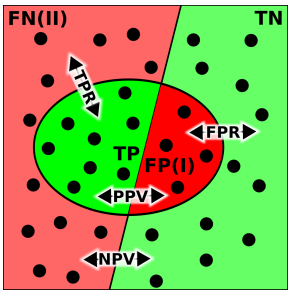

Figure 1.3: Binary Classification in ML
[4]

The existing research on classification for ER forecasts has incorporated the use of gen-

eralization techniques such as cross validation and early stopping in their ML workflows to prevent overfitting. However, the research on the crypto market is more limited compared to that on traditional markets such as foreign exchange (FX), stocks and commodities. Moreover, there are significant gaps in crypto ER forecasting research in terms of the development and evaluation of methods that aim to systematically close this generalization gap. This issue constitutes the main motivation for this study.

1.2 Problem Statement

Due to the non-deterministically chaotic nature of financial markets and the perpetual improvement in market efficiency with increased number of actors involved in the system and rapid advancements in AI, devising ER prediction methods that yield consistent positive returns at meaningful margins for long timeframes is of extreme difficulty. While possible via the recognition of utilizable patterns within that chaos, effective ER forecasts call for continual improvement of the models used and swift adaptation to out-of-distribution data to keep the performance high.

ML methods that do not generalize well to unseen data may lead the model to overemphasize the relevance of features that will not help accurately predict the ER movements in the future, effectively overfitting due to a number of reasons such as; the market dynamics shifting from one region to another, the relative frequency of the patterns identified in the processes lacking stationarity, and profitable strategies eventually being identified by multiple players in the system, causing their effectiveness to decrease over longer periods of time, and eventually converging to a gambler's ruin scenario.

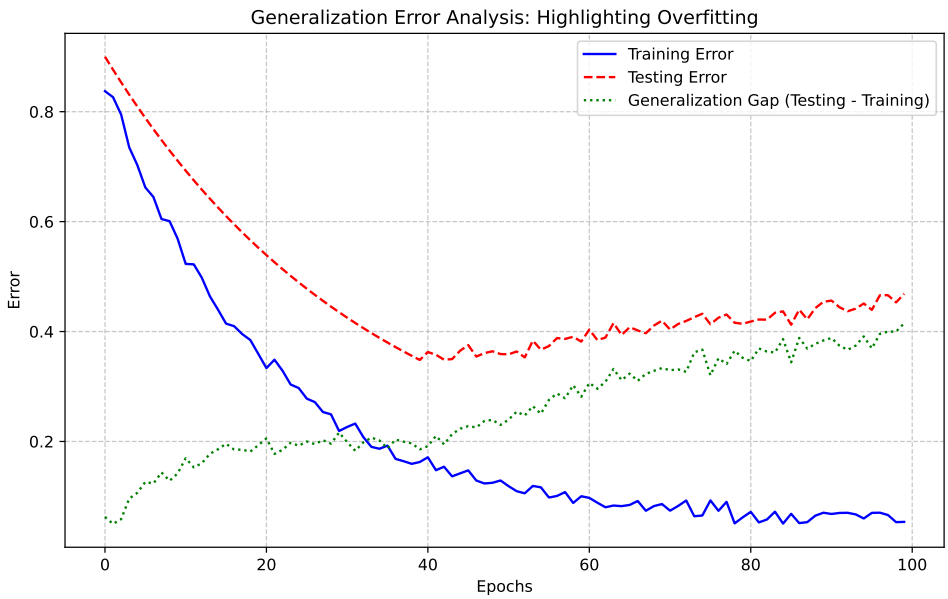

Figure 1.4: Generalization Gap in ML

ML robustness, however, is not just an optional consideration that will help improve profits in an ER forecasting setting. Due to the mathematics behind position sizing and

the market microstructure, non-robust methods lead to bankruptcy even in scenarios where the forecasting performance of the ML models significantly exceeds what is expected to make a profit. It is not only necessary that the methods deployed are accurate, but also that it is known precisely how accurate they are. For these reasons, the generalization gap (see Figure 1.4) should be closely monitored during the training phase in ER forecasts, treating performance as secondary to robustness.

1.3 Research Objectives

This study aims to propose methods to procedurally improve ML robustness in binary classification (BinClass) based ER forecasting for cryptocurrencies. Specifically, the statistical tools and measures of predictability borrowed from chaos theory and applied to financial markets and their effects on ML robustness is studied. The focus is on the utilization of these tools with a unique, robustness-first outlook, as well as procedures that have yet to be (or less extensively) explored on the crypto market, drawing inspiration from the existing literature on traditional financial markets.

There is an explicit aim to demonstrate and quantify the improvements in the generalization gap between the measures of error noted during the validation and testing phases in order to facilitate informed decisions regarding the use of these tools and inspire further research on the use of these tools in classification based crypto ER forecasting.

1.4 Scope

The data obtained from the selected CEX was assumed true and connectivity issues with websockets or REST APIs, as well as software architecture were kept out of scope of the analysis.

In line with the existing literature on BinClass-based ER forecasting, the return magnitude is not taken into consideration with the assumption that returns are normally distributed.

A random subset of recent listings in a CEX is considered with the assumption that the expected market efficiency would be lower. However, the recency was limited to include a minimum number of observations (outcomes) on the order or magnitude of 100,000 per process to counteract the potential bias introduced due to insufficient or unbalanced data.

The statistical significance of the findings were calculated under the assumption that each outcome would be an i.i.d. Bernoulli process (and outcome cluster a binomial).

The market microstructure model was based on the inner workings of the specific CEX selected (Binance) while small differences may be present based on the platforms used.

1.5 Contributions

This study contributes the following to the existing body of knowledge:

- The use of the Hurst exponent as a market-regime filter on ensemble classifiers ran on **synthetic** cryptocurrency ER processes,
- Demonstrating the **robustness impact** on multiple time frames of a given dynamical system, for multiple dynamical systems,
- Conducting BinClass based ER forecasting in the crypto market using **tick-based summarization** rather than time-based summarization.

1.6 Thesis Outline

This master's thesis presents the research conducted in 6 main chapters:

- **Chapter 1:** Introduction
  This chapter presents the research objectives and overall scope of the study, as well as summarizing the research framework.
- **Chapter 2:** Literature Review
  This chapter reviews the existing literature on the subject, including research on traditional financial markets and the crypto market, as well other non-linear systems. The gaps in the literature, such as the lack of robustness-focused research on classification based ER forecasting as an end in itself, as well as methods utilized in traditional markets that have not been tested in the crypto market, are highlighted.
- **Chapter 3:** Mathematical Foundations
  This chapter provides a deep-dive into the trading microstructure in the cryptocurrency market and derives the mathematical equations necessary to understand and properly evaluate machine learning performance and robustness in BinClass based ER forecasts. It extends itself to include the detailed study of synthetic processes, clearly demonstrating the mathematical necessity for robustness.
- **Chapter 4:** Methodology
  This chapter details the practical steps taken to carry out the experiments necessary for the study, including the machine learning models utilized, data collection and preprocessing steps unique to the contributions of the thesis, the cloud infrastructure used in carrying out the experiments and the statistical procedures used to evaluate the outcomes of the experiments.
- **Chapter 5:** Results and Discussion
  This chapter collects, provides visual representations for, and evaluates statistically the outcomes of the experiments, as well as reorganizing the findings of the study for conclusions to be drawn based on the empirical results.
- **Chapter 6:** Conclusions and Future Work
  This chapter provides a final synthesis of the findings, evaluates the potential impacts of the findings and notes the potential for future work to be conducted based on the findings of the study.

2 Literature Review

This chapter goes into the existing academic literature on classification based ER forecasting, the integration of tools from nonlinear dynamics and chaos into financial markets, robustness focused ER forecast research, the common ML models used in the literature and their performance metrics, ensemble models and research on traditional financial markets.

2.1 Introduction to ER Forecasting in Financial Markets

Outside of negotiations conducted between humans in exchanging goods and services since pre-historic civilizations and fluctuations in coin values with changing metal contents, there is no significant historical evidence that points to humans attempting to forecast exchange rates of currencies or other assets of value up until the classical era [5]. Moreover, systematic attempts at predicting exchange rates appear to have emerged in the medieval era with Italian merchant banking and the appearance of the first forward exchange contracts around the 13th century [5] [6].

In the early modern period, in addition to the legal recognition of the forward contracts (with the currency, exchange rate and settlement date information) in the 15th century, further methodical thought started to emerge in attempts to predict currency values and inflation, with evidence pointing to the presence of market letters (publications that contain information related to financial markets) and brokers [6].

Starting around the 1800s, exchange rate forecasts were conducted using quantitative information such as interest rates, costs of exporting gold and durations of delay in shipping [7]. In this period, with currencies being pegged to gold and silver there was still limited volatility present in the markets [7] [8]. However, FX volatility exploded with WW1 and ER forecasting started to become an integral part of international finance in the interwar period (1918-1939) due to the increased demand [9] [10]. Speculation among individual traders and economists were documented where economic indicators are used systematically [11].

With WW2 (1939-1944), which eventually led up to the Bretton Woods system, ER forecasts attracted more academic attention, where early theoretical analyses of currency movements were conducted [11]. With the Bretton Woods Agreement, FX rates were fixed to the US Dollar, in an occasionally adjustable manner, and the concepts of fundamental and technical analysis emerged in the period following this agreement, where economic indicators and political data became structurally relevant in ER forecasts [11] [12].

With the end of the US Dollar's gold-equivalence in 1971 and the end of the Bretton Woods system, FX rates were subject to increased volatility, which stimulated efforts for this volatility to be captured [11] [13]. In the two decades that followed, many statistical and quantitative methods were developed, significantly expanding the field of econometrics [14]. While computing was also on the rise in this period with personal computers coming into the picture in the 70s, computers started making a big difference in ER forecasts in

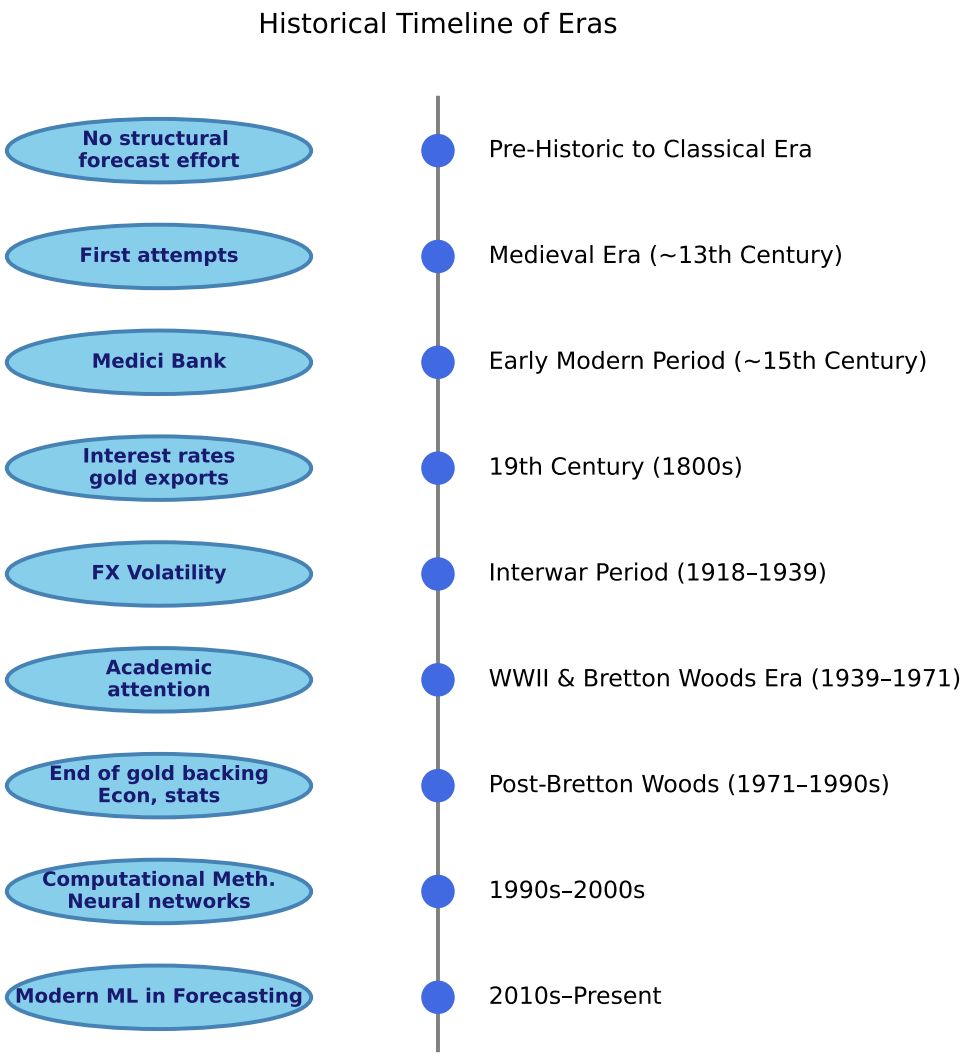

Figure 2.1: The Evolution of ER Forecasting

the 90s with higher frequency data and machine learning eventually resulting in a complete paradigm shift in how ER forecasts are carried out [15].

In the 1990s and 2000s, ML models such as neural networks (NNs) were utilized in attempting to capture ER dynamics programmatically [16] [17]. Moreover, technical analysis gained popularity since tools such as technical indicators could be easily adapted to be used in automated systems for trading, following computer simulations that attempted to assess model performance and risk [17]. However, limitations such as computing resources and data availability have, despite the improvements in comparison to the previous periods, relatively slowed down the progress of ER forecast research in this period.

In the 2010s, with major improvements in access to data and computing resources, as well as improved access to a wide range of financial markets through digitalization, ML and artificial intelligence (AI) based solutions started dominating the space, incorporating techniques such as sentiment analysis with natural language processing (NLP), high frequency arbitrage, using satellite data as features and much more [18] [19] [20]. Complex

ML methods such as deep learning and reinforcement learning were developed and there was significant diversification in the meta-level frameworks used in ER forecasts [18] [20].

2.2 ML Approaches in ER Forecasting

With improvements in computing power and the widespread availability of data, ML started dominating the ER forecasting space, both academically and privately. A brief summary of how ML can be applied to ER forecasts and the associated challenges can be explored via the following discussion points: *Supervised Learning, Unsupervised Learning, Reinforcement Learning, Hybrid and Ensemble Models, Feature Engineering and Data Preprocessing, Challenges and Considerations.*

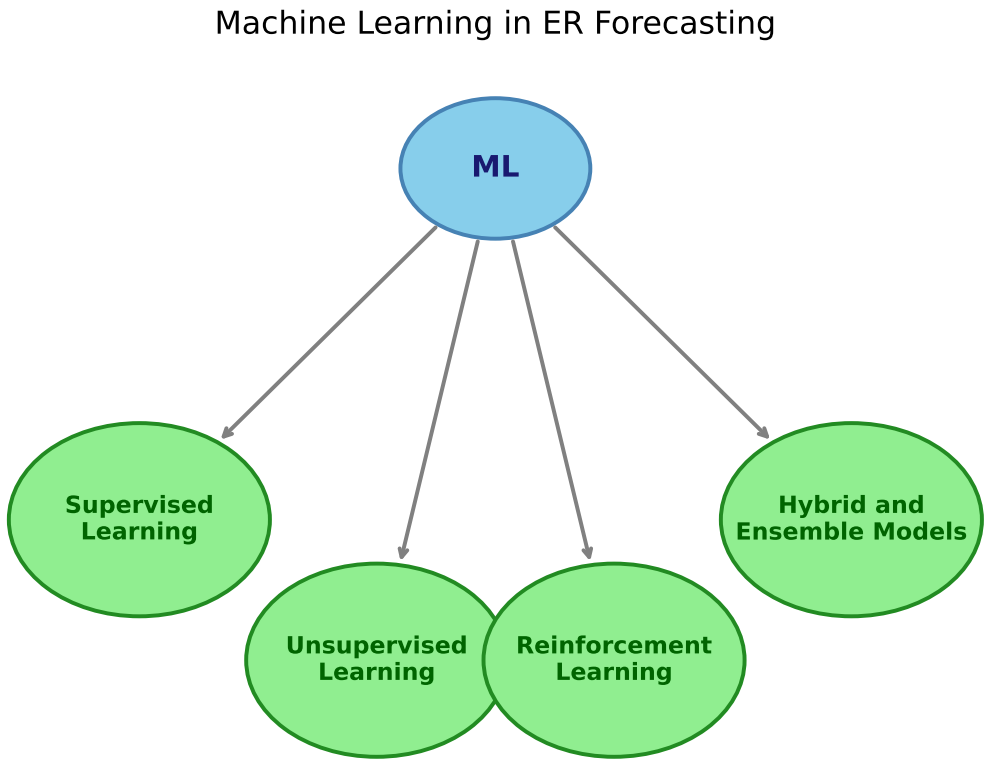

Figure 2.2: Machine Learning in ER Forecasting

2.2.1 Supervised Learning

Supervised learning (SL) is frequently seen in ER forecasting where ML models are trained using labeled historical data and later deployed real-time to forecast the dynamics of the process under investigation. Supervised learning typically refers to either *Regression or Classification.*

Regression Models in ER Forecasts

Regression is typically used to conduct continuous variable forecasts, such as predicting what the price of a cryptocurrency would be, as a floating-point number, in terms of the

base asset. In this scenario the performance is often measured using statistics that quantify how close the model can get to the ground truth, or what portion of the variability is explained by the model. Many regression models have been used in ER forecasts across a wide range of financial markets [21] [22].

Models as simple as LR remain relevant in the scientific literature on ER forecasting in many financial markets, including crypto, still to date [23] [24]. Many researchers use LR as a baseline model for regression performance comparisons [25], while others use LR in combination with modern methods, or even as the main ML model [24], depending on the study setup.

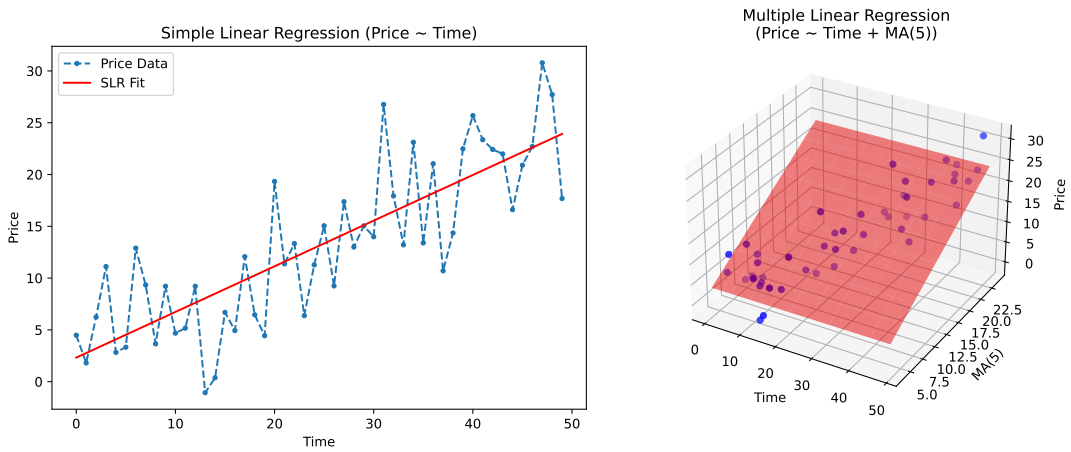

Figure 2.3: Linear Regression Fit (SLR / MLR)

For instance, both univariate (SLR) and multivariate (MLR) linear regression has been used as baseline models to neural networks in predicting cryptocurrency prices for coins such as Bitcoin (BTC), Ethereum (ETH), Litecoin (LTC) [26]. In this study, lagged prices and volumes were used as features, as they are in many other works in this area as well. LR has also been used extensively in combination with other methods, such as neural networks. For instance, researchers have used trend lines approximated by LR as features in multilayer perceptron (MLP) regression in the FX market [27].

Support vector regression (SVR) is another method that has been studied extensively in an ER forecasting setting. SVR attempts to fit functions to the price action that keep the error margin below a certain point and minimize the error outside of those bounds, instead of the errors everywhere (as seen in LR) [28]. For instance, SVR has been used alongside technical indicators for forecasting BTC prices [29]. B-share stock price forecasting (mainland China) has also been studied using SVR [30]. Commodity (gold) price forecasting using SVR has also been conducted where a hybrid model with a differential evolution optimizer was deployed [31].

Random (RF) Forest regression is another method that is commonly used in an ER forecasting context. RF's are originally used for classification but can be adapted to be used for regression tasks by taking the average of the multiple tree predictions [32]. RF regression has also been used in cryptocurrency ER forecasts alongside the FX market and commodities [33]. Technical indicators and candle patterns were often utilized as features in studies using RF regression, and the model was frequently compared to Long-Short

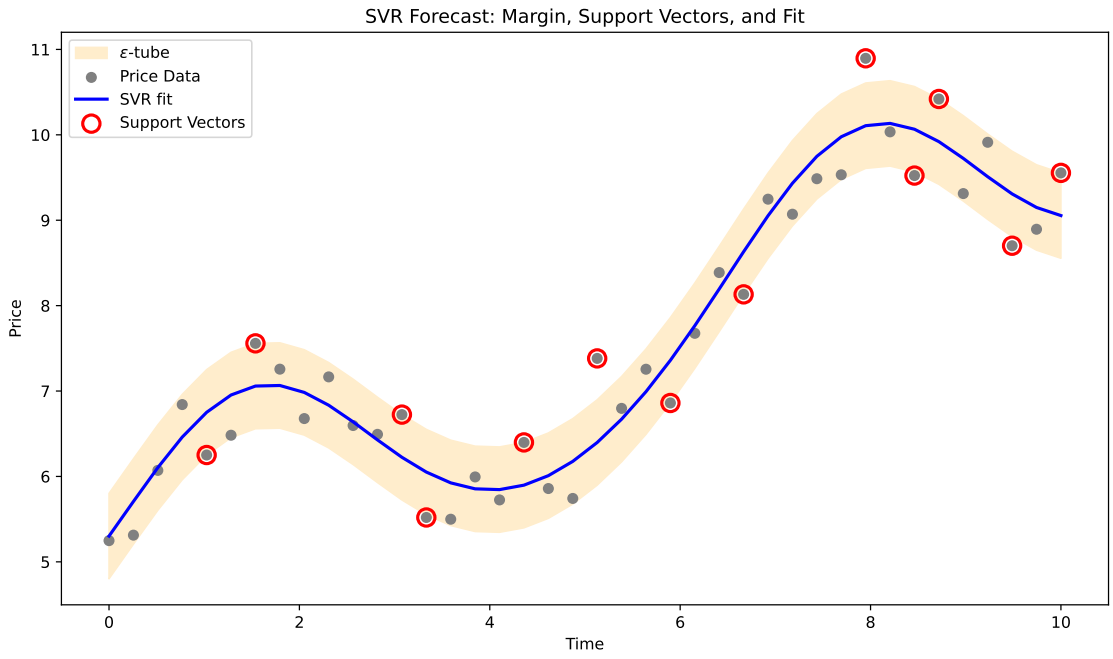

Figure 2.4: Support Vector Regression (SVR)

Term Memory (LSTM) and Autoregressive Integrated Moving Average (ARIMA) models, with varying degrees of success [33] [34].

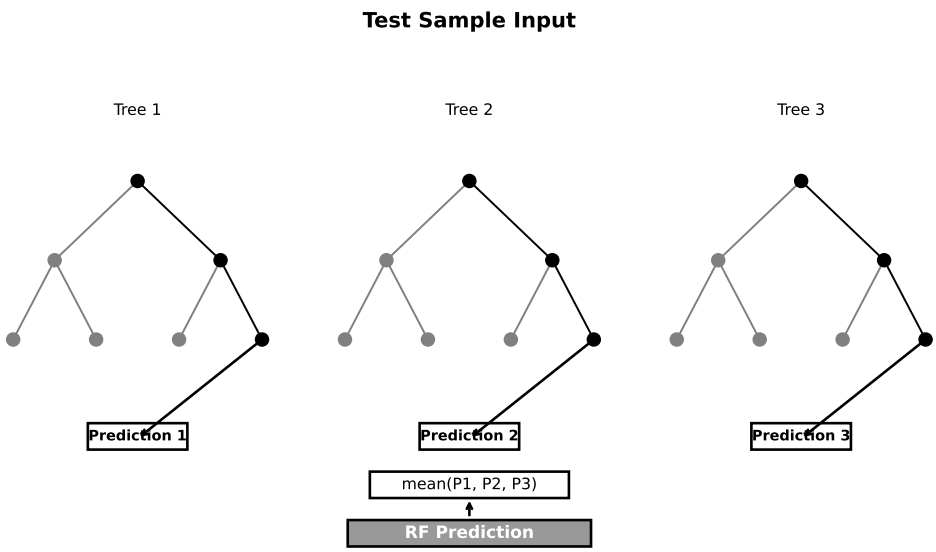

Figure 2.5: Random Forest Regression

Gradient Boosting Machines (GBMs) combine decision tree based predictions and create an ensemble model to minimize errors [35], and they have been used in predicting cryptocurrency ERs, in addition to FX, stocks and commodity prices [36]. LGBM and XGBoost are popular examples using GBMs [37], with their own separate tree growth strategies (XGBoost with level-wise or "vertical" tree building and LGBM with leaf-wise or "horizontal" tree building), with different compute efficiency and robustness profiles.

For instance, GBM regression has been used in electricity price forecasting and performed better than ARIMA models in normalized error outputs [38]. In the FX market, XGBoost regression has been used for ER forecasting on the USD/IDR pair, with outstanding results in RSME [39].

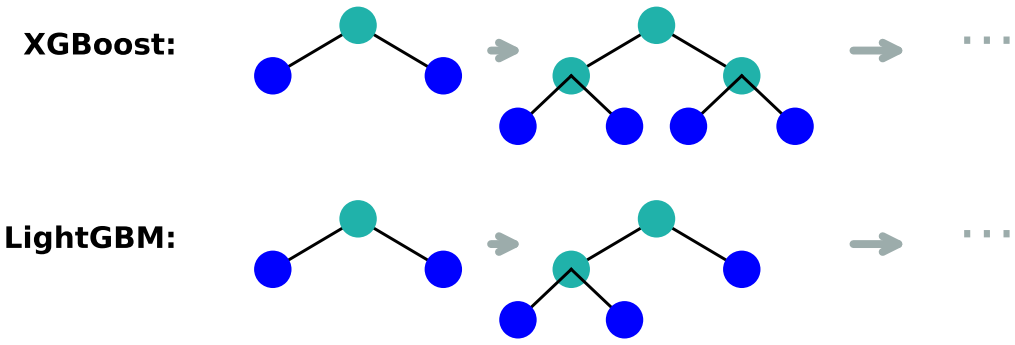

Figure 2.6: XGBoost vs. LGBM

XGBoost and LGBM have also been used in the crypto market for regression tasks instead of classification by various modifications. Similar features have been used, such as technical indicators and volume. One paper on XGBoost regression has used tweets and tweet sentiment as features to predict ZClassic (ZCL) prices [40], while another used worked on BTC and ETH prices and found the LGBM forecasts to be more robust than other methods [41].

Neural networks imitate the human brain's neuronal structure and can be used for a wide range of regression and classification tasks [42]. They have been used in ER forecasting for a very long time [43] and have evolved in their applications to the FX market, stocks, commodities and cryptocurrency markets [44] [45]. In terms of regression-based ER forecasting, feedforward neural networks (FFW NNs), convolutional neural networks (CNNs), recurrent neural networks (RNNs) as well as subtypes of these models (such as LSTM and GRU as subtypes of RNNs) have been used with varying degrees of complexity and success [45] [46]. Deep neural networks have also been shown to be able to outperform most traditional methods, as well as crypto buy-and-hold strategies, which are difficult to outperform given the explosion in crypto market caps [47].

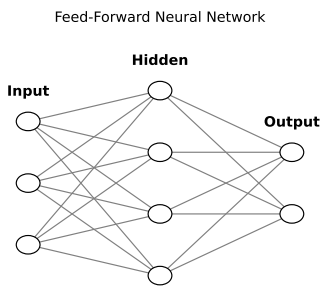

Figure 2.7: FFW Neural Network

For instance, FFW NNs (see Figure 2.7) have been used in a wide range of applications in FX, stocks and cryptocurrencies. FFW NNs were used to predict Borsa İstanbul (BİST) stock prices in Turkey and found that FFW NNs that were built using BRICS indexes performed well [48]. FFW NNs have also been used in the FX market as a baseline model on AUD/USD price forecasting, where their performance was compared to LSTM based methods [49]. FFW NNs have also been used in the cryptocurrency market, forecasting both BTC and altcoin prices, studying the impacts of volatility on market predictability and market efficiency [50].

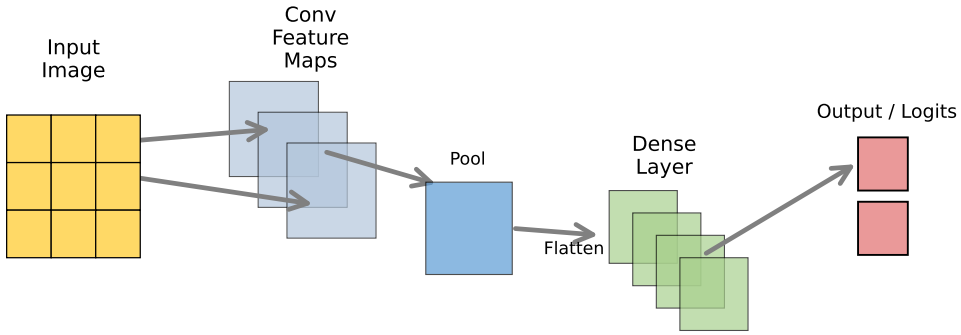

Figure 2.8: Simple CNN Diagram

Although mainly developed to process visual data, CNNs are also used frequently in ER forecasting [51]. These models can be used to extract features from images of market data and feed these features into other models, or be used on raw or transformed market data to directly make forecasts. Improving on related networks such as deep belief networks (DBNs), in the FX market, it has been demonstrated that CNNs have been successful in predicting a wide range of currency pairs [52]. CNNs have also been used in combination with LSTM in hybrid models in the FX market, reporting increased accuracy compared to individual CNN and LSTM based predictions [53] [54].

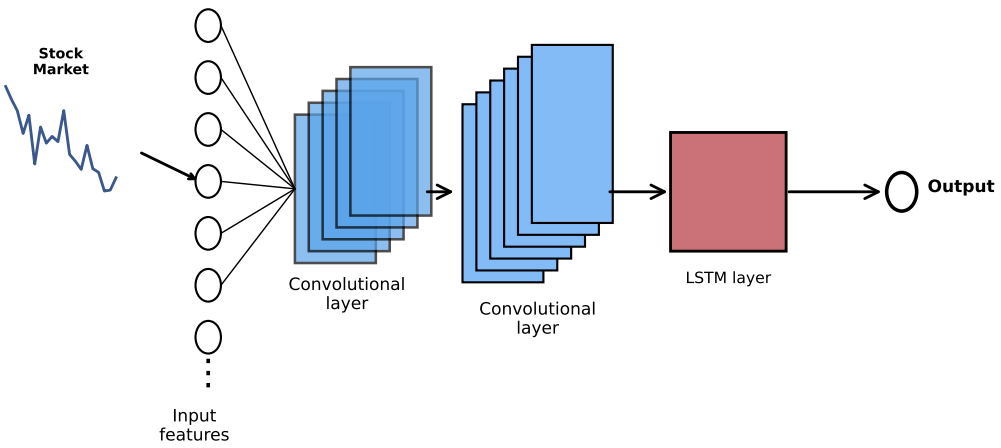

Figure 2.9: CNN-LSTM Hybrid Model for ER Forecasting of Stocks

The LSTM-CNN hybrid model has also been used extensively in stock price forecasting, where CNNs were used to extract features and feed them into the LSTM model for forecasting [55]. These models have outperformed traditional methods such as using ARIMA and SVR, as well as LSTM methods on their own. This hybrid model has also been used in crypto markets, predicting BTC, ETH and LTC prices, noting improved performance on the aforementioned traditional methods [56].

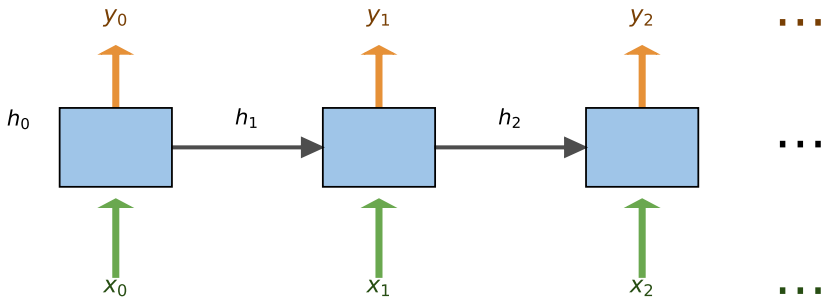

Figure 2.10: Recurrent Neural Networks

RNNs, on the other hand, are, to this day, a very frequently used NN architecture in ER forecasts, as they are built to deal with time-series data, and are able to utilize their "memory" state from historical data to make accurate predictions [57]. LTSM and GRU models have been found to outperform other RNN architectures in general and most of the literature on RNNs include LSTM and GRU based models.

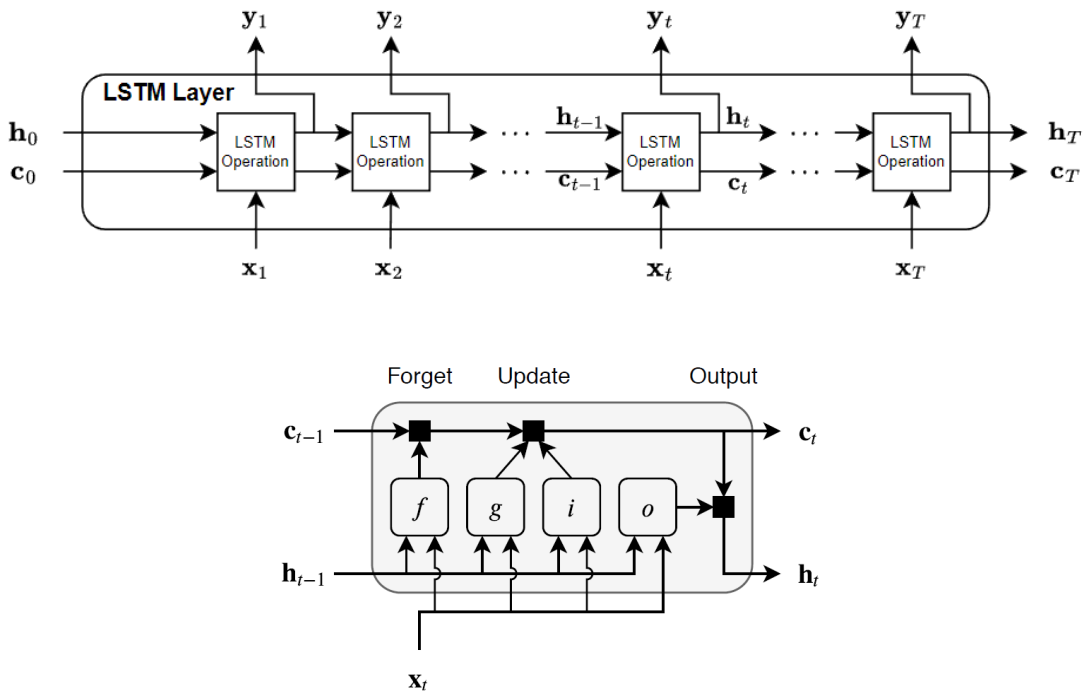

Figure 2.11: LSTM Layer Example / LSTM Operation [58]

For instance, In the stock market, RNNs have been used to predict index prices, finding that LSTM models outperformed other ANNs and various tree-based models, as well as basic RNN models [59]. GRU models have also been used in stocks and it was shown that they can yield similar performance to LSTM models and simple RNNs in individual stock prediction in regression error rates [60]. On average, LSTM models are expected to take more computing power and GRUs, since they have less parameters, can be a good alternative for low-latency applications as well [61]. LSTM and GRU based models have also been studied in the crypto market, outperforming traditional models in ER forecasting for the cryptocurrencies BTC, ETH and LTC [62]. They have also been shown to perform similarly in the FX market, depending on the setup [63]. However, the added complexity in the LSTM models can add notable boosts in performance and this model has attracted significant research attention due to performing very well in ER forecasting contexts.

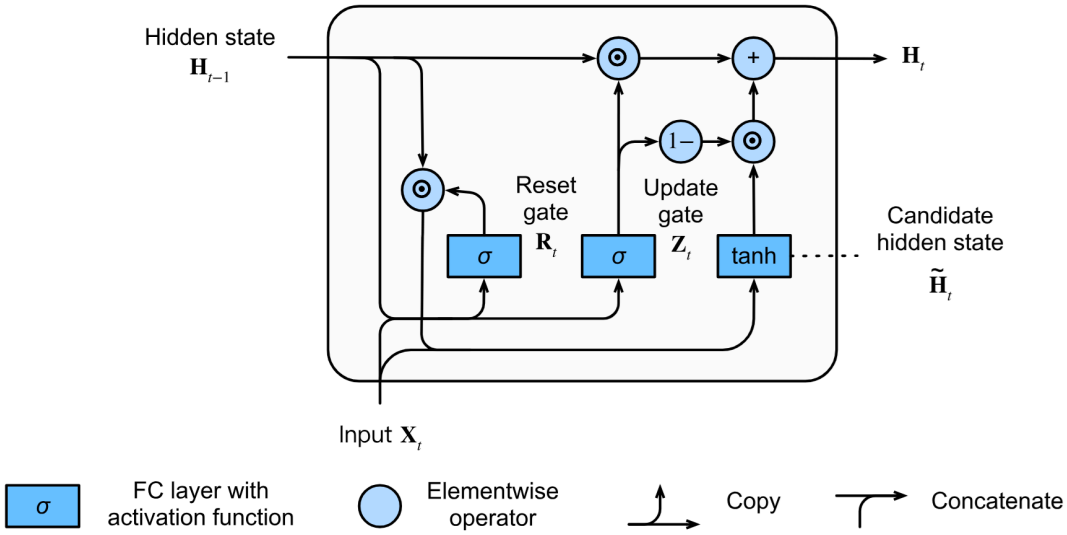

Figure 2.12: Gated Recurrent Unit (GRU) Operation [64]

Although advanced RNN architectures such as GRU and LSTM models have dominated the space for a while, their ability to counteract the vanishing gradient problem is, although impressive, limited at very long time frames. In order to solve this issue, transformers have been used with promising results. Popularized in the early 2020s, transformers have been used in natural language processing (NLP) and computer vision applications, and can perform extremely well in using "self-attention" to make use of historical patterns that go very far back in time [65].

Transformers can also process features in parallel and take less time to train on GPUs, as well as outperforming RNN based models in terms of accuracy in various ER forecasting contexts. For instance, transformers have been used in predicting stock prices, outperforming LSTM and GRU based models in metrics such as RMSE, where transformer based architectures did extremely well [66]. However, in some instances their performance has also been shown to be lower in general compared to RNNs in the crypto market in an ER forecasting setting [67].

In summary, regression based ER forecasting aims to predict where the price might go in the future on a continuous scale. It has evolved from basic moving average based tech-

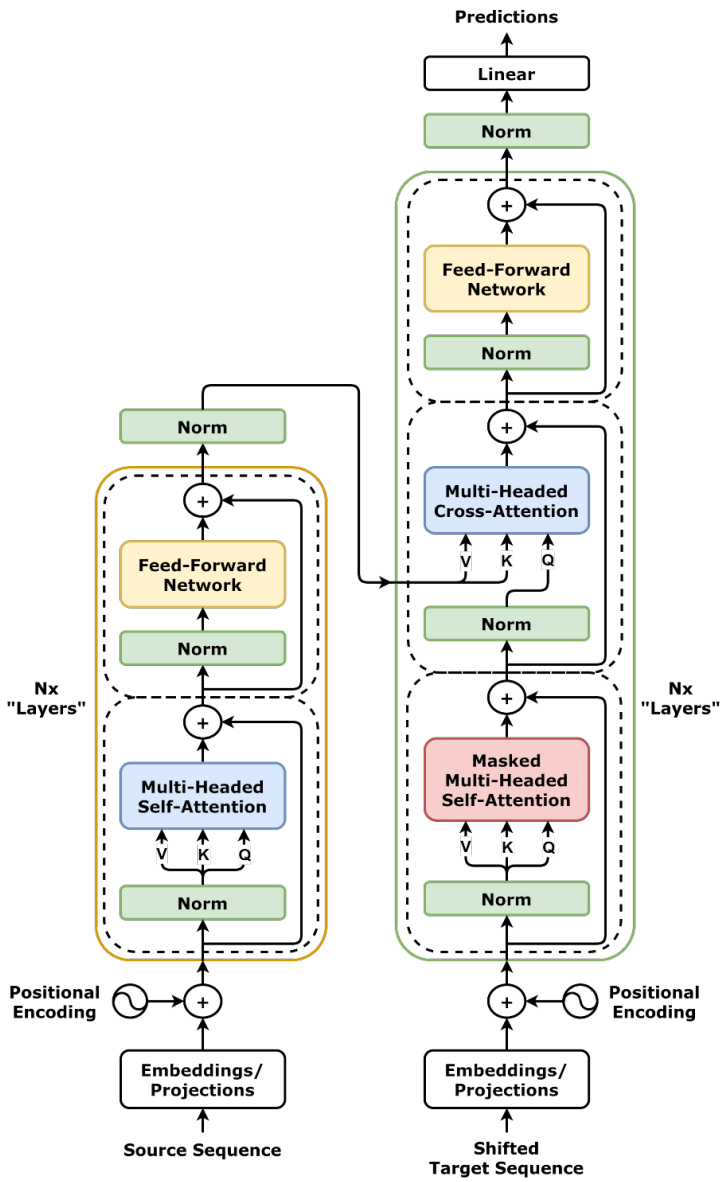

Figure 2.13: Example Transformer Architecture
[68]

nical analysis and linear regression to breakthrough models like advanced RNNs, as well as transformers, which are becoming the new state-of-the-art in the literature, and their effectiveness in capturing patterns in longer timeframes has improved over time.

Classification Models in ER Forecasts

Classification, on the other hand, is often used for dividing the data into a predetermined set of categories. For instance, it can be forecasted whether the market will be up or down in the next 24 hours, or whether a market movement of 5% would be observed in the upward or downwards direction first. The performance can then, for instance, be expressed using the relative frequency of correct predictions. Classification in ER forecasts can be multiclass or binary, as well as multidimensional, with single dimensional binary classification being the most common framework for ER forecasting, looking at directional

market movement.

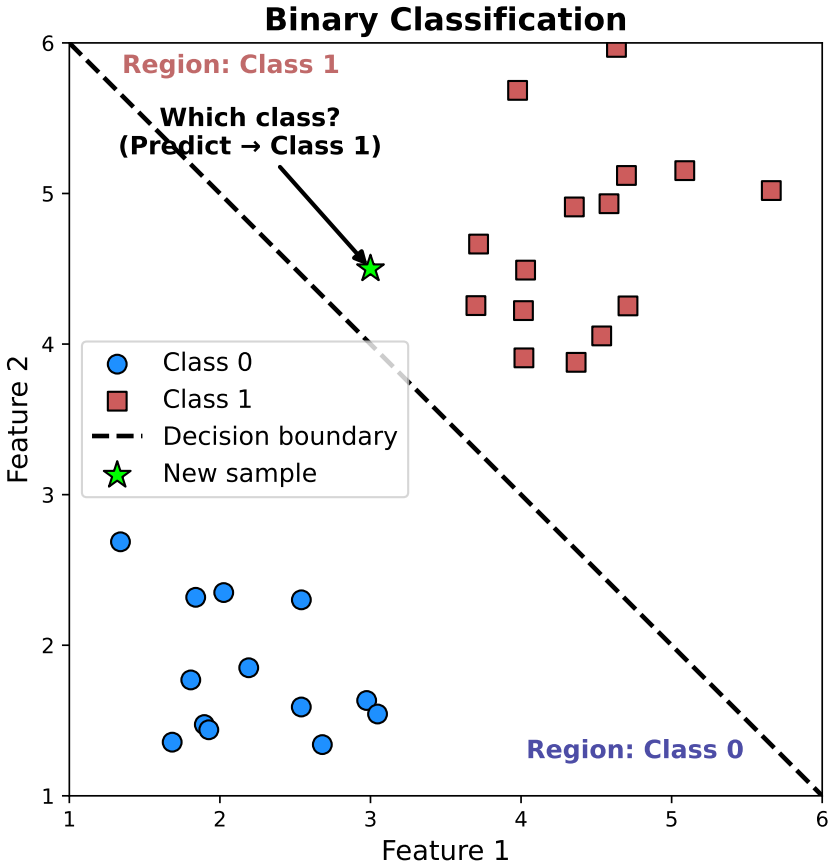

Figure 2.14: Binary Classification Model

Although not as popular as regression models, there is a growing body of research on using classification models for ER forecasting. For instance, logistic regression are utilized in stocks, commodities and the FX market as baseline and standalone models in ER forecasting, as well as in cryptocurrency ER forecasting applications, but they are often outperformed by tree based models and NN architectures [69] [70]. These methods will be discussed in their own section in far greater detail and are therefore skipped in this introduction.

2.2.2 Unsupervised Learning

Unsupervised learning applications can generally be described as ML application where the output dataset is not labeled, and the model identifies patterns in the dataset without explicit label-data from the user. Often used as a supporting tool to other ML methods, unsupervised learning applications have also been found to be used in ER forecasting applications. It has been proposed that an unsupervised learning paradigm can identify hidden patterns in financial data, in terms of the features used to make predictions as well as the very processes that are being forecasted, and prevent biases in ML.

Unsupervised learning methods can be utilized for outlier detection and segmentation of

market conditions; such as market sentiment analysis, price movement direction analysis like bullish/bearish regimes and relative volatility analysis in markets that grow rapidly and trading volume expands exponentially over time (such as the cryptocurrency market). For instance, in the stock market, unsupervised learning methods were used to build portfolios [71], perform data augmentation for increasing robustness [72], and extract features [73] to be later fed to various predictive models to increase model performance, as well as combined with technical indicators to classify data into oversold and overbought groups to facilitate accurate ER forecasts. They have also been used in the FX market to demonstrate how changing market conditions can affect user behavior [74].

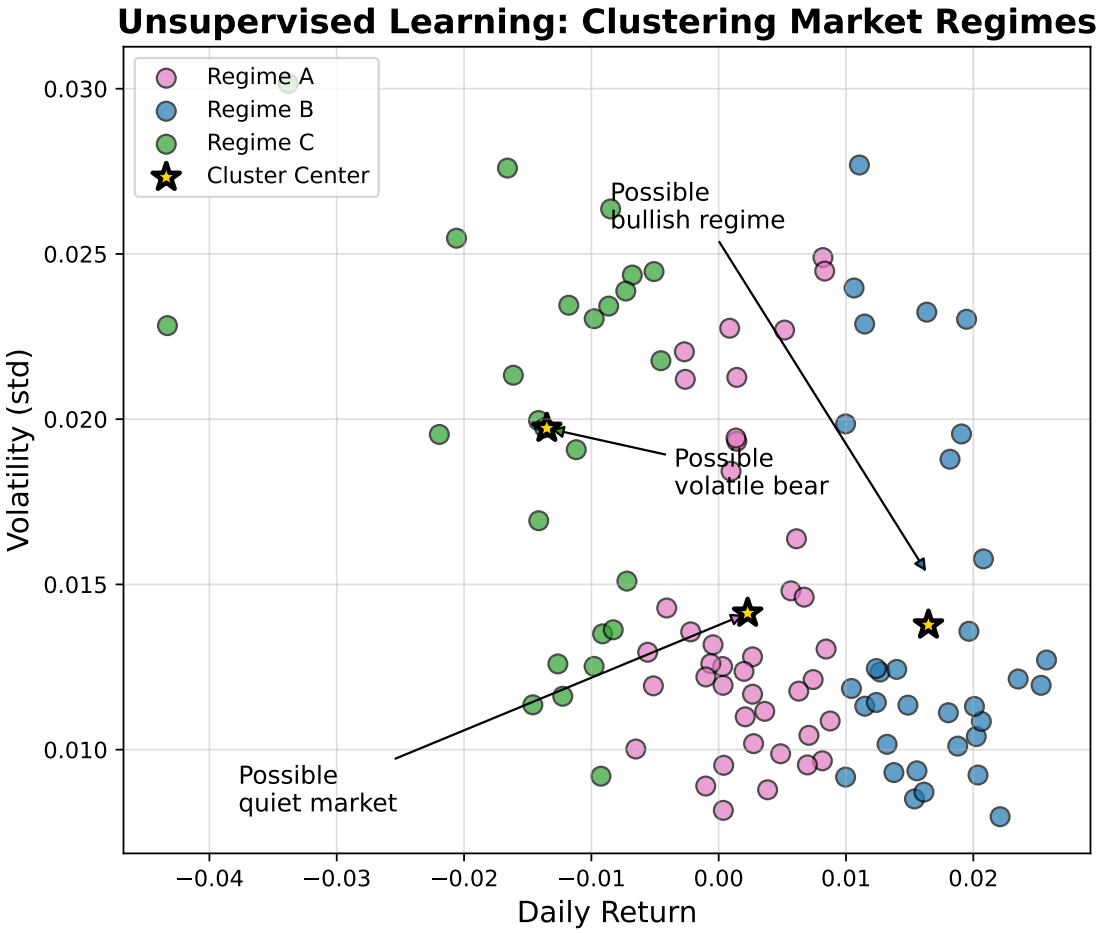

Figure 2.15: Clustering (Unsupervised) for Market Regime Identification

The cryptocurrency market has also seen the application of unsupervised learning methods with papers focusing on clustering to identify changing market regimes [75], as well as catching pump-and-dump schemes based on trading volume changes that are unusual [76]. These applications serve as both performance improvement tools and risk management and safety instruments, capturing details that may not be possible to catch in supervised learning applications.

2.2.3 Reinforcement Learning

The reinforcement learning (RL) paradigm often involves an agent that goes through a number of trials interacting with a system with the objective of learning from the rewards

built into the architecture when a desired action is taken by the model, improving the model's performance in the task it's being trained to do with each iteration. RL based models do a great job at building complex responses to complex input without the need to explicitly build the complexity into the ML model during training.

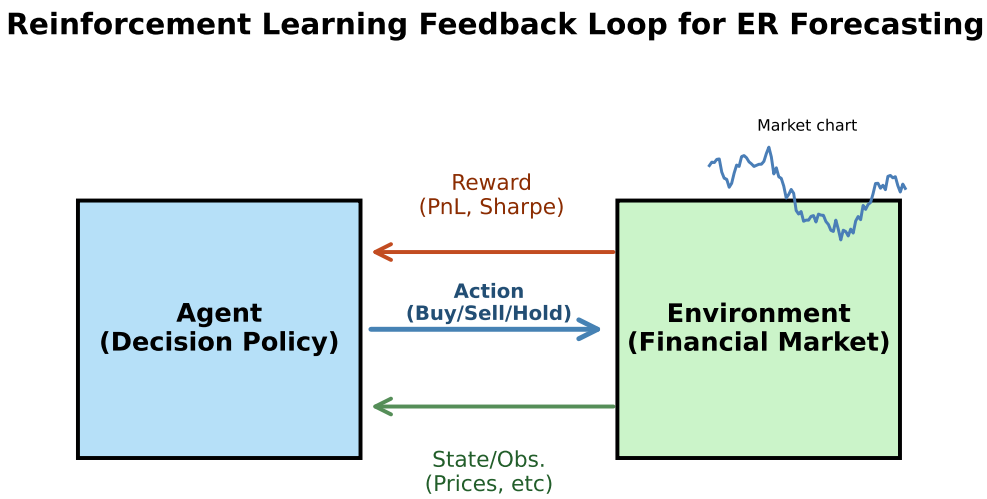

Figure 2.16: Reinforcement Learning for ER Forecasting

There are deep and non-deep RL approaches where non-deep approaches can be used to learn to play simple games such as tic-tac-toe, where the model updates the actions that are taken based on the result of the game, improving the player's performance after a large-enough number of trials have been completed. In ER forecasting however, often deep RL architectures are utilized, where deep NNs are used to approximate the functions used in the architecture for navigating a continuous probability space [47].

In RL applications used for ER forecasting, both value based and policy based methods may be utilized. For instance, a Deep Q-Network agent can be trained to perform buy and sell actions on a cryptocurrency in an attempt to maximize the expected profits from the trading strategy being built, given the features available. Later, when the real features are presented, the agent will suggest the action that has the highest expected value of financial profit. This is also called a "value-based" method in RL. Continuous outputs, such as portfolio weights using multiple assets, can be obtained with policy-based methods, on the other hand, where the market state predicts the weight-map of the portfolio.

RL has been applied in the stock market for ER forecasting [77], where single stock or index (such as the S&P 500) prices are forecasted using Q-Learning [78], portfolio weights were set using RL agents with a Deep Deterministic Policy Gradient (DDPG) where buy-and-hold outcomes were outperformed [79], and ensemble models combining multiple ML methods showing robust performance in stock trading outcomes [47]. These methods have also been applied in the FX and crypto markets, where profitable high-frequency trading algorithms were built using different RL agents [80] [81]. In the crypto market, RL agents have shown significant positive outcomes in a portfolio optimization setting where RL agents have outperformed buy-and-hold strategies of static portfolios [82].

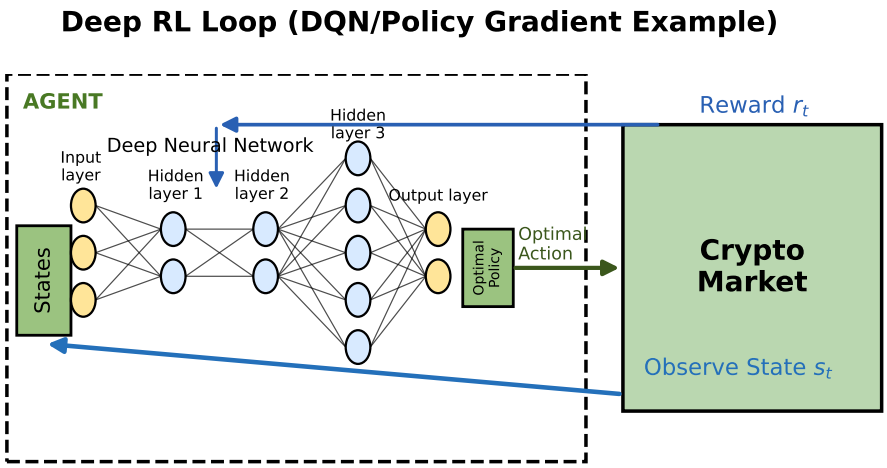

Figure 2.17: Deep Reinforcement Learning Using a Deep-Q-Network

2.2.4 Hybrid and Ensemble Models

Hybrid and ensemble models, in their essence, work by combining multiple methods and can perform better than the parts that make them up, individually, in an ER forecast setting. In this context, ensemble models often refer to multiple forecasts being combined or aggregated in some way, where hybrid models may refer to two models working sequentially, such as a CNN being used for feature extraction and an LSTM algorithm making the predictions.

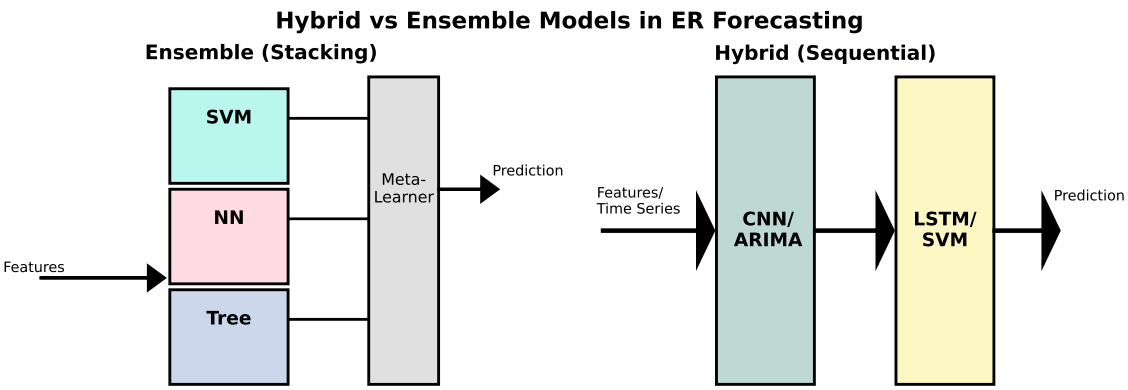

Figure 2.18: Hybrid Models vs. Ensemble Models

Hybrid and ensemble models have been studied in detail in ER forecasts, with notable articles showing improvements over using single methods in performance and robustness. Tree based models have been used very commonly in ensemble models, with stacking, where RNNs, and especially LSTM-based models, have been used very commonly in hybrid models following some form of feature engineering using another tool. Ensemble models have attracted a lot of research attention and it has been proposed that their prevalence would increase in the literature moving forward.

In the stock market, the CNN-LSTM hybrid model has been used extensively to show significant improvements to standalone deep neural network architectures [83]. In the

crypto market, hybrid models that combine traditional models like ARIMA with SVM have been used for ER forecasting of BTC [84]. Ensemble models have also been used in cryptocurrency ER forecasts, where tree-based models were utilized with a technique known as "stacking", where performance was enhanced significantly by combining multiple predictions using multiple classifiers [85].

There are other examples of the applications of hybrid models in ER forecasting, such as using ARIMA-based models to transform the ER dynamics into a smoother function to later use it with a NN-based prediction algorithm, using multiple base learners for conducting training with different segments of the raw data (feature or observation based), and much more [86]. In general, hybrid and ensemble models can provide cognitive diversity that may not be possible to achieve even with complex deep learning methods when used on their own, and improve the overall performance and robustness of the ER forecasting efforts in question.

2.2.5 Feature Engineering and Data Preprocessing

In various ML applications, including in ER forecasting, raw data goes through preprocessing steps being fed into ML models. This ensures that the ML models can function as intended as some standalone models can not give good results unless the inputs and outputs are scaled. In addition, financial data can be difficult to work with due to gaps and missing values in the data. For instance, in markets such as the FX market and the stock market, trading hours are limited and there is no trading on the weekends. This leads to gaps in the data regularly, which can become very difficult to deal with, especially during times of high volatility due to important news on the weekend, for instance (see Figure 2.19).

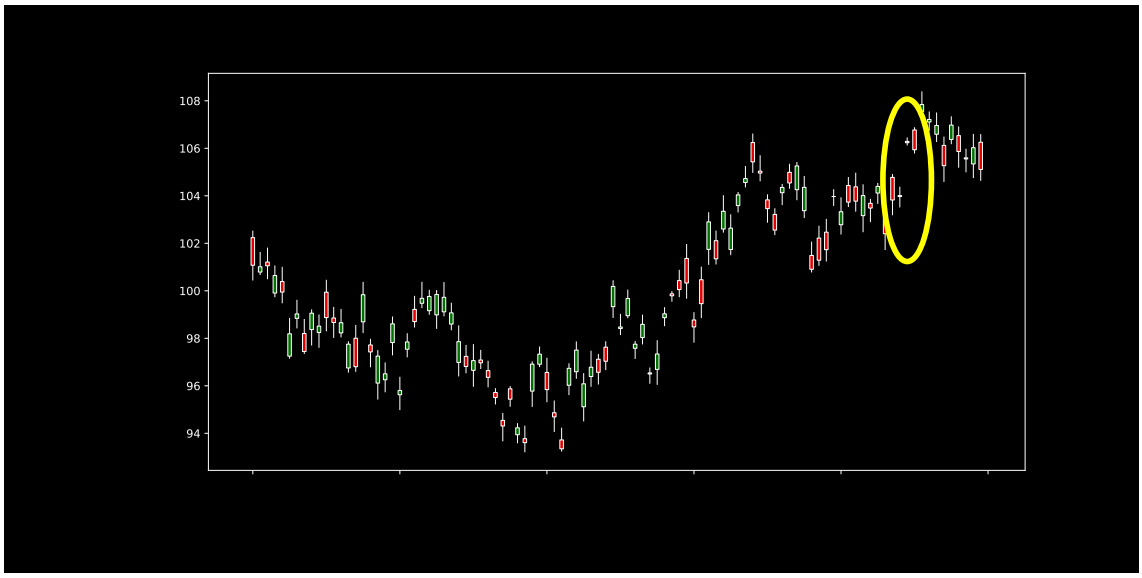

Figure 2.19: Weekend Gaps in Traditional Markets

The success of a research effort in an ER forecasting setting can depend heavily on the quality of the feature engineering performed, where lagging features and diverse sources of information can be considered [87]. In addition, mistakes during feature engineering, such as data leakage, can lead to serious trouble during deployment, and can be very difficult to detect in complex applications. Data leakage, in this context, refers to mistakes made

in the ML architecture that allows models to, during the training phase, get their hands onto data that would not be available to them until the testing phase, and conduct training with that data. This tends to result in extremely high performance in the lab, and disappointment in the real world. Very small mistakes in indexing or edge-case handling during the data wrangling process can lead to disastrous outcomes during deployment.

Unsupervised learning methods can also be utilized during the feature engineering process, as exemplified earlier in the literature review. Various market "regimes" can be defined through clustering, which can be used to influence how the forecasting agents decide to interact with the market. In addition, unsupervised learning techniques can be used to reduce the dimensionality (the number of input features) of a problem as well, improving training costs and potentially enhancing robustness. One example of this application is using Principal Component Analysis (PCA) in stock market ER forecasting to reduce the dimensionality of the problem and feeding the updated features to a tree-based algorithm, such as GBM [88] [89].

2.2.6 Challenges and Considerations

ER forecasting in financial markets remains a difficult task, despite the advanced ML models and rapidly evolving infrastructure available. ERs are non-stationary time series that are practically impossible to predict beyond a certain level of accuracy, and challenges with data collection, noisy data and volatility [90]. This non-stationarity renders most traditional statistical models obsolete and even may result in biased learning when ML models are not trained within time periods of sufficient length. In addition, the dynamics of the system can be very difficult to interpret with periodic gaps in financial data, due to dynamics inherent to the system such as weekend closures.

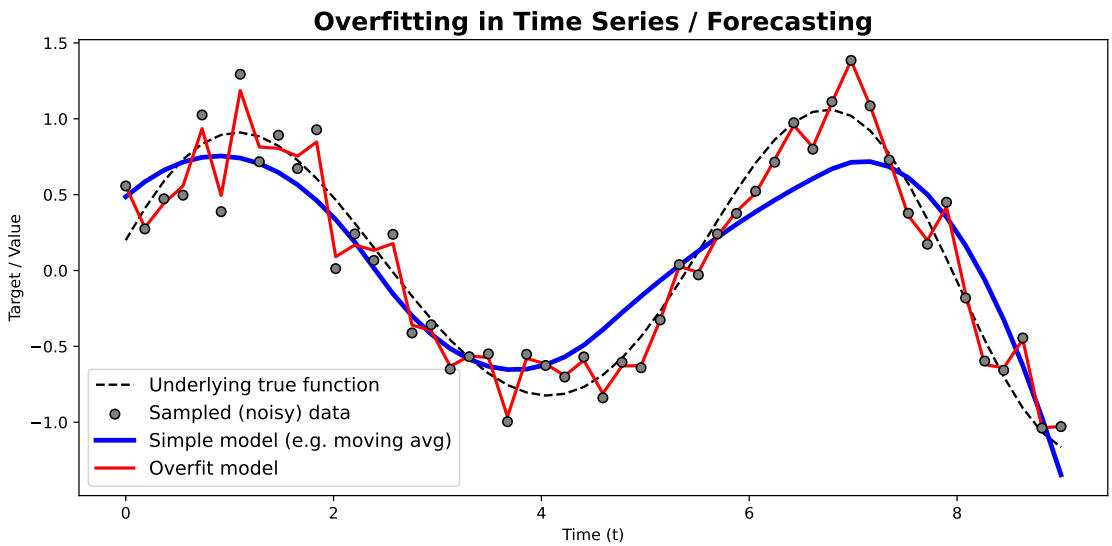

Figure 2.20: Overfitting in Forecasting Applications

The competitive nature of the ER forecasting ecosystem also makes it difficult to create methods that continue to work over long periods of time, making it necessary that the methods used are continuously improved. As profitable trading strategies are shared among a larger pool of market participants, the competitiveness decreases gradually [91]. Moreover, due to conflicts of interest, it can be speculated that it would be relatively diffi-

cult to arrive at a transparent and democratized research environment in an ER forecasting setting. In addition, errors in software architecture or faulty hypothesis formulation in an ER forecasting context may lead to overfitting (see Figure 2.20) and data leakage issues that can be extremely hard to detect.

In terms of data leakage, future data (from the testing set) may “leak” into the training environment where the ML models incorporate information that they would not have access to in a live trading setup, leading to disappointing results after deployment [92]. In addition, due to the ever-changing dynamics of the system, patterns that are captured using ML models may disappear from the market dynamics entirely, leading to past performance not being translated to future profits. In this context, developing robust methods that stay relatively resistant to out-of-distribution data, as well as comparing model performance to smart benchmarks such as buy-and-hold strategies in the cryptocurrency market where backtests are run in an environment where the coins may see extreme increases in value consistently, is critical in order to ensure the success of the ER forecasting efforts in question.

2.3 ER Forecasting with Binary Classifiers

Binary classifiers have also been explored in ER forecasting in a number of financial markets for predicting the directional movement of the market and classifying the overall state of the market into regimes. It is also possible to create more complex trading setups using binary classification with use cases in futures and options trading. Binary classifiers have been suggested to be easier to track and modify, as well as provide higher trading performance than regression by some researchers, and there is growing research on classification for ER forecasting, separate from market segmentation, despite most research still focusing on regression in ER forecasting.

The data preprocessing steps in binary classification are crucial, not only for the data preparation and feature engineering steps, but also for defining the “outputs” of the process under investigation. For instance, if the binary classifier is predicting whether the market would go up or down after a certain time period following a prediction, the price differences must be collected in a periodic fashion. If the binary classifier is predicting the directional market movement in a chronological fashion given a threshold, then it may be necessary for the researcher to deal with a larger amount of data (down to the “tick data” level), instead of simply using summarized data (presented in the form of candles).

There are additional considerations in summarizing ER data, where the price action can be summarized in a time-based, tick-number-based or a volume-based fashion. Research has shown that non-traditional ER summarization methods may allow for hidden patterns to be recognized and result in improved forecasting outcomes [93], with some research found in the literature using these "tick-candles" [94], although most of the research continues to focus on "time-candles". In this context, the selection of the size of the groups become relevant and dependent on the process under investigation as well.

It must be noted that, since most of the ER forecasting research available focuses on regression, and since cryptocurrencies are still a small part of the whole algo-trading ecosystem with markets being largely unregulated and trading volumes being low [95], there are significant gaps in the literature in terms of methods that leverage classification in predict-

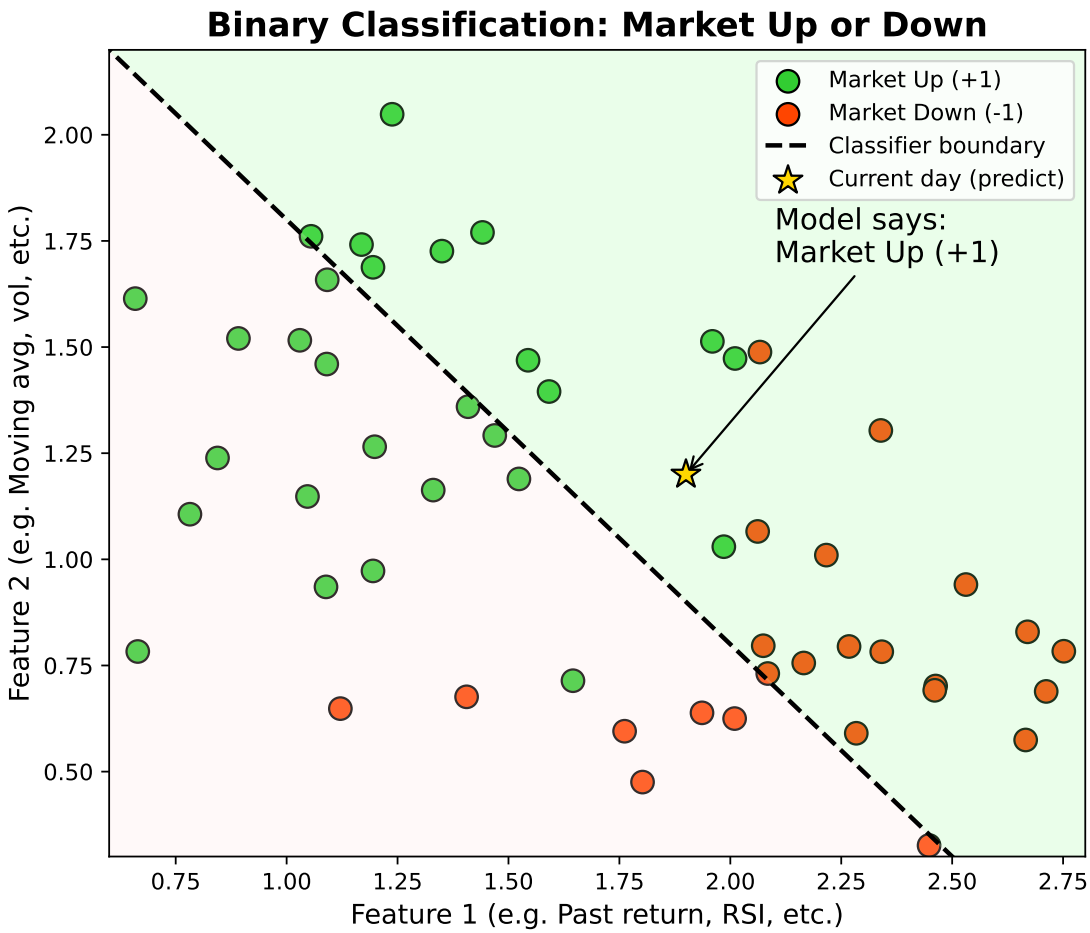

Figure 2.21: BinClass-based ER Forecasting

ing the movements of the crypto market as a whole or single cryptocurrencies as separate dynamical systems. However, there is more research present on other financial markets and it can be summarized as follows.

Basic binary classifiers, such as the SVM classifier, have been used as standalone models in the FX market to predict directional movement in the EUR/USD pair with some success [96]. In this article, the outputs were defined as "1" for any magnitude of increase or stagnation in price, and "0" for any magnitude of decrease in the price. This is a common trend in BinClass-based ER forecasting applications, and the risks of predicting smaller magnitudes of desired movement more frequently is largely dismissed with the assumption that returns would have a similar distributions in cases of correct or false predictions.

Though seen as standalone models, these basic models, such as SVM and Logistic Regression are often used as benchmarks to more complex methods. In one article focusing on the Korean stock market, tree based models such as random forest and GBM were shown to significantly outperform simpler models such as Logistic Regression and k-Nearest Neighbours (kNN) [97]. In this article, similarly, no movement magnitude was set in either directions.

In an effort to set a magnitude for profitability, a slightly different binary encoding

framework was suggested, where increases of a minimum of 0.3% was suggested for a "1" classification and "0" otherwise [98], however this reduces the interpretability of classification measures such as precision. The static threshold will not only create unbalanced classes at the outputs, but also create a different level of unbalance based on stock volatility.

Another article has compared the performance of deep neural networks and transformers in a BinClass setting in the US, China and Hong Kong stock markets, with directional accuracies approaching but remaining below 60% [99]. Although switching from the continuous regression framework to the discrete classification framework was highlighted and regularization loss is used in this article, and the models proposed are tested across different cycles (bull and bear) in different markets, the generalization property of the models are not put to a solid test as single market indices are used and there is potential that the results could be incidental.

A more advanced binary encoding framework was suggested, this time in an article focusing on the cryptocurrencies BTC, ETH, BCH, XRP, DASH, and ETC [100]. Here, the magnitude of the desired movement would increase with increasing time after the last observation for a "1" class to be assigned. An RNN (LSTM) layer was utilized with k-fold cross-validation to improve robustness. The article used the f1-score to compare the predictability of various cryptocurrencies, however, interpretability was also low in this work due to the binary encoding methods employed.

Another article using ANNs, Logistic Regression, SVM and a few other models focused on cryptocurrencies, using the typical symmetrical binary encoding method with no magnitudes, has reached accuracy levels of slightly above 60% as well [101]. However, the applicability of the methods in the article would yet be in question due to the selection of the trading period coinciding with the highly volatile 2017-2018 peak in cryptocurrencies, as well as with lack of interpretability in terms of comparing model performance against a buy-and-hold strategy.

Another article that focuses on BinClass in cryptocurrency ER forecasting compares the performance of various classifiers [102]. Although containing promising performance given the models used, the duration of the data only being 1 month and no justification being present for the selection of pairs that may be considered low-liquidity may point to risks of data snooping.

Given the contents of the works discussed above, although high quality studies are present in the literature, it can be argued that there is a significant gap in literature where cryptocurrency ER forecasting is studied with robustness in mind. It should also be noted that the cryptocurrency studies discussed above has made their CEX selections as Bitfinex or Binance.

2.4 The Problem of Generalization and Robustness in Financial ML

Due to the number of actors in the system, predicting the future value of financial assets and instruments does not come without error. It is very difficult to make predictions that are profitable over longer periods of time and procedures that are capable of achieving consistent profitability tend to provide diminishing returns with increasing market efficiency and competition.

Since financial markets largely produce non-stationary and complex time-series, the ML methods used generalize poorly, not performing well in real-life testing scenarios [103]. It has been shown in many studies that past performance does not guarantee future returns due to problems such as overfitting and unpredictable regime changes in market dynamics [104]. The returns expected in studies using historical data are rarely representative of real-life returns, which is a major concern for academics and private market participants.

Overfitting, rather than underfitting, is a very big concern in ER forecasts because financial data is often noisy and the dynamics of the underlying system can be buried under high levels of noise, rendering the signal-to-noise ratio low and overfitting very easy in ER forecasting [105].

The issues of poor generalization may also be caused by data snooping, an issue that occurs when multiple different models are used for ML on a single dataset. This may lead to a situation where if one tries enough models, there will eventually be some models that perform well enough to show robust profits in historical data, despite poor predictive capability in a live trading environment [106]. Data snooping has also been studied in the cryptocurrency market [107].

For these reasons, the criticality of robust frameworks for measuring historical performance and the use of ML models that generalize well has been discussed in literature, with the incorporation of **regularization techniques** such as dropout and early stopping to reduce overfitting [108]. In addition, robust loss functions have been proposed for improved estimations to be made in model performance [109].

Novel statistical methods have also been suggested to improve robust predictions, such as the use of cointegration through synthetic pairs in the FX market, where the underlying idea is that the relationship between two assets that are similar to each other in a certain manner (such as two tech stocks in the same sector) may be exploited as they would be expected to behave similarly given a set of market conditions, potentially canceling out the noise present in the system when a function of the two spreads are considered [110] [111].

While the field continues to evolve, research that directly focuses on robustness is overshadowed by research that focuses on improving testing performance using historical data, and methods such as using synthetic pairs are often not explored in smaller areas of research like the cryptocurrency market. Moreover, there are no studies with synthetic pairs being used in cryptocurrency research focused on BinClass-based ER forecasting.

2.5 Applications of Chaos Theory in Financial Forecasts

Often introduced through the analysis of non-linear differential equations, chaos theory studies unpredictable deterministic and non-deterministic processes that are sensitive to perturbations or small differences in initial conditions [112]. Chaos theory has been used in physics, hydrology and many other fields for various purposes, and has found use in financial market forecasting as well [113].

Exchange rates in financial markets tend to follow non-linear patterns and are very difficult to predict in the long run, which are characteristic of system that are chaotic [114].

This has inspired the use of tools developed in chaos theory, such as the Lyapunov exponent and the Hurst exponent, for ER forecasting in the short term.

The Lyapunov exponent studies changes in trajectory based on small differences in initial conditions, as well as the speed at which this occurs in chaotic systems, which may give clues regarding the predictability of systems in the short term. Positive maximal Lyapunov exponents suggest that the system is chaotic and is unpredictable in the long term. However, this does not mean that structure can not be found in the chaos to make short term predictions regarding market movements. Lyapunov exponent estimations have been studied in the stock market as indicators of predictability [115].

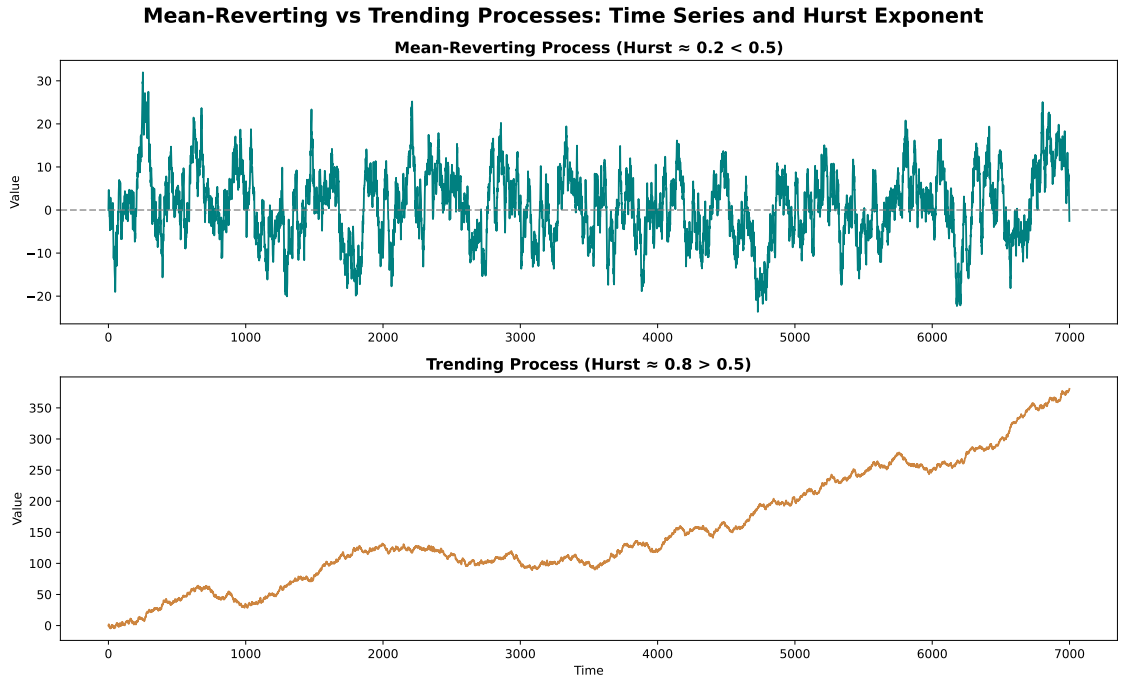

Figure 2.22: Hurst exponent

Another measurement of predictability borrowed from chaos theory and has found significant use in ER forecasting is the **Hurst exponent**. The Hurst exponent attempts to identify whether a time series is trending or mean reverting, based on the Hurst value being greater or less than 0.5, with values close to 0.5 indicating randomness. Similarly, the Hurst exponent has found use in the stock market, studying SP500 stocks and Dow-Jones indices [116] [117].

These methods have not seen much use in the cryptocurrency market, and especially in a BinClass-based ER forecasting setting, which is a significant gap in the literature that can inspire further research.

2.6 Summary

ER forecasting in financial markets, at its current stage of evolution, is a field that is dominated by ML applications. Regression based ER forecasting remains the most popular

approach to the research while BinClass-based applications are also being developed. In this context, cryptocurrency prices are also studied using BinClass-based ER forecasting workflows. However, the literature on other markets such as FX and stocks are richer and there are significant gaps in the crypto market research such as the lack of work on synthetic pairs, application of measures of randomness and predictability from chaos theory and robustness focused experiments.

3 Mathematical Foundations

In order to accurately assess BinClass-based ER forecast outcomes, as well as understand the critical role that robustness plays in this regard, the mathematics behind the system must be studied closely. This chapter will derive the mathematical equations behind the growth factor of an account that's trading with a BinClass-based strategy, the implications of position sizing and the deleterious outcomes associated with ML models with poor robustness profiles.

3.1 Market Interaction Outcomes in BinClass-based ER Forecasting

The profitability of a complete trading strategy that runs primarily through a classification workflow depends on the setup. This study will analyze the profitability of trading strategies that operate under the following constraints:

- The forecasting agent of interest interacts with the market by outputting buy or sell signals.
- The agent operates on native or synthetic pairs of cryptocurrencies in the stablecoin based section of the futures trading market (the base assets are stablecoins).
- There are two potential outcomes of market interaction: wins and losses.
- The market interaction outcomes can be fixed in relative magnitude to the value of the pair, or fluctuate based on the fixed duration set by the trading strategy.
- Trading fees, hidden costs (such as slippage) and losses due to data quality issues are not ignored and must be included in the calculation.
- The order in trading fee collection is calculated based on the CEX selected.
- A multiplicative position sizing protocol is embraced, where the returns from a single market interaction is compounded into the next market interaction (leading to exponential growth/decay or "non-ergodic" returns).

Within these boundaries, calculations regarding the criteria for profitability in BinClass-based ER forecasting can be built up in the following sequence; for long and short strategies in native pairs, and finally for synthetic pairs.

3.1.1 Long Strategy, BinClass-based Outcomes

At each market interaction, the total value allocated to a trading strategy would change. This change can be expressed by a multiplier that is applied to the total value of the initial allocation at each interaction. This multiplier will be referred to as the growth factor of

the market interaction of interest.

This growth factor can be calculated by factoring in the costs of trading, in the following 6 steps, to arrive at the final result.

Step 1. Initial Calculation

Initial balance x dollars, 0 coins, 0 debt. / Price: a dollars per 1 coin,

—> $\frac{x}{a}$ coins bought at price a,

Balance: $\frac{x}{a}$ coins, 0 dollars, 0 debt.

—> Price goes to (b dollars per 1 coin)

—> $\frac{x}{a}$ coins sold at b dollars per 1 coin.

$$\begin{cases} 1 \text{ coin} = b \text{ dollars} \\ 2 \text{ coins} = 2b \text{ dollars} \\ \frac{x}{a} \text{ coins} = \frac{x}{a} b \text{ dollars} \end{cases}$$

Balance:

$$\frac{x}{a} \cdot b = \left(\frac{b}{a}\right) x \text{ dollars.}$$

$$\boxed{\text{gfac} = \frac{b}{a} \left\{ = \frac{exitprice}{entryprice} \right\}}$$

Step 2. With Trading Fees

Initial balance:
x dollars.
0 coins
0 debt
/ Price : a dollars per 1 coin
/ Trading fees are f where,
$\{ \ f = 0.1\% = 0.001$

—> $\frac{x}{a}$ coins bought at price a

Balance: 0 dollars, $\frac{x}{a}$ coins, 0 debt.

—> CEX charges: $\left(\frac{x}{a} \text{ coins}\right) \cdot f$

Balance:
0 dollars
$\left(\frac{x}{a}\text{ coins}\right) - \left[\left(\frac{x}{a}\text{ coins}\right) f\right] = \frac{x}{a}\text{ coins} - \frac{x}{a} f\text{ coins}$
$= \frac{x}{a}\left[\text{coins} - f \cdot \text{coins}\right] = \frac{x}{a}(1-f)\text{ coins}$
0 debt

Balance: $\frac{x}{a}(1-f)$ coins, 0 dollars, 0 debt.

—> Price goes to b dollars per coin

—> $\frac{x}{a}(1-f)$ coins sold at b dollars per 1 coin

Balance: $\frac{x(1-f)b}{a}$ dollars, 0 coins, 0 debt

—> CEX charges: $\left(\frac{x}{a}(1-f)b\text{ dollars}\right) f$

Balance:

$$\left[\frac{x(1-f)b}{a} - \left(\frac{x(1-f)b}{a}\right) f\right]\text{ dollars} = [c - (cf)]\text{ dollars} = c(1-f)\text{ dollars}$$

where $c = \frac{x(1-f)b}{a}$

Balance: $c(1-f)$ dollars $= \left(\frac{x(1-f)b}{a}\right)(1-f)$ dollars $= x\left(\frac{b}{a}\right)(1-f)^2$ dollars

Balance: $x\left[\frac{(1-f)^2 b}{a}\right]$ dollars $= x\left(\frac{b}{a}\right)(1-f)^2$ dollars

$$\text{gfac} = \frac{x\left(\frac{b}{a}\right)(1-f)^2}{x} = \left(\frac{b}{a}\right)(1-f)^2$$

$$\boxed{\left(\frac{b}{a}\right)(1-f)^2 = \frac{\text{exit price}}{\text{entry price}}(1-\text{fees})^2 = \text{gfac}}$$

Step 3. With Trading Fees and Position Sizing

Initial Balance:
x dollars
0 coins
0 debt
/ Price : a dollars per coin
/ Leverage: L
/ Trading fees: f

$$\begin{cases} L = 3 \text{ means I have } x \text{ dollars and I need to borrow 2x dollars} \\ \text{to open a posn. worth } 3x \end{cases}$$

—> To open a position worth Lx dollars, $(L-1)x$ dollars borrowed.

Balance:
Lx dollars
0 coins
$(L-1)x$ dollars debt

—> $\frac{Lx}{a}$ coins bought at price a dollars per 1 coin

Balance:
0 dollars
$\frac{Lx}{a}$ coins
$(L-1)x$ dollars debt.

—> CEX charges: $\left(\frac{Lx}{a}\right) f$ coins

Balance:

$$0 \text{ dollars}, \qquad \left(\frac{Lx}{a} - \frac{Lx}{a} f\right) = \frac{Lx}{a}(1-f) \text{ coins}, \qquad (L-1)x \text{ dollars debt}$$

—> The price goes to b dollars per 1 coin.

—> $\frac{Lx}{a}(1-f)$ coins sold at a price b dollars per 1 coin.

Balance:

$$\frac{Lx}{a}(1-f)b \text{ dollars}, \quad 0 \text{ coins}, \quad (L-1)x \text{ dollars debt}$$

—> CEX charges: $\left(\frac{Lx}{a}(1-f)b\right) f$ dollars

So,

$$\left(\frac{Lx}{a}(1-f)b\right) - \left(\frac{Lx}{a}(1-f)b\right) f = \frac{Lx}{a}(1-f)b(1-f) \text{ dollars}$$

$$0 \text{ coins}, \quad (L-1)x \text{ dollars debt}$$

—> Debt of $(L-1)x$ dollars paid.

Balance:

$$\frac{Lx}{a}(1-f)b(1-f) - (L-1)x$$

dollars, 0 coins, 0 debt

gfac:

$$\text{gfac} = \frac{\frac{Lx}{a}(1-f)b(1-f) - (L-1)x}{x} = \frac{L(1-f)b(1-f)}{a} - (L-1)$$

or

$$\text{gfac} = \frac{L(1-f)^2 b}{a} - (L-1) = L\frac{b}{a}(1-f)^2 - (L-1)$$

$$= L\frac{b}{a}(1-f)^2 - L + 1$$
$$= L\left[\frac{b}{a}(1-f)^2 - 1\right] + 1$$
$$\boxed{1 + L\left[\frac{b}{a}(1-f)^2 - 1\right] = \text{gfac}}$$

(1) $\frac{b}{a}(1-f)^2$ is the non-leveraged (L=1 or "original") gfac of a trading-fees-only buy/sell multiplet.

(2) $1 + L\,[\text{original_gfac} - 1]$ *structure*

Step 4. With Hidden Costs Only

Intitial balance
x dollars
0 coins
0 debt
/ Price: a dollars per 1 coin
/ Assuming market orders signalled by last price,

$$\begin{cases} \text{If I press buy when price is } A\text{, I end up buying at } A + \text{slip}_{\text{buy}} = A + s_b. \\ \text{If I press sell when price is } A\text{, I end up selling at } A - \text{slip}_{\text{sell}} = A - s_s. \\ s_b \text{ and } s_s \text{ are often positive, though they may rarely be 0 or } < 0. \end{cases} \quad \left\{ \begin{smallmatrix}\text{due to}\\ \text{slippage}\end{smallmatrix} \right.$$

——> Desired market buy x dollars worth of the coin at price a.
——> Buy order sent when the price (last) is a.
——> The available entry on the books is at $a + s_b$ by the time the request reaches the server and is executed.
Balance:

$$\begin{array}{l} 0 \text{ dollars} \\ \dfrac{x}{a+s_b} \text{ coins} \\ 0 \text{ debt} \end{array}$$

——> Price goes to b dollars per coin.
——> Desired market sell x dollars worth of the coin at price b,
——> Sell order sent when the last price is b.
——> The available entry on the books is $b - s_s$ by the time the request reaches the server and is executed.

Balance:

$$(b - s_s) \cdot \frac{x}{a+s_b} \text{ dollars,} \qquad 0 \text{ coins,} \quad 0 \text{ debt.}$$

$$\text{gfac} = \frac{x \cdot (b - s_s)}{x \cdot (a + s_b)} = \frac{b - s_s}{a + s_b} = \boxed{\frac{\text{exit price} - \text{slip}_{\text{sell}}}{\text{entry price} + \text{slip}_{\text{buy}}} = \text{gfac}}$$

If $f_{ss} = \frac{s_s}{b}$ and $f_{sb} = \frac{s_b}{a}$,

$$\text{gfac} = \frac{b - bf_{ss}}{a + af_{sb}} = \frac{b(1 - f_{ss})}{a(1 + f_{sb})} = \boxed{\frac{b}{a}\left(\frac{1 - f_{ss}}{1 + f_{sb}}\right) = \text{gfac}}$$

Step 5. With Trading Fees and Hidden Costs

Initial Balance:
x dollars
0 coins
0 debt
Price: a dollars per 1 coin
trading fees: f
$f_{ss} = \frac{s_s}{b}, \quad f_{sb} = \frac{s_b}{a}$

—> Desired market buy x dollars worth of the coin at price a.

—> Buy signal sent when the last price is a. The available entry slips to $a+s_b = a(1+f_{sb})$.

$$\text{Balance: 0 dollars}$$
$$\frac{x}{a+s_b} = \frac{x}{a(1+f_{sb})} \text{ coins}$$
$$\text{—> CEX charges: } \left(\frac{x}{a(1+f_{sb})}\right) f \text{ coins.}$$
$$\text{Balance 0 dollars, } \left(\frac{x}{a(1+f_{sb})}\right)(1-f) \text{ coins, 0 debt.}$$

—> Price goes to b dollars per coin.

—> Desired market sell $\dfrac{x}{a(1+f_{sb})}(1-f)$ coins at price b.

—> Sell order sent when the price is b dollars per coin.
—> The available exit slips to $b - s_s = b(1-f_{ss})$.

$$\text{Balance: } \frac{x}{a(1+f_{sb})}(1-f)\cdot b(1-f_{ss}) = \frac{x(1-f)b(1-f_{ss})}{a(1+f_{sb})} \text{ dollars}$$
$$\text{0 coins, 0 debt}$$

—> CEX charges: $\left(\frac{x(1-f)b(1-f_{ss})}{a(1+f_{sb})}\right) f$ dollars.

$$\text{Balance: } \frac{x(1-f)b(1-f_{ss})}{a(1+f_{sb})}(1-f) \text{ dollars, 0 coins, 0 debt.}$$

$$\text{gfac} = \frac{b(1-f_{ss})}{a(1+f_{sb})} \cdot (1-f)\cdot(1-f)$$

$$\boxed{\text{gfac} = \frac{b}{a}\left(\frac{1-f_{ss}}{1+f_{sb}}\right)(1-f)^2 = \left(\frac{b-s_s}{a+s_b}\right)(1-f)^2}$$

(1) Factors $(1-f)^2$ and $\frac{1-f_{ss}}{1+f_{sb}}$ can be applied separately.

Step 6. With Trading Fees, Hidden Costs and Position Sizing

- Initial Balance:
  - x dollars
  - 0 coins
  - 0 debt
- Price: a dollars per 1 coin
- Leverage: L
- Trading fees: f
- $f_{ss} = \frac{s_s}{s}$
- $f_{sb} = \frac{s_b}{a}$

—> $(L-1)x$ dollars borrowed.
Balance:

- Lx dollars
- 0 coins
- $(L-1)x$ dollars debt

—> Desired market buy Lx dollars' worth of the coin at price a dollars per 1 coin.

—> Buy order sent when the last price is a.

—> The available entry slips to:

$$a + s_b = a(1 + f_{sb})$$

Balance:

- 0 dollars
- $\dfrac{Lx}{a(1+f_{sb})}$ coins
- $(L-1)x$ dollars debt

—> CEX charges:

$$\frac{Lx}{a(1+f_{sb})} f \quad \text{(coins)}$$

Balance:

- 0 dollars

- $\dfrac{Lx(1-f)}{a(1+f_{sb})}$ coins
- $(L-1)x$ dollars debt

——> Price goes to b dollars per coin.

——> Desired market sell of

$$\frac{Lx(1-f)}{a(1+f_{sb})} \text{ coins at price } b \text{ dollars per 1 coin.}$$

——> CEX charges: (dollars) f.
——> Sell order sent when the price is b dollars but the available exit slips to:

$$b - s_s = b(1 - f_{ss})$$

Balance:

- $\dfrac{Lx(1-f)b(1-f_{ss})(1-f)}{a(1+f_{sb})}$ dollars
- 0 coins
- $(L-1)x$ dollars debt

——> Debt of $(L-1)x$ dollars paid.
Balance:

$$\frac{Lxb(1-f_{ss})(1-f)^2}{a(1+f_{sb})} - (L-1)x \text{ dollars, 0 coins, 0 debt.}$$

$$\begin{aligned} g_{fac} &= \frac{Lxb(1-f_{ss})(1-f)^2}{a(1+f_{sb})} - (L-1)x \\ &= L\left(\frac{b}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2 - L + 1 \\ &= 1 + L\left[\left(\frac{b}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2 - 1\right] \end{aligned}$$

Long Strategy Final Result

$$g_{fac} = 1 + L\left[\left(\frac{b}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2 - 1\right] = 1 + L\left[\frac{b-s_s}{a+s_b}(1-f)^2 - 1\right]$$

Note: $1 + L[\text{orig.} - 1]$ observed again as this is the no-leverage ($L = 1$) g_{fac} of a trading fee and slippage buy/sell multiplet.

3.1.2 Short Strategy, BinClass-based Outcomes

Similarly, the same steps can be taken to arrive at the final growth factor of the short strategy.

Step 1. Initial Calculation

Initial balance: x dollars, 0 coins, 0 debt. (Price: a dollars per 1 coin)

—> x dollars worth of coins borrowed ($\frac{x}{a}$ coins)

Balance: x dollars, $\frac{x}{a}$ coins, $\frac{x}{a}$ coins of debt.

—> $\frac{x}{a}$ coins sold at price a dollars per 1 coin (for x dollars).

Balance: $x + x$ dollars, 0 coins, $\frac{x}{a}$ coins of debt.

—> Price moves to b dollars per 1 coin.

—> $\frac{x}{a}$ coins bought at price b dollars per 1 coin.

Balance: $x + x - x\frac{b}{a}$ dollars, $\frac{x}{a}$ coins, $\frac{x}{a}$ coins of debt.

—> $\frac{x}{a}$ coins of debt paid.

Balance $x + x - x\frac{b}{a}$ dollars, 0 coins, 0 debt.

$$x + x - x\frac{b}{a} = x\left(2 - \frac{b}{a}\right)$$

$$g_{\text{foc}} = \frac{x + x - x\frac{b}{a}}{x} = \frac{2x - x\frac{b}{a}}{x} = 2 - \frac{b}{a} = 1 - \left(\frac{b}{a} - 1\right) = g_{\text{foc}}$$

$$\boxed{2 - \frac{b}{a} = 1 - \left(\frac{b}{a} - 1\right) = g_{\text{foc}}}$$

Step 2. With Trading Fees

Initial balance: x dollars, 0 coins, 0 debt. / Price: a dollars per 1 coin, trading fees are f.

—> x dollars worth of coins borrowed ($\frac{x}{a}$ coins)

Balance: x dollars, $\frac{x}{a}$ coins, $\frac{x}{a}$ coins of debt.

—> $\frac{x}{a}$ coins sold at price a dollars per 1 coin (for x dollars).

—> CEX charges: xf dollars of trading fees.

Balance $x + x - xf$ dollars, 0 coins, $\frac{x}{a}$ coins debt.

—> Price moves to b dollars per 1 coin.

—> $\left(\frac{1}{1-f}\right)\left(\frac{x}{a}\right)$ coins bought at price b dollars per 1 coin.

—> CEX charges: $\left(\frac{1}{1-f}\right)\left(\frac{x}{a}\right) f$ coins of trading fees.

Balance

$$x + x - xf - \left[\left(\frac{1}{1-f}\right)\left(\frac{x}{a}\right) b\right] \text{ dollars, } \left(\frac{1}{1-f}\right)\left(\frac{x}{a}\right) \text{ coins, } \frac{x}{a} \text{ coins debt.}$$

—> Debt of $\frac{x}{a}$ coins paid.

Balance:

$$x\left[1 + 1 - f - \frac{b}{(1-f)a}\right] \text{ dollars,0 coins,0 debt.}$$

$$\boxed{g_{\text{fac}} = 2 - f - \frac{b}{(1-f)a}}$$

Step 3. With Trading Fees and Position Sizing

Balance: x dollars, 0 coins, 0 debt. / Price: a dollars per 1 coin, trading fees: f, leverage: L.

—> Lx dollars worth of coins borrowed ($\frac{Lx}{a}$ coins)

Balance: x dollars, $\frac{Lx}{a}$ coins, $\frac{Lx}{a}$ coins debt.

—> $\frac{Lx}{a}$ coins sold at price a dollars per 1 coin.

Balance: $x + Lx$ dollars, 0 coins, $\frac{Lx}{a}$ coins debt.

—> CEX charges: $Lx \cdot f$ dollars of trading fees.

Balance: $x + Lx - Lxf$ dollars, 0 coins, $\frac{Lx}{a}$ coins debt.

—> Price moves, b dollars per 1 coin.

—> $\left(\frac{1}{1-f}\right)\frac{Lx}{a}$ coins bought at price b dollars per 1 coin.

Balance: $x + Lx - Lxf - \left(\frac{1}{1-f}\right)\frac{Lx}{a}b$ dollars, $\left(\frac{1}{1-f}\right)\frac{Lx}{a}$ coins, $\frac{Lx}{a}$ coins debt.

—> CEX charges: $\left(\frac{1}{1-f}\right)\frac{Lx}{a}f$ coins of trading fees.

Balance: $x+Lx-Lxf-\left(\frac{1}{1-f}\right)\frac{Lx}{a}b$ dollars, $\left(\frac{1}{1-f}\right)\frac{Lx}{a}-\left(\frac{1}{1-f}\right)\frac{Lx}{a}f$ coins, $\frac{Lx}{a}$ coins debt.

Balance: $x\left[1 + L - Lf - \left(\frac{1}{1-f}\right)\left(\frac{b}{a}\right)L\right]$ dollars, 0 coins, 0 debt.

$$
\begin{aligned}
g_{\text{foc}} &= 1 + L - Lf - \left(\frac{1}{1-f}\right)\left(\frac{b}{a}\right)L \\
&= 1 + L\left[1 - f - \frac{b}{a(1-f)}\right] \\
&\boxed{= 1 + L\left(2 - f - \frac{b}{(1-f)a} - 1\right)}
\end{aligned}
$$

Note: $1 + L[\text{orig.} - 1]$ observed once more!

Step 4. With Hidden Costs Only

Initial Balance:
x dollars 0 coins 0 debt

Price: a dollars per 1 coin

$fss_{SH} = \frac{s_s}{a}$ $fsb_{SH} = \frac{s_b}{b}$

—> x dollars worth of coins borrowed ($\frac{x}{a}$ coins)

Balance: x dollars $\frac{x}{a}$ coins $\frac{x}{a}$ coins debt

—> Desired market sell of allocated coins when the price is a dollars per 1 coin,
—> Sell order given at price=a
—> The available order on the books is at $a - s_s$ by the time the sell request reaches the server and is executed
—> Sell order executed at $a - s_s$

Balance: $x + x \cdot \left(\frac{a-s_s}{a}\right)$ dollars 0 coins $\frac{x}{a}$ coins debt

$x + x \cdot (\frac{a-s_s}{a}) = x + x \cdot (1 - fss_{SH})$ dollars 0 coins $\frac{x}{a}$ coins debt

—> Price moves to b dollars per 1 coin
—> Desired market buy $\frac{x}{a}$ coins when the price is b dollars per 1 coin
—> The available entry on the books is at $b + S_b$ by the time the request reaches the server and is executed

—> Buy order executed at price $b + S_b$

Balance: $x + x \cdot (1 - fss_{SH}) - x \cdot \left(\frac{b+s_b}{a}\right)$ dollars $\frac{x}{a}$ coins $\frac{x}{a}$ coins debt

—> Debt of $\frac{x}{a}$ coins paid

Balance: $x + x \cdot (1 - fss_{SH}) - x \cdot \left(\frac{b+s_b}{a}\right) = x + x \cdot (1 - fss_{SH}) - x \cdot (\frac{b \cdot (1+fsb_{SH})}{a})$ dollars 0 coins 0 debt

$$g_{fac} = 1 + (1 - fss_{SH}) - \frac{b}{a}(1 + fsb_{SH}) = (2 - \frac{b}{a}) - fss_{SH} - \frac{b}{a} fsb_{SH}$$

$$= 2 - \frac{b}{a} - \frac{s_s}{a} - \frac{b}{a} \cdot \frac{s_b}{b} = 2 - \left(\frac{1}{a}\right)(b) - (\frac{1}{a})(s_s + s_b) = 2 - \frac{b + s_s + s_b}{a} = g_{fac}$$

$$\boxed{2 - \frac{b + s_s + s_b}{a} = g_{fac}}$$

Step 5. With Trading Fees and Hidden Costs

Initial Balance: x dollars 0 coins 0 debt / Price: a dollars per 1 coin / trading fees: f
$fss_{SH} = \frac{s_s}{a}$ $fsb_{SH} = \frac{s_b}{b}$

—> x dollars worth of coins borrowed $\left(\frac{x}{a} \text{ coins}\right)$

Balance: x dollars, $\frac{x}{a}$ coins, $\frac{x}{a}$ coins debt

—> Desired market sell of $\frac{x}{a}$ coins at price a dollars per 1 coin.
—> Sell order given when the last price is a.
—> The available entry slips to $a - s_s = a \cdot (1 - fss_{SH})$

Balance: $x + \left(\frac{x}{a}\right)(a \cdot (1 - fss_{SH}))$ dollars, 0 coins, $\frac{x}{a}$ coins debt
$= x + x(1 - fss_{SH})$ dollars, 0 coins, $\frac{x}{a}$ coins debt

—> CEX charges: $x(1 - fss_{SH})f$ dollars of trading fees.

Balance: $x + x(1 - fss_{SH}) - x(1 - fss_{SH})f$ dollars, 0 coins, $\frac{x}{a}$ coins debt

Balance: $x + (x(1 - fss_{SH}))\,(1 - f)$ dollars, 0 coins, $\frac{x}{a}$ coins debt

—> Price moves to b dollars per 1 coin.

—> Desired market buy of $\frac{1}{1-f}\frac{x}{a}$ coins at price b dollars per 1 coin.

—> Buy order sent when the last price is b.

—> The available exit slips to $b + s_b = b(1 + fsb_{SH})$

Balance: $x + \left[x(1 - fss_{SH})(1 - f) - \left(\frac{1}{1-f}\frac{x}{a}\right) b(1 + fsb_{SH})\right]$ dollars
$\left(\frac{1}{1-f}\frac{x}{a}\right)$ coins
$\frac{x}{a}$ coins debt

CEX charges: $\left(\frac{1}{1-f}\frac{x}{a}\right) f$ coins as trading fees.

Balance: $x + \left[x(1 - fss_{SH})(1 - f) - \left(\frac{1}{1-f}\frac{x}{a}\right) b(1 + fsb_{SH})\right]$ dollars
$= \left(\frac{1}{1-f}\frac{x}{a}\right)(1 - f) = \frac{x}{a}$ coins
$\frac{x}{a}$ coins debt

—> Debt of $\frac{x}{a}$ coins paid.

Balance:

$$x\left[1 + (1 - fss_{SH})(1 - f) - \frac{b}{a}\frac{(1 + fsb_{SH})}{1 - f}\right] \text{ dollars}$$

0 coins
0 debt

$gfac = 1 + 1 - f - fss_{SH} + ffss_{SH} - \frac{b}{a} \cdot \frac{1+fsb_{SH}}{1-f}$
$= 2 - f - fss_{SH}(1 - f) - \frac{b(1+fsb_{SH})}{a(1-f)}$
$= 2 - f\left(1 - \frac{s_s}{a}\right) - \frac{b+s_b+s_s(1-f)}{a(1-f)}$

$$\boxed{gfac = 2 - f\left(1 - \frac{s_s}{a}\right) - \frac{b + s_b + s_s(1-f)}{a(1-f)}}$$

t.f. only gfac: $2 - f \cdot (1 - \frac{s_s}{a}) - \frac{b+s_b+s_s(1-f)}{a(1-f)}$
slip only gfac: $2 - f \cdot (1 - \frac{s_s}{a}) - \frac{b+s_b+s_s(1-f)}{a(1-f)}$ forms preserved!

Step 6. With Trading Fees, Hidden Costs and Position Sizing

- Initial balance:
  - x dollars
  - 0 coins
  - 0 debt
- Price: a dollars per 1 coin
- Leverage: L
- Trading fees: f
- $fss_{SH} = \frac{s_s}{a}$
- $fsb_{SH} = \frac{s_b}{b}$

—> Lx dollars worth of coins borrowed

$$\left(\frac{Lx}{a} \text{ coins}\right)$$

Current balance:

- x dollars
- $\frac{Lx}{a}$ coins
- Lx coins debt

—> $\frac{Lx}{a}$ coins at price a dollars per 1 coin sold.

—> Sell order given at price a. The available entry slips to

$$a - s_s = a(1 - f_{ss,sh})$$

Current balance:

- $x + xL(1 - f_{ss,sh})$ dollars
- 0 coins
- Lx coins debt

—> Exchange charges $xL(1 - f_{ss,sh})f$ dollars of trading fees.

Current balance:

- $x + xL(1 - fss_{SH}) - xL(1 - fss_{SH})f$ dollars
- 0 coins
- Lx coins debt

$$\text{Current balance: } x + [xL(1 - fss_{SH})]\,(1 - f) \text{ dollars}$$

—> Price moves to b dollars per 1 coin.

—> $\frac{Lx}{a}$ coins at price b dollars per 1 coin market bought

—> Buy order given when last price is b.
—> The available exit slips to

$$b + s_b = b(1 + f_{sb,sh})$$

$$\text{Current balance: } x + [xL(1 - f_{ss,sh})]\,(1 - f) - \left(\frac{1}{1-f}\right)\left(\frac{Lx}{a}\right) b(1 + f_{sb,sh}) \text{ dollars}$$

—> CEX charges:

$$\left(\frac{1}{1-f}\right)\left(\frac{Lx}{a}\right) f \text{ coins as trading fees}$$

$$\text{Current balance: } x + [xL(1 - f_{ss,sh})]\,(1 - f) - \left(\frac{1}{1-f}\right)\left(\frac{Lx}{a}\right) b(1 + f_{sb,sh}) \text{ dollars,}$$

$$-\left(\frac{1}{1-f}\right)\left(\frac{Lx}{a}\right) f \text{ coins,}$$

$$Lx \text{ coins debt.}$$

$$\text{Current balance: } x + \left[xL(1 - f_{ss,sh})(1 - f) - \frac{Lx}{a} b(1 + f_{sb,sh})\frac{1}{1-f}\right] \text{ dollars}$$

—> Debt of Lx coins (including the trading fees on buy) paid:

$$\text{Current balance: } x\left[1 + L(1 - f_{ss,sh})(1 - f) - \frac{Lb(1 + f_{sb,sh})}{a(1 - f)}\right] \text{ dollars.}$$

Short Strategy Final Result

$$g_{fac} = 1 + L(1 - fss_{SH})(1 - f) - \frac{Lb(1 + fsb_{SH})}{a(1 - f)}$$

$$g_{fac} = 1 + L\left[(1 - f_{ss,sh})(1 - f) - \frac{b(1 + f_{sb,sh})}{a(1 - f)}\right]$$

$$g_{fac} = 1 + L\left[2 - f - (1 - f_{ss,sh}) - \frac{b}{a(1 - f)} - \frac{bf_{sb,sh}}{a(1 - f)}\right]$$

Short Strategy Final Result - Refactored

$$g_{fac} = 1 + L\left[\left(2 - f(1 - \frac{s_s}{a})\right) - \frac{b + s_b + s_s(1 - f)}{a(1 - f)} - 1\right]_{\text{(orig)}}$$

Note: $1 + L[\text{orig} - 1]$ at play once more!

3.2 Strategy Success based on Overall Model Performance

Given the final growth factor formulas found for the long and the short cases, and the probability of a successful (winning) market interaction (precision for positive predictions and negative predictive value (NPV) for negative predictions) can be expressed as follows:

Winning case for strategies that long on a "1" signal:

The growth factor of a winning trade is

$$g_{\text{fac, win}} = 1 + L\left[\left(\frac{b_1}{a}\right)\frac{(1 - f_{ss})}{1 + f_{sb}}(1 - f)^2 - 1\right]$$

where $b_1 > a$.

The growth factor of a losing trade is

$$g_{\text{fac, loss}} = 1 + L\left[\left(\frac{b_2}{a}\right)\frac{(1 - f_{ss})}{1 + f_{sb}}(1 - f)^2 - 1\right]$$

where $b_2 < a$.

Given that the precision of the binary classifier is P, the expected value of the gfac of one trade becomes:

$$E[gfac_{LONG}] = (g_{\text{fac,win}})^P \cdot (g_{\text{fac,loss}})^{1-P}$$

which makes the profitability condition

Profitability Condition - Long

$$(1+L\left[\left(\frac{b_1}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2-1\right])^P\cdot(1+L\left[\left(\frac{b_2}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2-1\right])^{1-P}>1$$

Or, similarly, winning case for strategies that short on a "0" signal:

The expected value of the gfac of one trade is:

$$E[gfac_{SHORT}] = (g_{\text{fac,short-win}})^{NPV}\cdot(g_{\text{fac,short-loss}})^{1-NPV}$$

which makes the profitability condition

Profitability Condition - Short

$$(1+L\left[\left(2-f(1-\frac{s_s}{a})\right)-\frac{b_2+s_b+s_s(1-f)}{a(1-f)}-1\right])^{NPV}\cdot$$
$$(1+L\left[\left(2-f(1-\frac{s_s}{a})\right)-\frac{b_1+s_b+s_s(1-f)}{a(1-f)}-1\right])^{1-NPV}>1$$

Or, in their short formats;

Profitability Conditions - Overall

$$\boxed{(g_{\text{fac,long-win}})^{P}\cdot(g_{\text{fac,long-loss}})^{1-P}>1}$$

$$\boxed{(g_{\text{fac,short-win}})^{NPV}\cdot(g_{\text{fac,short-loss}})^{1-NPV}>1}$$

Visualizing these profitability conditions will make it clear how important it is to not just have ML models that have great performance but also solid robustness.

Assuming the presence of machine learning models that are able to predict directional market movement with sufficient precision or NPV, these equations can be plotted against the "L" variable to conduct position sizing/margin optimization (see Figure 3.1).

Following the ML workflow, a rational forecasting agent would calculate the most profitable position sizing strategy to employ based on trading fees and additional costs (such as costs of cloud computing and data sourcing). There are multiple ways to optimize the size of the position (leveraging or deleveraging the market interaction instead of always using an L factor of 1), and using the optimum “Kelly bet” maximizes the expected value of market interaction outcomes in a multiplicative trading framework.

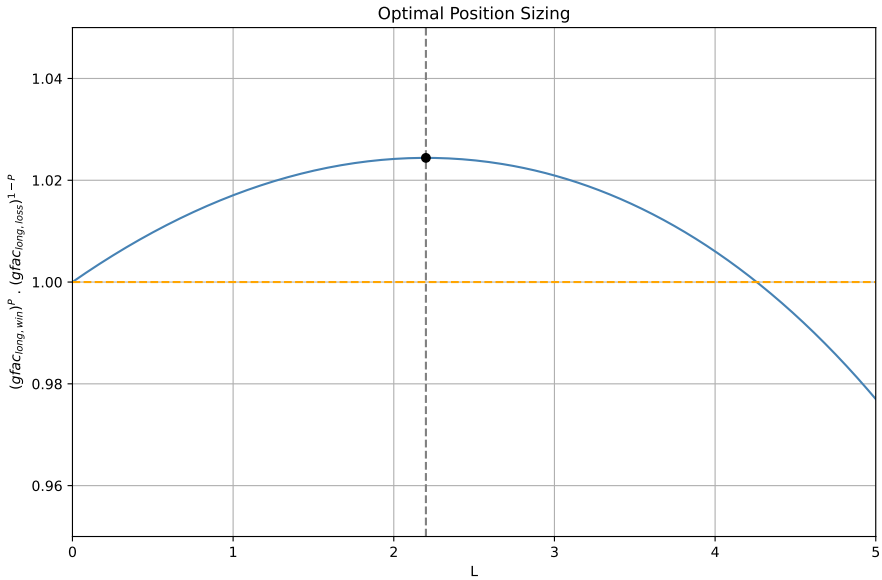

Figure 3.1: Profitability Condition Visualized

Based on how profitable the strategy is, the trading fees and the hidden costs associated with the state of the market and the CEX used, there will appear a unique profitability curve that changes the expected growth multiplier of the initial allocation with each trade. There will be an "L" value that corresponds to the global maxima of this curve, that should be selected to maximize the expected growth rate of the initial allocation, assuming perfect robustness.

However, this L selection can lead to problems when the predictions do not come to life as expected. For instance, as seen in Figure 3.2, if the ML model deployed performs well enough to make a profit but not as well as expected, the "L" selection based on the validation precision is not only no longer profitable, but is also leading to bankruptcy.

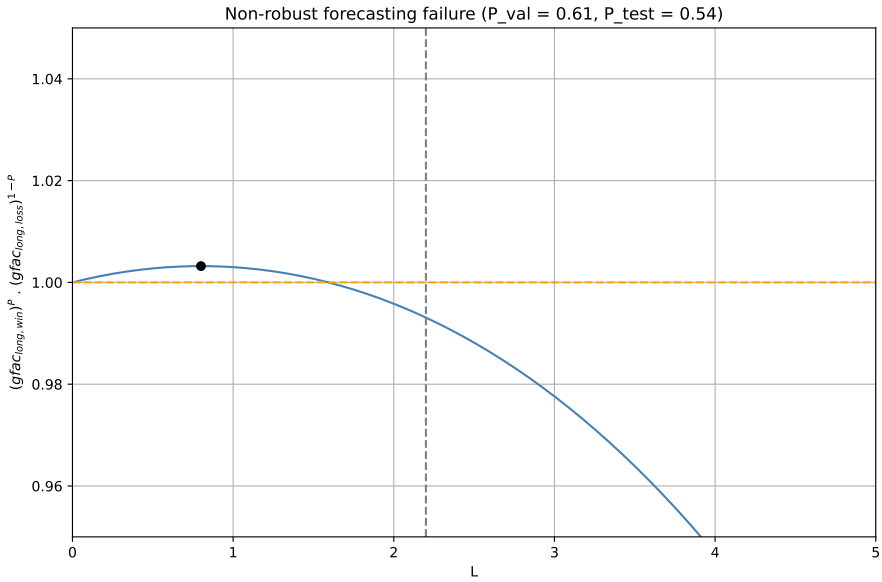

Figure 3.2: Non-robust forecasting failure

More importantly, this robustness-related failure is so critical that ML models that perform exceptionally well will also lead to complete losses. Depending on the problem, an ML workflow expected to deliver 80% precision delivering only 70% precision will not only be outcompeted by another workflow expecting to deliver 55% that really gives out 54%, it will go bankrupt despite having orders of magnitude more edge on the market.

This renders the issue of robustness extremely critical and calls for robustness-first approaches in BinClass-based ER forecasting.

4 Methodology

This chapter details the data sources, processing workflows and experimental methods used in the study.

4.1 Data Sourcing

Given the quality and availability of their data in the futures market, as well as the larger pool of cryptocurrencies listed, the CEX was selected as Binance. There are two types of futures markets in Binance - *COIN Margined Futures* for pairs where the quote asset is a cryptocurrency and *USDS-M Margined Futures* for pairs where the quote asset is a stablecoin, often pegged to the US Dollar. For the synthetic-pair paradigm we are building, two USDS-M Margined Future contracts would be selected to create one synthetic pair.

There are various data sourcing options in Binance, such as using the web databases or using the API, which contains more data, as well as more recent data compared to the web databases, of Binance. Since the experiments concerning this thesis would be conducted using historical tick data, and since historical tick data is available on the web databases in its entirety, it was decided to use the web databases for I/O efficiency. The database can be found at

https://data.binance.vision/

and the futures tick data for each pair can be found at

https://data.binance.vision/?prefix=data/futures/um/monthly/trades/

. Given these data sources, data can be programmatically downloaded for multiple months since the pair listing date.

4.1.1 Pair Selection

Given the highly competitive nature of algorithmic trading, pairs that have been listed for longer periods of time, as well as pairs with higher trading volumes will tend to be more efficient, and it will be harder to predict market movements of larger magnitudes in these pairs, although exceptions are present. On the other hand, pairs that are recently listed or have very small trading volume do not have sufficient amounts of data to allow for proper ML to take place.

For these reasons, as well as to minimize biases, a random selection of pairs that were listed in 2025 that contain a minimum of 5,000,000 rows (per pair) were made. The pairs were listed in chronological order using data from the Binance REST API, and the download links for the files were constructed using the URL structure from the web database. The data was later downloaded in .zip format and saved to disk in .csv format after extraction.

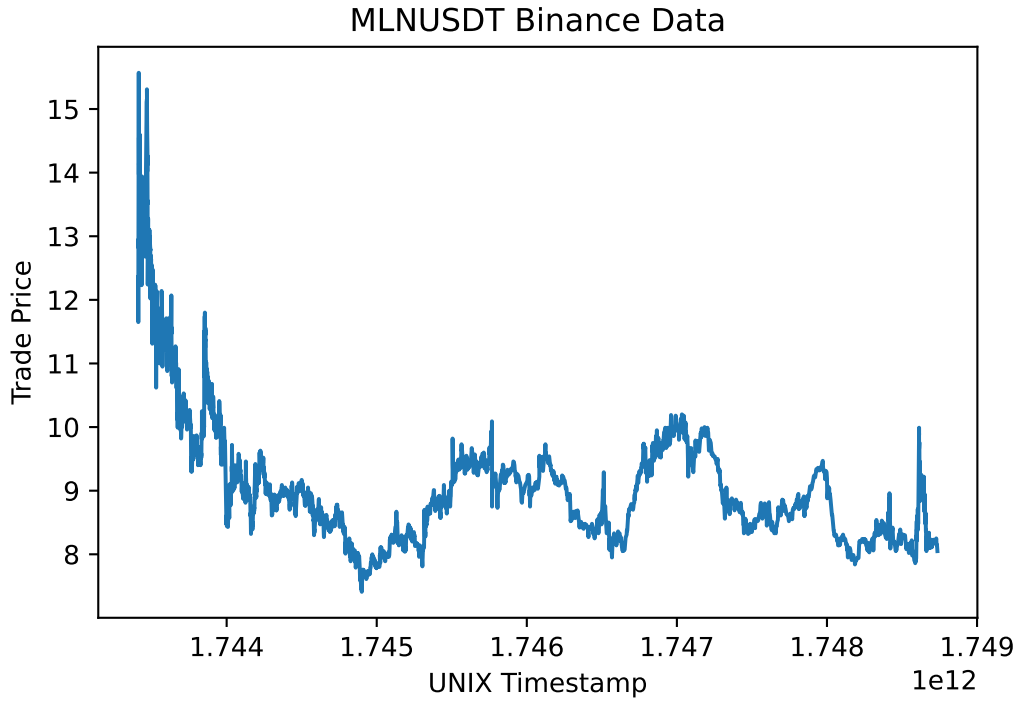

Figure 4.1: Selected Pairs Visualized: MLNUSDT

Given the random selection, the three (native) pairs were selected as given in the following table:

Table 4.1: Selected Trading Pairs and Trading Period

Pair	Trading Period
MLNUSDT	2025-03 to 2025-05
PLUMEUSDT	2025-03 to 2025-05
SIRENUSDT	2025-03 to 2025-05

4.2 Data Preprocessing and Feature Engineering

The downloaded .csv files were joined using the *pandas* library in python, per each pair, for easier organization and quality checks. Data quality checks were ran on the .csv files organized to ensure that non-chronological data, repeating rows or irrational values (such as negative-prices or zero-trading volumes) were not encountered.

Upon confirmation that the data was clean, the "joined data" was used to generate synthetic pairs, as discussed earlier in this work. These synthetic pairs are a basic division between the prices of the individual "native" pairs selected.

4.2.1 Generation of the Synthetic Pairs

The algorithm that generates these synthetic pairs can be summarized as follows:

1. It's determined which of the pairs is listed first (has the oldest datapoint in the database) and that pair is called the "old pair", and the other the "new pair".

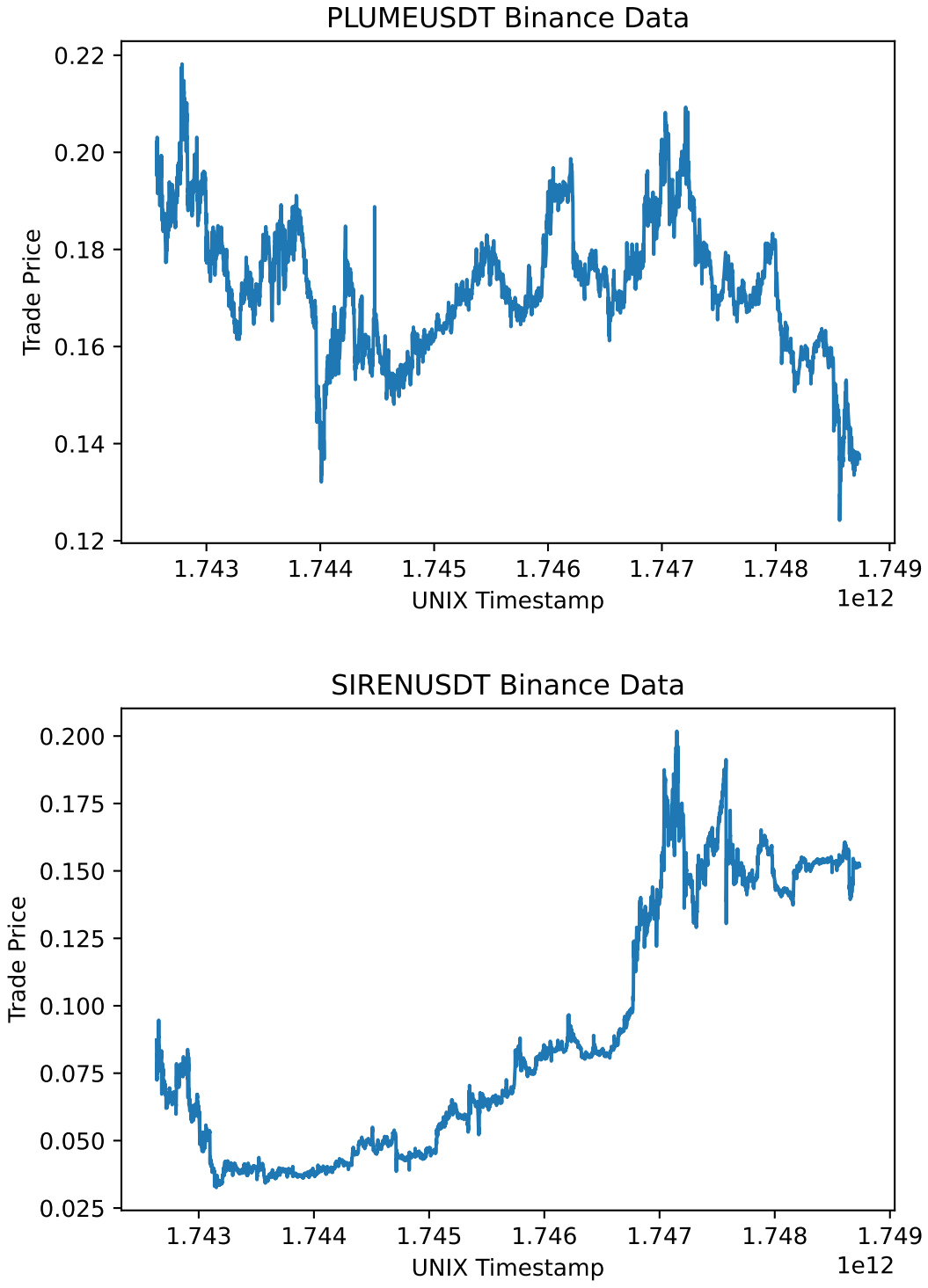

Figure 4.2: Selected Pairs Visualized: PLUMEUSDT and SIRENUSDT

2. Any portion of data belonging to a UNIX timestamp before the oldest datapoint of the new pair is deleted.

3. Starting with the first datapoints in each pair, one synthetic datapoint is created for each datapoint-pair sharing a UNIX timestamp, where the price is the division of the old pair price by the new pair price.

4. For cases where only one of the pairs were traded on a given UNIX timestamp, a synthetic datapoint is created for that UNIX timestamp where the price is the division of the latest (at or closest to the current UNIX timestamp) old pair price by the new pair price.

5. The resulting time/price data is returned as an array.

Given these rules and the three native selections, the synthetic pairs become;

Table 4.2: Synthetic Trading Pairs and Trading Period

Synthetic Pair	**Trading Period**
MLNUSDT_SIRENUSDT	2025-03 to 2025-05
MLNUSDT_PLUMEUSDT	2025-03 to 2025-05
SIRENUSDT_PLUMEUSDT	2025-03 to 2025-05

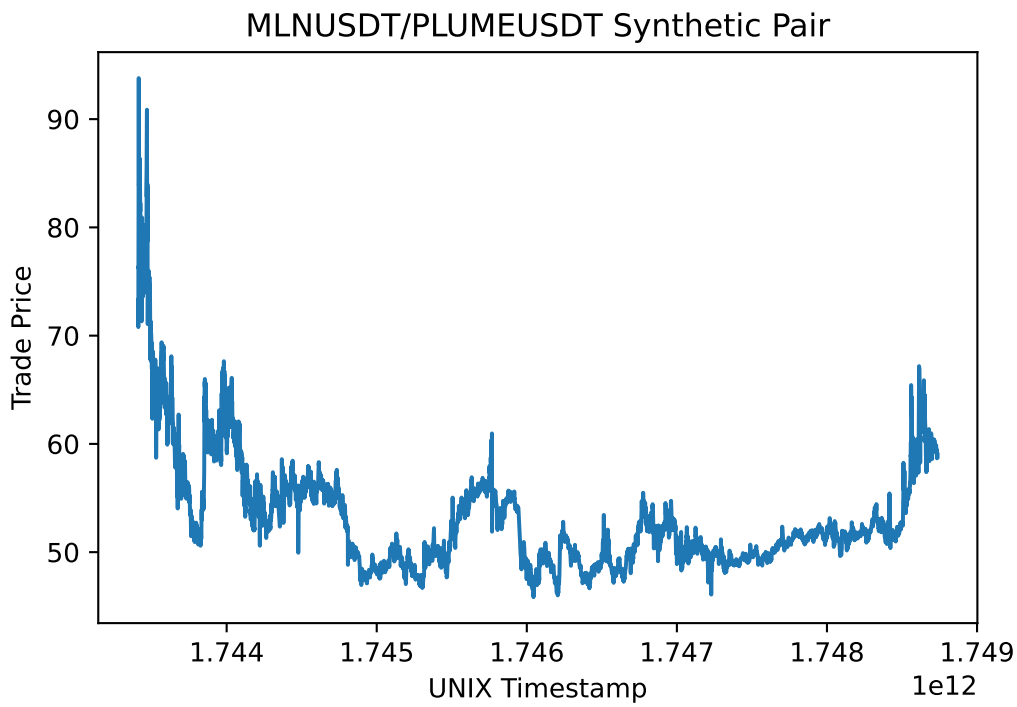

Figure 4.3: Synthetic Pair MLNUSDT/PLUMEUSDT

4.2.2 Data Grouping

Data grouping is a crucial component of data preprocessing in ER forecast workflows as the total size of data must be reduced for meaningful analyses to be conducted while keeping the cloud infrastructure costs manageable.

There are different methods to group tick data in ER forecast workflows, the most common method working by grouping all datapoints observed in a pre-defined time interval into a single group and reporting only the first, maximum, minimum, and last elements of that group, for all time intervals possible within the range. This grouping method produces

what's called *time candles* or *time OHLC candles*.

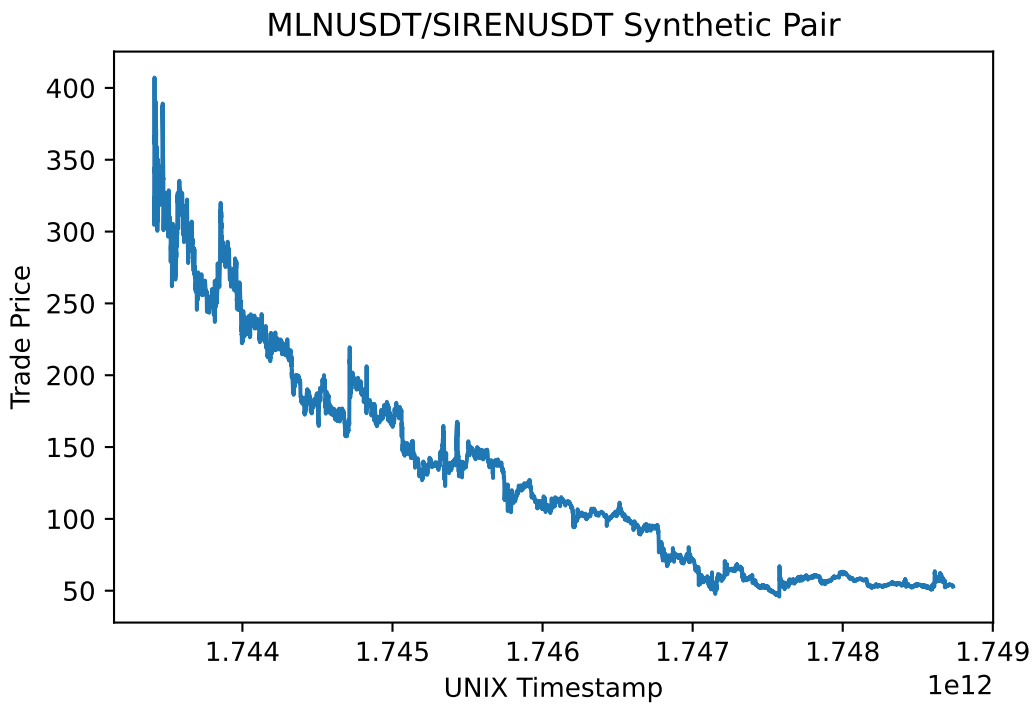

Figure 4.4: Synthetic Pair MLNUSDT/SIRENUSDT

There are other candle types and candle grouping methods. This study will be using tick grouping instead of time grouping. This method groups tick data using a predetermined number of ticks instead of for a time period. This method is more robust to extreme increases in short term trade volume as it allows the capturing of very quick market movements without delay in cases of tweets or formal announcements moving the market quickly. However, this method also has its disadvantages in cryptocurrencies that have their trade frequency change in orders of magnitude over long periods of time, since the relative change in price per candlestick can change dramatically, impacting ML performance. Given the period selected for this study, tick grouping was selected.

In order to study the impact of the methods discussed on robustness **procedurally instead of incidentally**, it was decided to run the experiments in multiple timeframes that are different in order of magnitude. The group tick-sizes were selected as follows:

	MLNUSDT	**PLUMEUSDT**	**SIRENUSDT**
# of ticks per group, 1	20	20	20
# of ticks per group, 2	50	50	50
# of ticks per group, 3	1000	1000	1000

Table 4.3: Candlestick group tick-size for each pair

This selection allows the method to be tested in a day-trading / scalping context, and an HFT context, as well as looking at the ultra-low timeframes for trading fee minimization use cases.

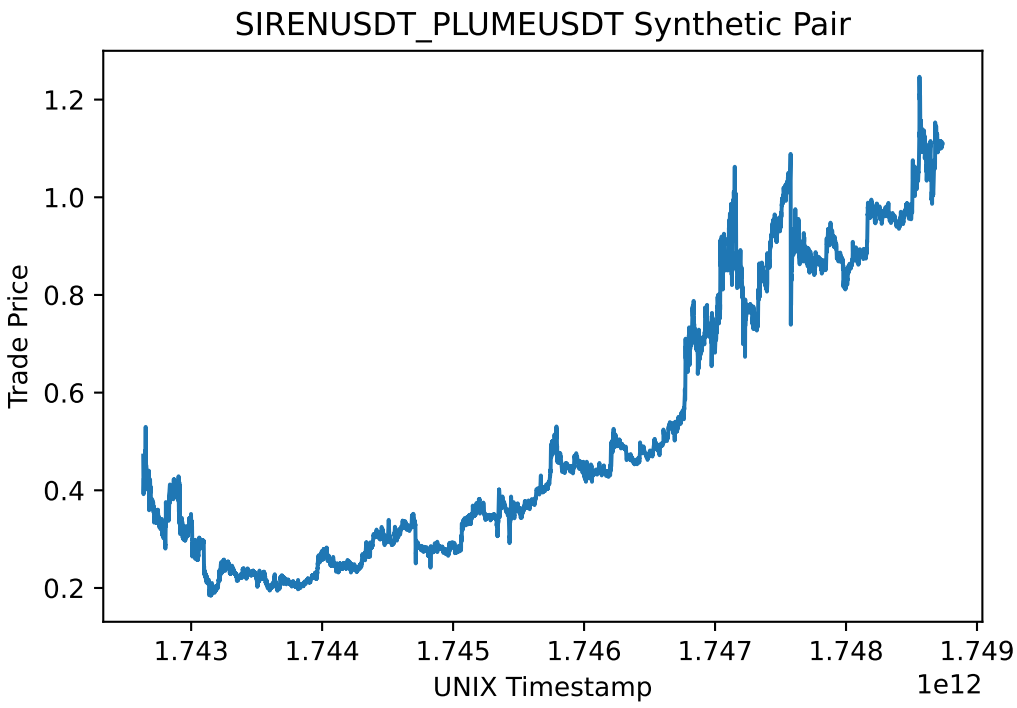

Figure 4.5: Synthetic Pair SIRENUSDT/PLUMEUSDT

4.2.3 Feature Engineering

Having generated candle data for the synthetic pairs, feature engineering could be conducted on the data present. As discussed earlier, the common *up-or-down* binary encoding method was used in creating the outputs, with one output being defined per candlestick. Moreover, in line with some of the literature already discussed, lagged features, as well as technical indicators, were included in the features of this dataset.

The lagged features of two columns (close_price and hlc3) from the grouped data were used, as shown in the table below.

Column Name	**Lag Context Window**
close_price	50
hlc3	50

Table 4.4: Input Features and Their Lag Context Window

The column hlc3 refers to the arithmetic mean of the maximum, minimum and last price of the group, h representing the high_price column, l representing the low_price column and c representing the close_price column.

The technical indicators were selected to include a momentum indicator and a trend indicator, with 3 different settings each, as seen in the table below.

As seen above, a price-only framework was preferred to keep the focus of the work on robustness. However, one must note that it should be possible to improve the performance of these models quite dramatically by incorporating other features in the dataset, such as features extracted from the orderbook data, funding rates based on the market conditions,

Technical Indicator	Period 1	Period 2	Period 3
EMA (Exponential Moving Average)	14	30	50
RSI (Relative Strength Index)	14	30	50

Table 4.5: Technical Indicators and Their Period Settings

and social media powered market sentiment, just to name a few, all of which are readily available.

4.3 Implementation of Chaos-Theory Inspired Techniques

This study has chosen to use the Hurst exponent as a filter to classify whether the data has been demonstrating behaviors that are closer to that of a mean-reverting process (H<0.5) or a trending process (H>0.5). More importantly, it was decided that this method would be combined with a custom ML layer to ensure that the predictions made by the ensemble model would be in parallel with the expectations from the Hurst exponent.

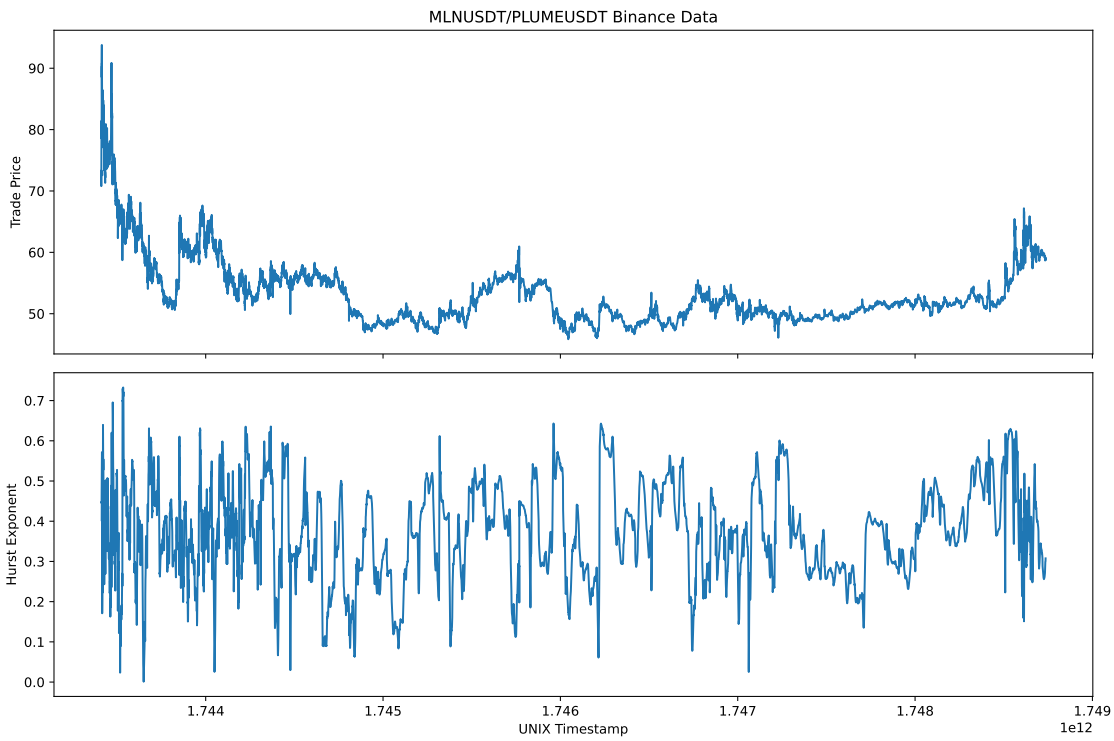

Figure 4.6: Hurst Exponent for MLNUSDT/PLUMEUSDT

The calculation of the Hurst exponent was limited to include the 100 rows leading to the current datapoint, in order to keep a balance between optimizing the computing resources and maintaining a longer context window, and a moving-Hurst exponent estimation was made. The estimation of the Hurst exponent was carried out using an approach similar to the rescaled range (R/S) analysis, and the algorithm used is summarized as follows:

- For each **lag_now** value in the list [2, 3, 4, ..., 20], calculate the standard deviation of the differences between:

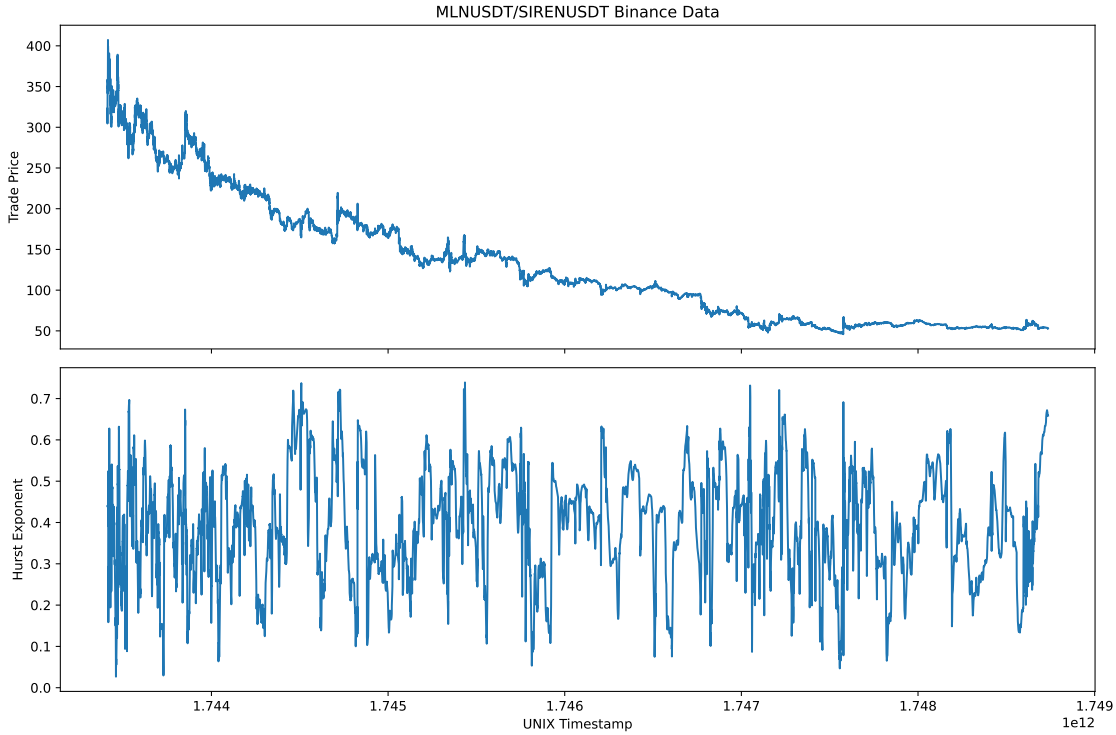

Figure 4.7: Hurst Exponent for MLNUSDT/SIRENUSDT

1. All elements in the input array except for the first **lag_now** elements
2. All elements in the input array except for the last **lag_now** elements

- Collect each of these standard deviations in a list of key-value pairs, with each standard deviation corresponding to the lag of interest.
- Fit a 1-D polynomial using:
  1. The list of all keys (lags)
  2. The list of all values (standard deviations)
- Return the slope of the polyfit as the Hurst exponent.
- (Repeat the above steps in a rolling-fashion for each 100-row window of the input array)

This Hurst filter would then be applied on the entire period observed, only allowing trading to take place within windows where H>0.4, or above/below another threshold determined.

It should also be noted that this filter is applied before any ML starts (before training/testing splits are made, even) to ensure that no survivorship bias is present and that the ML models will only be learning from data they are supposed to be learning from.

4.4 Machine Learning Workflow

An ensemble model is used as the final ML component of this study. There are two parts to this ensemble model, the LGBM classifier and the additional custom ML layer.

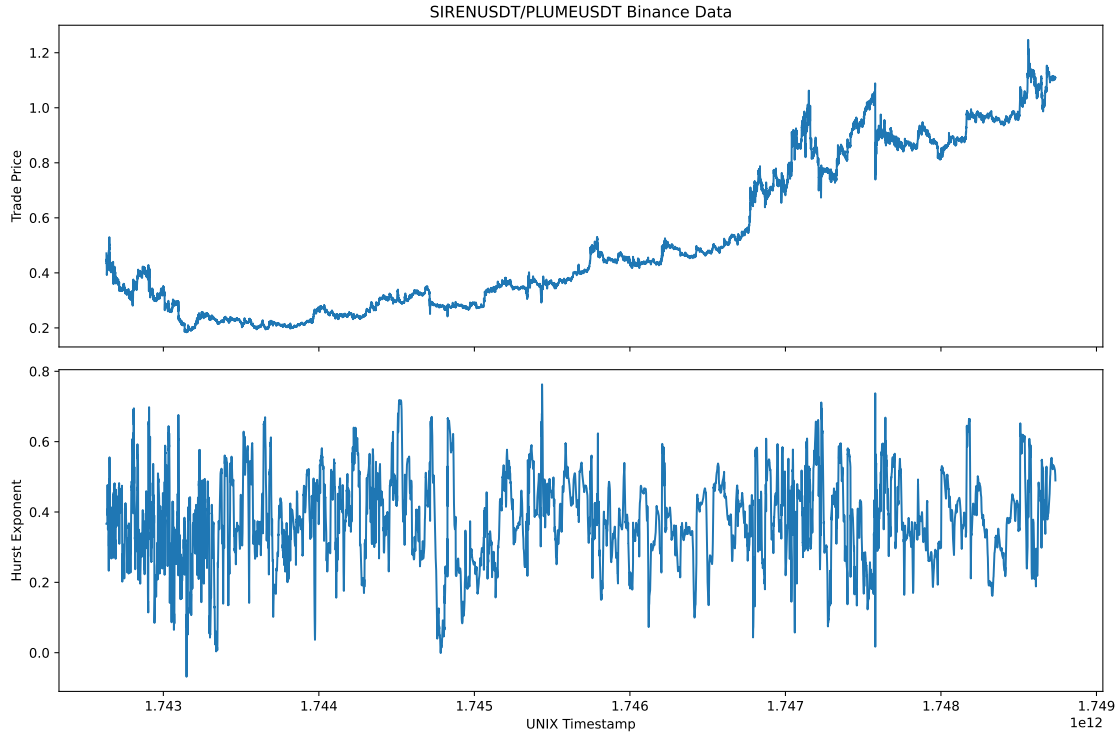

Figure 4.8: Hurst Exponent for SIRENUSDT/PLUMEUSDT

4.4.1 LGBM Classifier

Light Gradient Boosting Machine, or LGBM is a tree based framework that was designed to handle datasets with larger sizes with improved compute and memory efficiency. There are multiple reasons why LGBM was selected for this problem, as well as why it is such a good candidate for BinClass-based ER forecasting applications in general.

First of all, it is critical that algorithmic trading applications, even if not arbitrage based, to have low latency. This is mainly due to the fact that, if a model has edge, and that edge is captured by any other player in the market, statistically speaking, the longer the time between the appearance of the pattern and the execution of the market interaction necessary to exploit it, the greater the losses due to the changes in the orderbook in the meantime.

There are many tree-based ML models that perform extremely well in predicting ERs, and tree based models are considered the state-of-the-art, even in 2025, in classification based ER forecasting. However, as tree based models get more complex, the runtime of the algorithms can increase dramatically, rendering the edge the models may have obsolete due to delays, both in training (leading to lower frequency of model updates and potentially less theoretical performance) and prediction (leading to more time until action and potentially increased slippage and hidden costs).

Since the LGBM framework is built to handle larger datasets, and it possesses sufficient complexity to handle ER prediction, it is a perfect candidate for ER forecasting applications. It can be deployed in small VMs, train and predict in lightning speed, and provide excellent forecasting performance out-of-the-box.

4.4.2 Custom ML Layer

There is a custom layer added on the LGBM classifier, using an *AND-gate*, that acts as a trend continuation cue to support the mean-reversion filtering of the Hurst exponent.

The Hurst exponent, as discussed earlier, can be used to determine whether a process is behaving like a mean-reverting process (H<0.5) or a trending process (H>0.5). In this context, instead of making predictions simply when H is far away from 0.5, the Hurst expectation of reduction in non-stationarity can be supercharged by identifying periods in the market where the price is starting to trend and combining those trend-beginning cues with the Hurst expectations of a trending market.

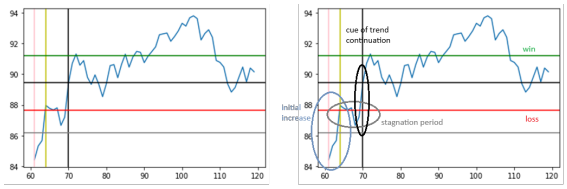

Figure 4.9: Trend Continuation Cue with Custom Layer

In this context, a custom layer can be developed that acts as an ensemble classifier, returning binary outcomes based on the preceding price action. This custom layer was manually developed and its inner workings can be summarized as follows:

- Identify all non-interrupted surges in value above a predetermined threshold as a percentage of the starting value of the asset
- Filter for surges that happen under a predetermined sequence length (period)
- Identify all non-interrupted slumps in value below a predetermined portion of the surge of interest, following each surge that was selected
- Filter for all post-surge slumps that last over a predetermined sequence length (period)
- Identify all re-surges that re-establish the trend by breaking above the upper limit of the first surge
- return 1 at each breaking point identified at the last step, and 0 everywhere else

4.4.3 Ensemble Model and Hyperparameter Tuning

The LGBM classifier and the custom layer is joined using a filter mask for numpy arrays in python. It should be noted that this filter masking is made to be completed before the training and testing splits are made, and that there is no data leakage from the testing set into the ensemble model.

The joining logic is a simple *AND-gate*, as discussed, to where the ensemble model can only predict a "1" class if both the LGBM classifier layer and the additional custom layer outputs a 1 prediction.

Hyperparameter tuning is conducted for the LGBM classifier within the below restrictions.

Table 4.6: Hyperparameter Tuning Setup for `LGBMClassifier`

Hyperparameter	**Search Space**
`n_estimators`	100 to 1000 (10 values, evenly spaced)
`num_leaves`	20 to 150 (10 values, evenly spaced)
`learning_rate`	{0.01, 0.05, 0.1, 0.2}
`max_depth`	3 to 15 (13 values, evenly spaced)
`min_child_samples`	5 to 30 (6 values, evenly spaced)
Search Method: RandomizedSearchCV (30 iterations) **Cross-validation**: TimeSeriesSplit (3 splits) **Scoring Metric**: Precision (positive class) **Random State**: 42 **Parallelism**: `n_jobs=-1`	

It should be noted that hyperparameter tuning takes place during the training/validation phase of the ML workflow, and that the model trains without any data leakage from the testing set. The LGBM classifier tunes its hyperparameters given the data that has already been filtered by the trend continuation cue.

4.4.4 Training, Validation, Testing Process

The train-validation-test splits are made to reflect the below table:

Table 4.7: Time Series Dataset Splits

Split	**Proportion**	**Description**
Training Set	60%	Used for model training
Validation Set	20%	Used for hyperparameter tuning
Test Set	20%	Used for final model evaluation

During these three stages, the LGBM classifier uses the first 60% of the entire dataset to train, and the following 20% of the dataset to run a preliminary test on how well the training was done. For instance, given a set of hyperparameters, if the initial training performance was high but the preliminary testing performance was low, those hyperparameters would not be selected for the final testing.

After the best hyperparameters are decided on, the first 80% portion of the entire dataset is now used for training, and the very last 20% of the dataset, which was not directly or indirectly available to the LGBM classifier until now, is used to get the real testing results. This paradigm of model optimization helps improve the robustness of ML models by preventing overfitting. However it should be noted that no paradigm is perfect as there are still risks of data snooping and leakage due to man-made errors.

4.5 Cloud Infrastructure and Experiment Setup

The experiment is set to run using cloud infrastructure for effective parallelization and time management. The main portion of the experiment ran is training the ML model and forecasting the binary outputs of the synthetic pairs generated for a range of Hurst exponent thresholds and candlestick tick-sizes.

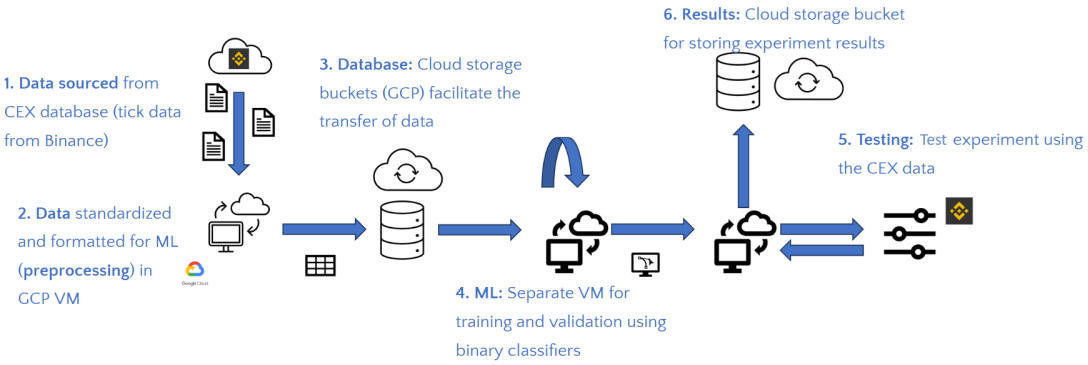

Figure 4.10: Experiment Workflow on the Cloud

These experiments are ran on multiple virtual machines (VMs) on the cloud, where a separate summary file of the True Positives, False Positives, True Negatives and False Negatives are recorded, alongside the tuned LGBM hyperparameters and strategy metadata (synthetic pair and candle data info, etc.).

Table 4.8: Compute and Memory Specifications of the VM

Specification	**Value**
Machine Type	`e2-highmem-4`
vCPUs	4
Memory	32 GB
CPU Platform	Intel Broadwell
Architecture	x86_64
GPU	None
Preemptibility	Off (Standard Provisioning Model)

The recorded experiment results are later grouped and read into memory using python again, and the results are programmatically re-organized and visualized for interpretation purposes.

4.6 Evaluation

The results are evaluated by comparing the unfiltered and the filtered generalization gaps across the multiple dimensions of the experiment and quantifying the improvements in the robustness of the BinClass-based forecasting agents deployed. However, it should be noted that a low generalization gap can certainly be deemed unsuccessful in cases where the model performance is equal to the theoretical expectation for an i.i.d. Bernoulli process, as a completely random buy-sell strategy would be expected to have no edge and slowly lose money due not winning or losing money except for losing to trading fees (a gambler's ruin scenario), which would be expected to have no generalization gap.

The precision of a BinClass problem can be expressed as follows:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

, and this would be used as the main evaluation metric regarding the problem.

5 Results and Discussion

This chapter details the results of the experiments conducted and quantifies the impact of the proposed procedural additions to BinClass-based ER forecasting in the crypto market for increased robustness. First, the results are listed by grouping size and then pairwise comparisons and overall visualizations are presented.

5.1 Results Listed by Grouping

This section details the results for all synthetic pairs, grouped by tick size.

5.1.1 1000-Tick Candlestick Group

The results for the candles with a tick size of 1000 can be summarized as follows:

Table 5.1: Model Evaluation Metrics, 1000

Pair	Val Prec. (Unf.)	Test Prec. (Unf.)	Gap (Unf.)	Val Prec. (Filt.)	Test Prec. (Filt.)	Gap (Filt.)
MLNUSDT_SIRENUSDT	0.571	0.500	0.071	0.611	0.636	0.025
MLNUSDT_PLUMEUSDT	0.423	0.520	0.097	0.636	0.600	0.036
SIRENUSDT_PLUMEUSDT	0.542	0.509	0.033	0.407	0.379	0.028

For instance, the MLNUSDT_SIRENUSDT pair with a 1000-tick grouping expected 57% precision following the hyperparameter tuning of the LGBM classifier. However, in testing, the precision drops down to 50%, demonstrating significant overfitting, since the theoretical break-even expectation for a strategy that has no edge is 50% (see Figure 5.1).

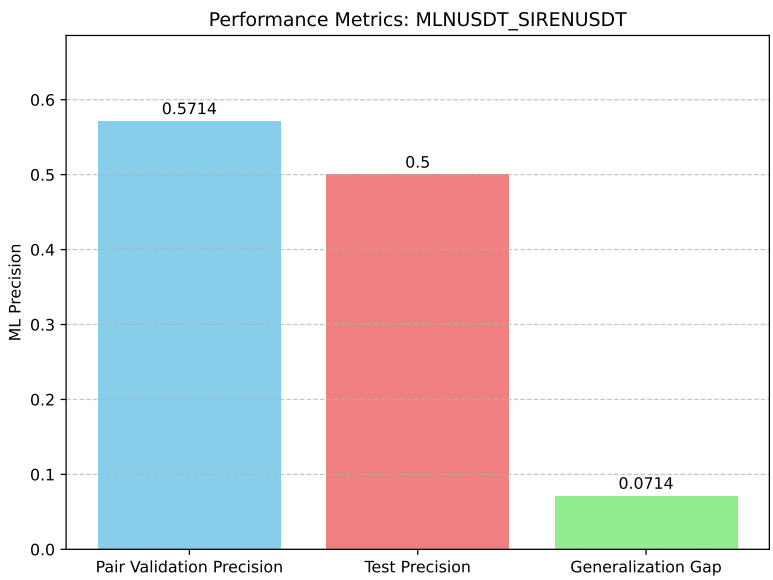

Figure 5.1: MLNUSDT/SIRENUSDT pair, 1000, unfiltered

However, after Hurst exponent mean-reversion filtering, the generalization gap is closed (see Figure 5.2), and the expected validation precision is tracked with very small error in the testing phase.

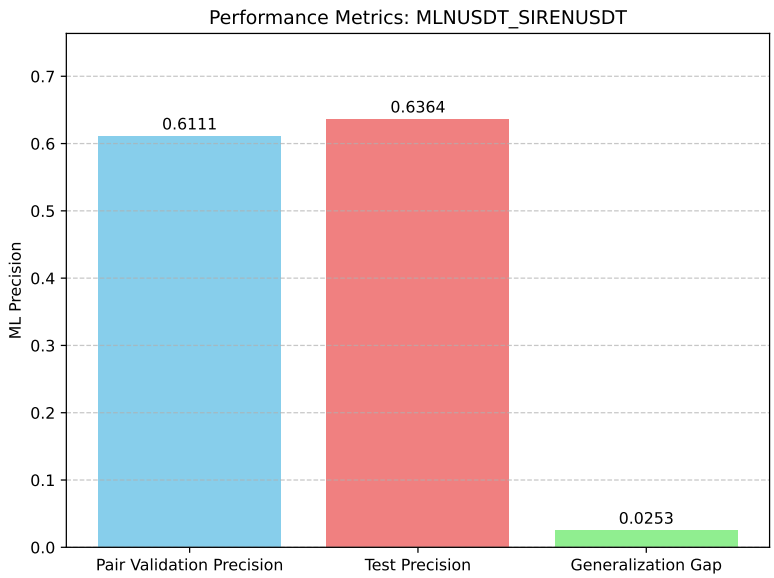

Figure 5.2: MLNUSDT/SIRENUSDT pair, 1000, filtered

Alternatively, the improvements may also be observed by the ROC curve and the increased Area-Under-[the]-Curve (AUC).

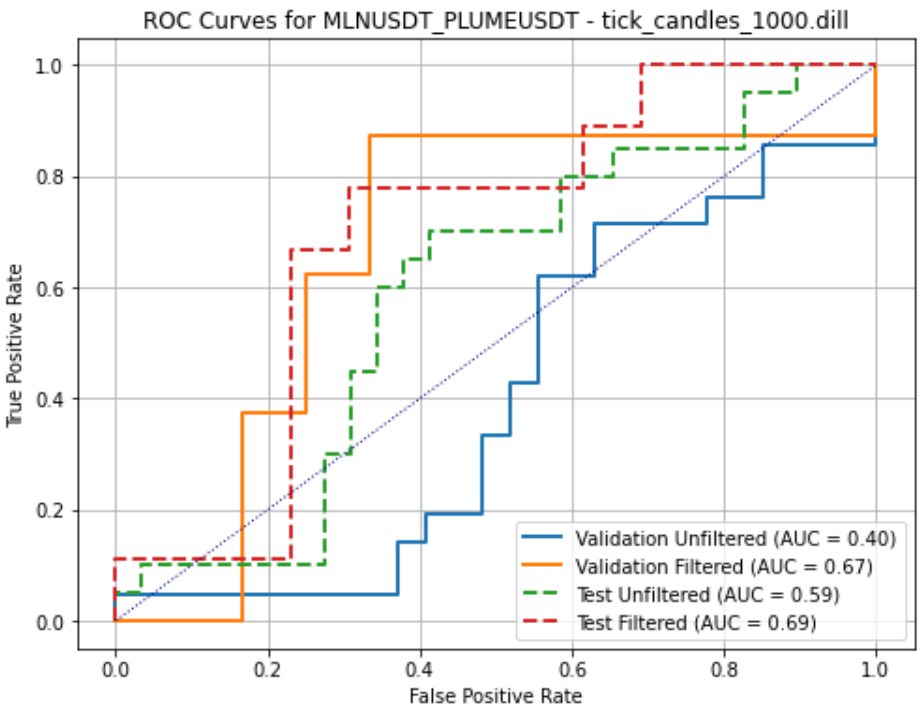

Figure 5.3: ROC Curve for MLNUSDT/PLUMEUSDT

Given the objectives of this thesis, the improvements will be expressed using bar charts, highlighting the generalization gap. The improvement in the MLNUSDT/SIRENUSDT pair is visualized in Figure 5.4, highlighting the dramatic improvement in robustness.

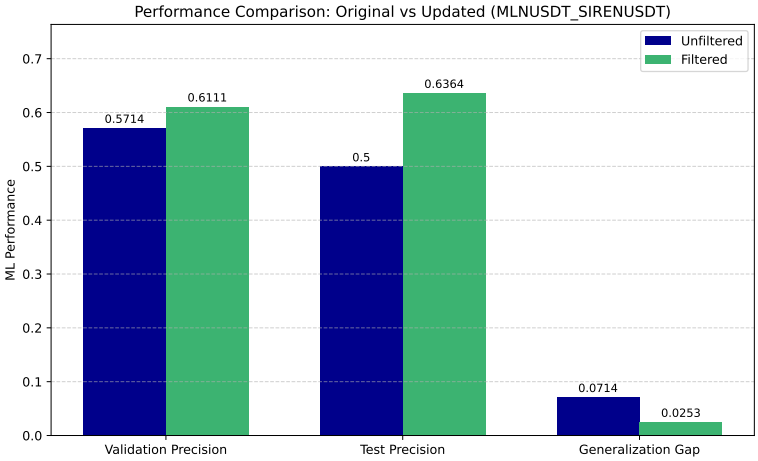

Figure 5.4: MLNUSDT/SIRENUSDT pair, 1000, improvement

Considering all three synthetic pairs, as seen on the first table, in the unfiltered case, validation precisions that are sufficiently far from the gambler's ruin scenario (at 50% precision) to make a profit, where two pairs would be carrying out a long strategy and one pair a short strategy, are observed. However, the testing precision moves toward 50%, as is the case for many non-robust applications, leading to gambler's ruin.

When the Hurst exponent filter is applied at 40% though, the generalization gap is consistently closed significantly, and the validation precisions expressed after the training and validation phases are completed are much more representative of what would be expected in a live-trading setting.

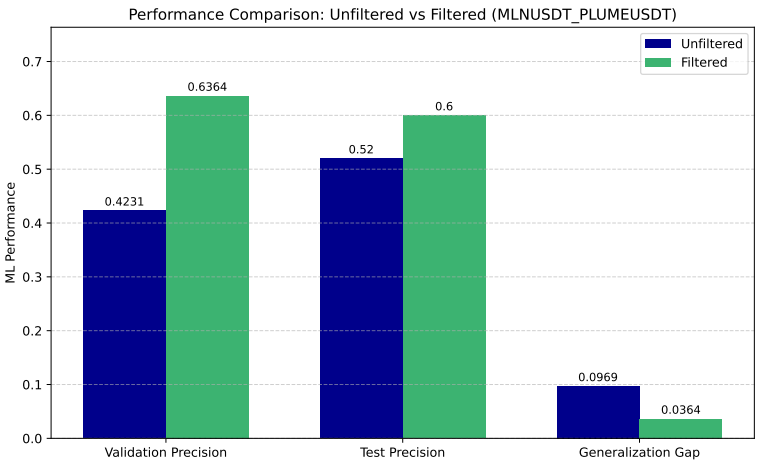

Figure 5.5: MLNUSDT/PLUMEUSDT pair, 1000, improvement

The second significant observation is the improve in performance. The combination of the Hurst exponent mean-reversion filter and the trend continuation cue is able to increase the testing precision very dramatically, having 10-15% improvement on the theoretical break-even expectation of 50%.

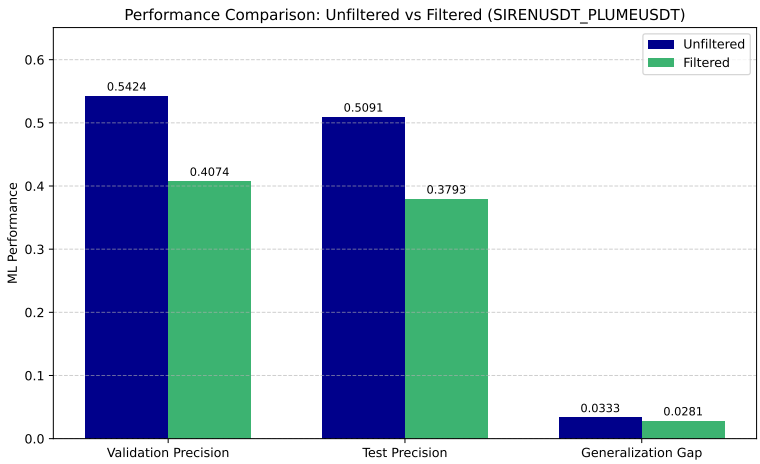

Figure 5.6: SIRENUSDT/PLUMEUSDT pair, 1000, improvement

In the SIRENUSDT/PLUMEUSDT pair, the model is predicting a precision level much below 50%, which can be used to short the market instead of longing it, given that robustness would be there. This might be counterintuitive but nevertheless the model's output is a better estimation of what should be expected in the testing period, given the improvement in the generalization gap (see Figure 5.6).

Finally, the improvement in the generalization gap can be expressed for each pair as follows (see Figure 5.7).

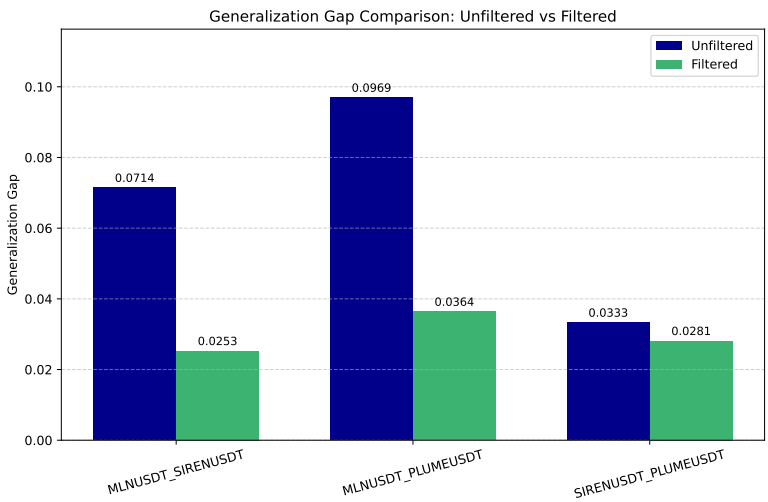

Figure 5.7: Robustness Improvements, All Pairs, 1000

5.1.2 50-Tick Candlestick Group

Looking at the MLNUSDT/SIRENUSDT pair;

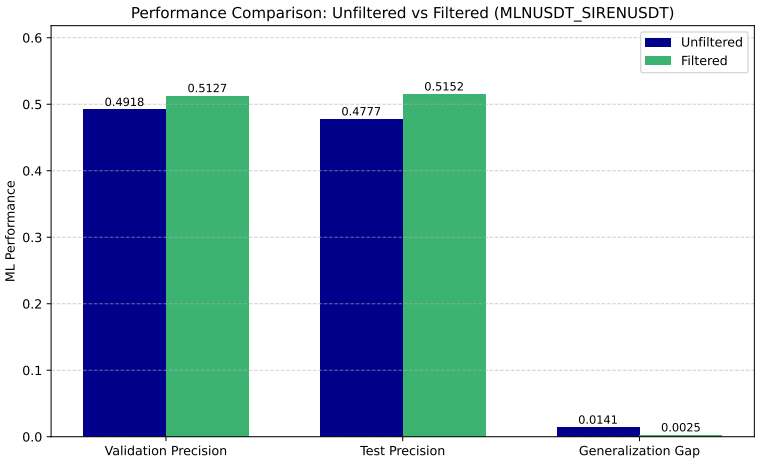

Figure 5.8: MLNUSDT/SIRENUSDT pair, 50, improvement

it is found that the methods remain effective dealing with a grouping that exists in a much smaller timeframe, where the robustness improvements are very significant and consistent among the pairs. This demonstrates the procedural durability of the methods across timeframes.

Similarly, looking at the MLNUSDT/PLUMEUSDT pair;

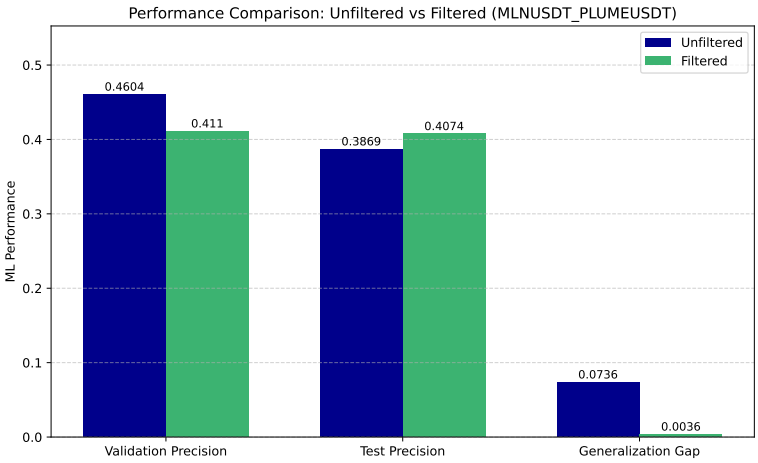

Figure 5.9: MLNUSDT/PLUMEUSDT pair, 50, improvement

it's seen that the generalization gap vanishes.

Finally, looking at the SIRENUSDT/PLUMEUSDT pair;
the improvement in the generalization gap can be seen. The numeric results of all three pairs are listed in the following table, and the improvements in the generalization gap can

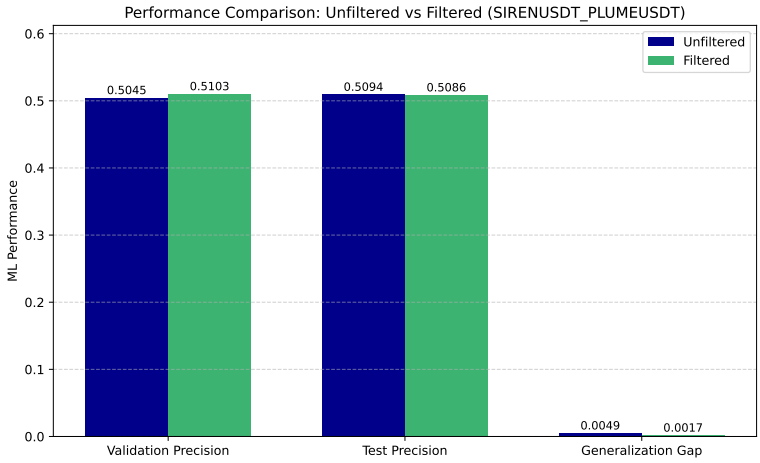

Figure 5.10: SIRENUSDT/PLUMEUSDT pair, 50, improvement

be summarized in Figure 5.11.

Table 5.2: Model Evaluation Metrics, 50

Pair	Val Prec. (Unf.)	Test Prec. (Unf.)	Gap (Unf.)	Val Prec. (Filt.)	Test Prec. (Filt.)	Gap (Filt.)
MLNUSDT_PLUMEUSDT	0.4604	0.3869	0.0736	0.4110	0.4074	0.0036
MLNUSDT_SIRENUSDT	0.4918	0.4777	0.0141	0.5127	0.5152	0.0025
SIRENUSDT_PLUMEUSDT	0.5045	0.5094	0.0049	0.5103	0.5086	0.0017

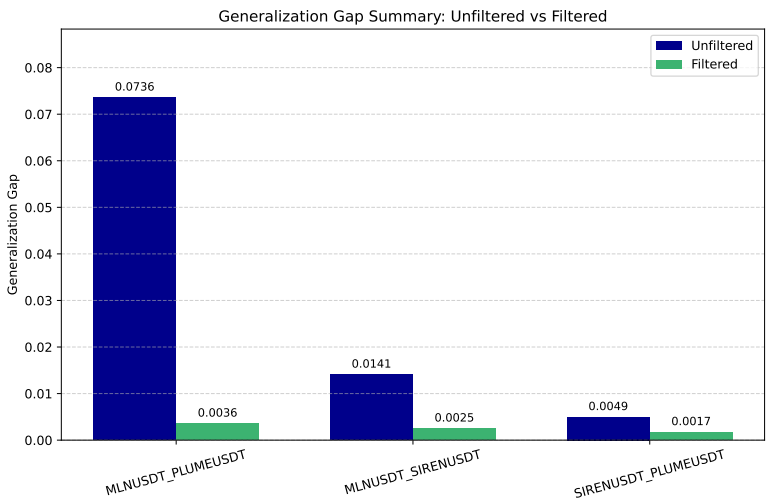

Figure 5.11: Robustness Improvements, All Pairs, 50

5.1.3 20-Tick Candlestick Group

The results are listed for the 20-tick group similarly. The robustness is significantly improved in cases where the model predicts there to be any edge (see Figure **??**), and in no-edge expectations, most likely due to the extreme zoom in effect in terms of the timeframe, similar results are obtained between the unfiltered and the filtered versions.

Table 5.3: Model Evaluation Metrics, 20

Pair	Val Prec. (Unf.)	Test Prec. (Unf.)	Gap (Unf.)	Val Prec. (Filt.)	Test Prec. (Filt.)	Gap (Filt.)
MLNUSDT_PLUMEUSDT	0.3551	0.3866	0.0315	0.3958	0.3939	0.0019
MLNUSDT_SIRENUSDT	0.4894	0.4836	0.0058	0.5125	0.5254	0.0128
SIRENUSDT_PLUMEUSDT	0.5166	0.5073	0.0093	0.5192	0.5047	0.0145

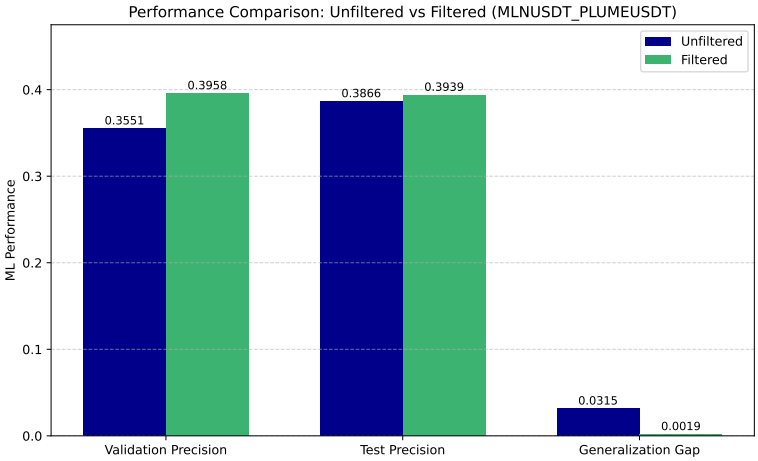

Figure 5.12: MLNUSDT/PLUMEUSDT pair, 20, improvement

It is observed that for all pairs / all groupings the generalization gap is closed to a significant degree, with an average closing of the generalization gap by more than 60% (see Figure 5.13).

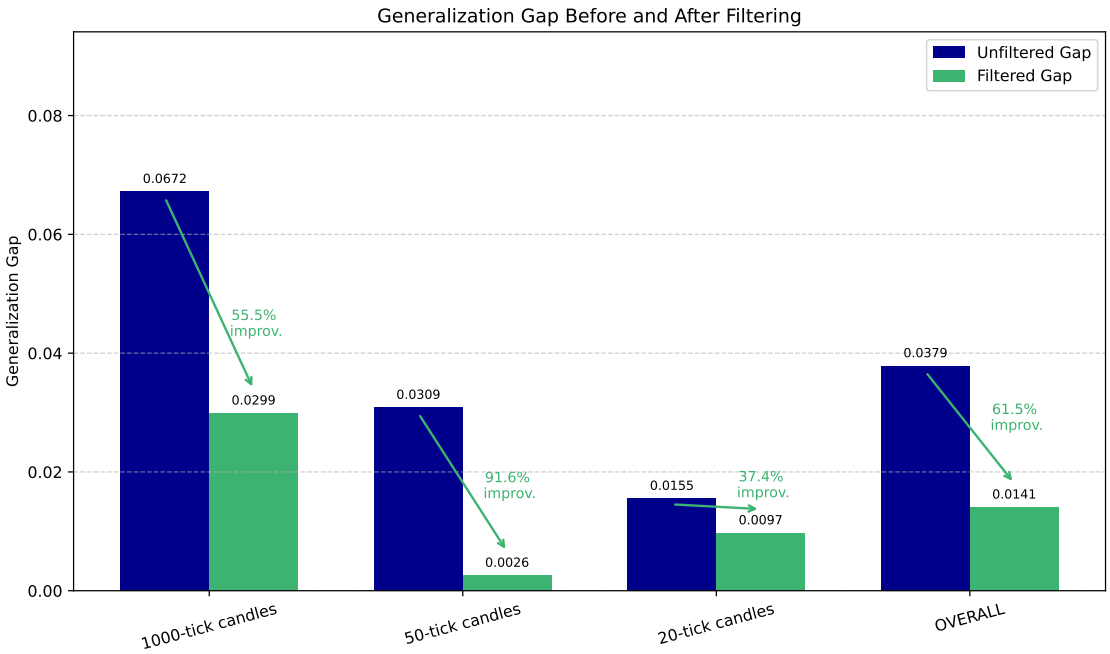

Figure 5.13: Improvements in the Generalization Gap, Overall

The biggest improvement is observed in the 50-tick candles, which is the middle group.

Due to small differences in the data and randomness, there are differences between groups in the method's effectiveness, but there is no clear trend observed such as the methods working less well in larger or smaller timeframes. However, there is a possibility that there is a peak level of effectiveness given the parameters used in this study and, as a next step, the parameters can be tuned to see if improvements can be achieved.

The improvements per each group are listed in the table below.

Table 5.4: Generalization Gap Improvement Across Tick Group Sizes

Group Size	**Gap (Unfiltered)**	**Gap (Filtered)**	**% Improvement**
1000-tick candles	0.0672	0.0299	55.51%
50-tick candles	0.0309	0.0026	91.62%
20-tick candles	0.0155	0.0097	37.36%
OVERALL	0.0379	0.0141	61.50%

5.2 Pairwise Results

Focusing on the ML robustness and performance for each pair for all groupings, the following figures are created to illustrate the improvements in robustness for each dynamical system. As seen in Figure 5.14, the robustness improvements are consistent not only among each grouping but also among each dynamical system. This finding further validates the procedural approach taken to the problem of robustness in ER forecasting.

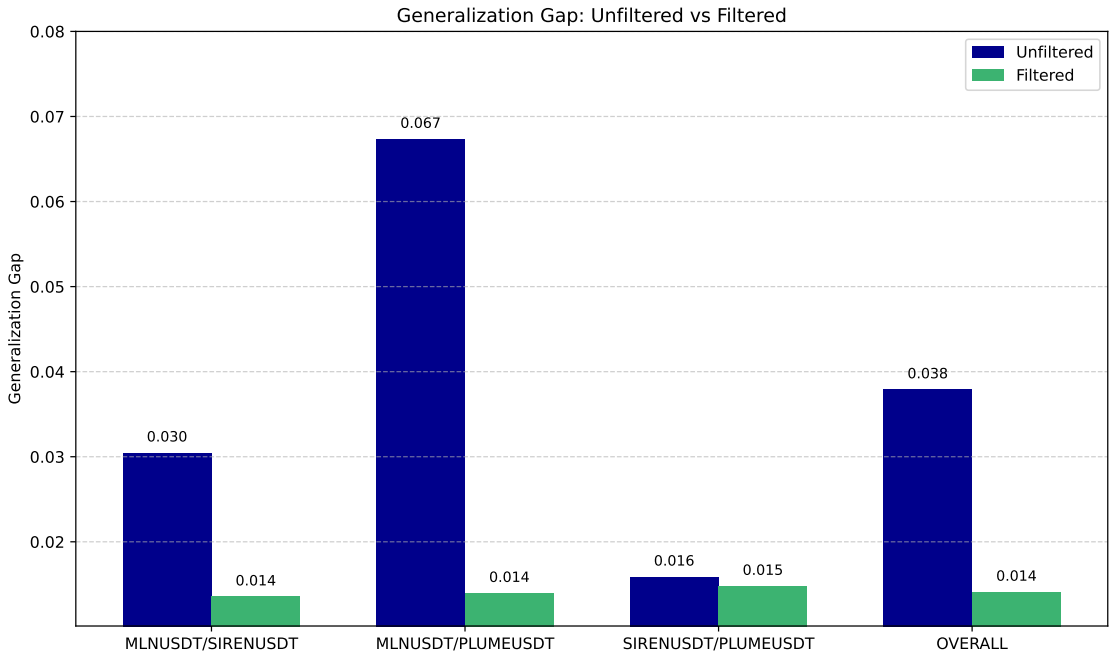

Figure 5.14: Improvements in the Generalization Gap, Pairwise

5.3 Best Hyperparameters for the LGBM Classifier

The best hyperparameters of the LGBM classifier, given the hyperparameter tuning, is shown in the below tables.

5.3.1 1000-Tick Candlestick Group

The best hyperparameters for the 1000-tick group is as follows.

Table 5.5: Best Hyperparameters, 1000

Pair	num_leaves	n_estimators	min_child_samples	max_depth	learning_rate
MLNUSDT_SIRENUSDT	92	700	15	13	0.01
MLNUSDT_PLUMEUSDT	63	300	5	14	0.05
SIRENUSDT_PLUMEUSDT	20	1000	30	11	0.01

5.3.2 50-Tick Candlestick Group

The best hyperparameters for the 50-tick group is as follows.

Table 5.6: Best Hyperparameters, 50

Pair	num_leaves	n_estimators	min_child_samples	max_depth	learning_rate
MLNUSDT_SIRENUSDT	63	400	5	7	0.01
MLNUSDT_PLUMEUSDT	63	300	5	14	0.05
SIRENUSDT_PLUMEUSDT	34	600	15	11	0.01

5.3.3 20-Tick Candlestick Group

The best hyperparameters for the 20-tick group is as follows.

Table 5.7: Best Hyperparameters, 20

Pair	num_leaves	n_estimators	min_child_samples	max_depth	learning_rate
MLNUSDT_PLUMEUSDT	77	900	25	8	0.05
MLNUSDT_SIRENUSDT	63	400	5	7	0.01
SIRENUSDT_PLUMEUSDT	20	300	25	10	0.20

The five hyperparameters tuned can be visualized for the three cases in Figure 5.15.

5.3.4 Summary of Hyperparameter Tuning

It was observed that the LGBM Classifier hyperparameters are quite different for each of the three synthetic pairs, within each grouping. This can be a sign that the dynamics of the underlying processes are best captured using slightly different tree-based structures, and also that the methods utilized for improving robustness remain valid throughout this range presented.

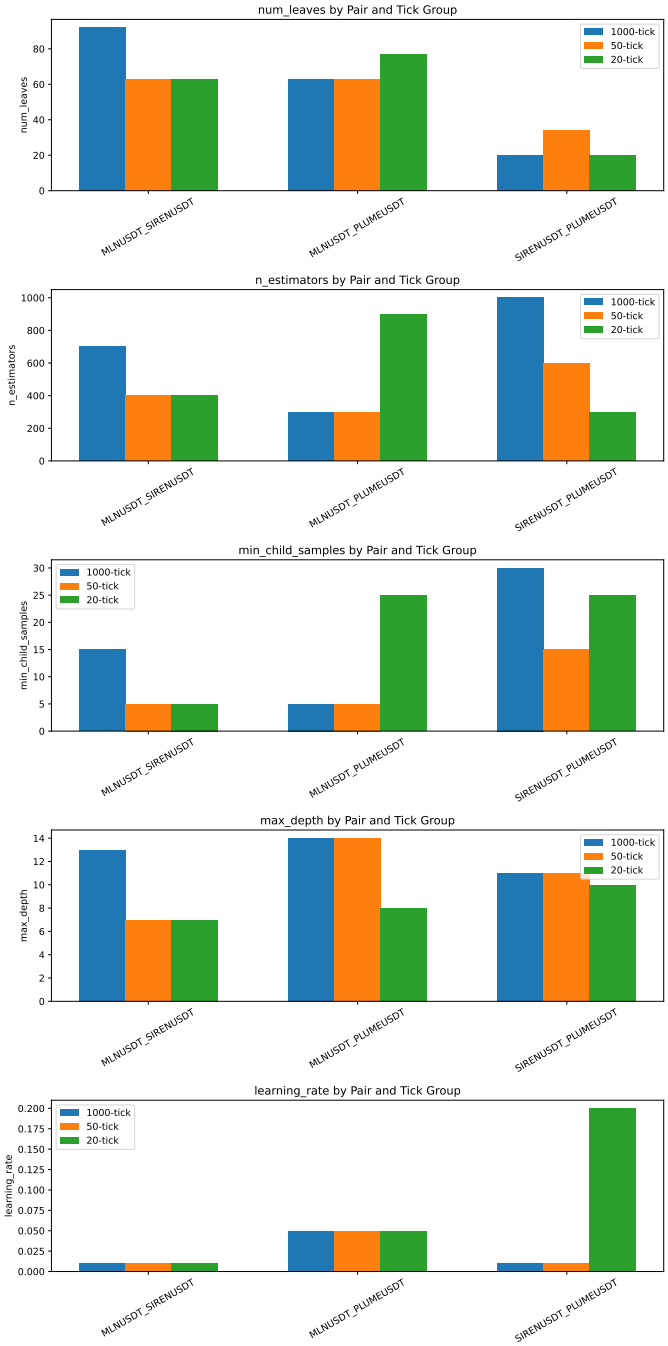

Figure 5.15: LGBM Classifier Hyperparameters, Tuned, Pairwise

5.4 Portfolio Impact of the Results

Exemplifying the impact of this study on the profitability of a trading strategy, using the 1000-tick grouped MLNUSDT/PLUMEUSDT outcomes, given the profitability condition;

Profitability Condition - Long

$$(1+L\left[\left(\frac{b_1}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2-1\right])^P\cdot(1+L\left[\left(\frac{b_2}{a}\right)\frac{(1-f_{ss})}{1+f_{sb}}(1-f)^2-1\right])^{1-P}>1$$

and the parameters,

- L = 23
- a = 1
- b1 = 1 + 0.0032
- b2 = 1 - 0.0032
- f_ss = 0.01/100
- f_sb = 0.01/100
- f = 0.01/100
- p = 0.6

the expression evaluates to;

Profitability Condition - Long

$$1.00288 > 1$$

which leads to the account balance of

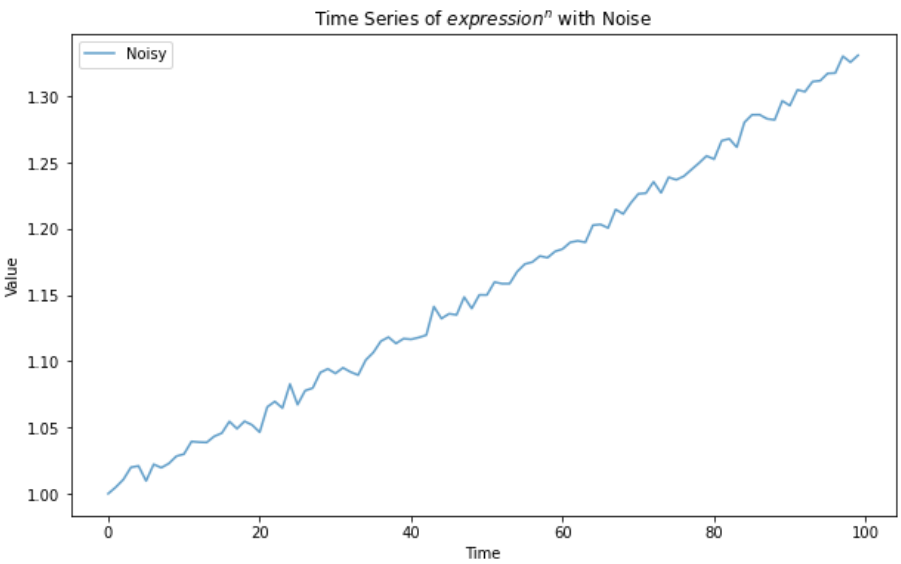

Figure 5.16: Account Growth with Robust Predictions

while the parameters for the unfiltered case

- L = 23

- a = 1
- b1 = 1 + 0.0032
- b2 = 1 - 0.0032
- f_ss = 0.01/100
- f_sb = 0.01/100
- f = 0.01/100
- p = 0.52

results in the profitability condition of;

Profitability Condition - Long

$$0.99101 < 1$$

leading to the account balance of

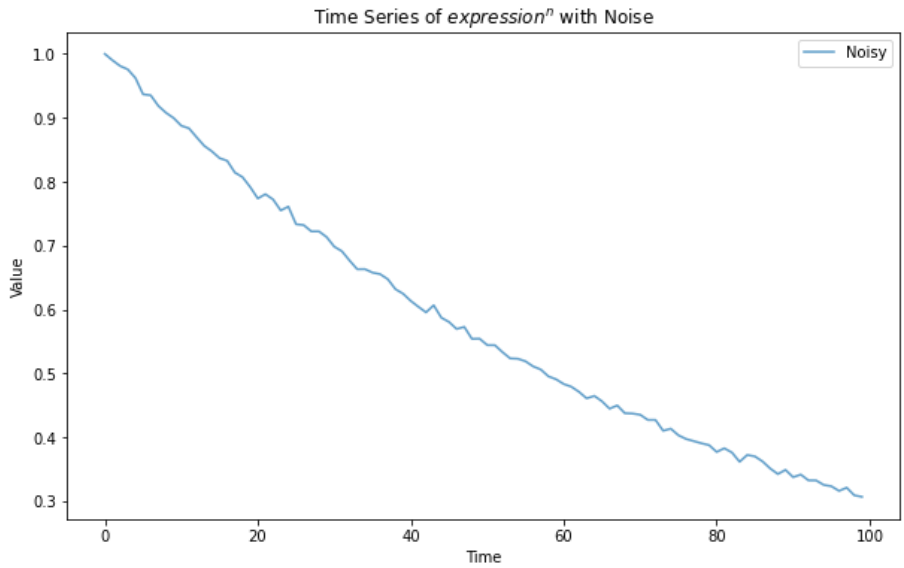

Figure 5.17: Non-robust case, account balance loss

, each over the course of 200 trades. It should be noted that hidden fees increase (in percentage) with increasing account balances (due to increasing slippage) and a more detailed analysis is needed to get the exact account growth rate at each iteration, but these two curves are a good estimator of how dramatic the differences between the two cases can be.

6 Conclusions and Future Work

This chapter concludes the thesis, summarizing the study and proposing pathways through which the work presented in this thesis can be built upon.

6.1 Conclusions

This thesis attempted to propose procedural methods to improve the robustness of binary classifiers used in an ER forecasting setting in the cryptocurrency market. The results of the experiments using the proposed methods imply that robustness can be improved significantly using the ensemble model proposed in combination with a Hurst exponent mean-reversion filter.

The highlights of the work can be summarized as follows;

- This thesis proposed a workflow that aimed to directly measure and improve the robustness of binary classifiers through the precision gap, which is a novel approach to the problem, as most studies aim to improve model performance.

- In this study, synthetic cryptocurrency pairs were used with candlesticks made of tick-number based groups in an attempt to increase the signal-to-noise ratio of the dynamical systems under investigation in an ER forecasting setting. This was not previously studied (or at least not extensively) in a BinClass-based ER forecasting setting in the crypto market and is a key novel significance of the work.

- The Hurst Exponent was utilized to function as a mean-reversion filter, in combination with a trend continuation cue, which successfully improved model performance while simultaneously closing a significant portion of the generalization gap. This combination is another key reasonn why the thesis has novel significance.

- The improvements in robustness can be quantified by improvements of from 37% to 92%, based on the grouping, and a mean improvement of 61.5% in the generalization gap. This is a very significant finding and displays how robustness-first approaches in ER forecasting can yield excellent results.

- Moreover, the Hurst exponent is able to improve the performance of the model on top of the robustness improvements discussed. An (additive) improvement in precision (defined as movement away from the theoretical breakeven expectation) of 5-6% was observed on average, with levels approaching 10-15% in groupings of longer period. This suggests that the combined use of the mean-reversion filter with a trend continuation cue may be underexplored by market participants globally in the cryptocurrency market.

- The improvements in both robustness and performance have been observed in timeframes that are largely different than each other. This further asserts the success of the proposed methods.

- The findings were benchmarked against losses due to trading fees and hidden costs and it was found that the improvements in robustness and performance are significant enough to make a dramatic difference in the profitability condition of trading strategies, and it was proven that the improvements were not trivial.
- The LGBM-based ensemble model was the only model used in this research and no data snooping is present. In addition, the probabilities of data leakage is also very small as the feature engineering was conducted carefully, and the poor performance of the standalone LGBM classifier is testament to the validity of the ML workflow as far as data leakage is concerned.

With these points in mind, it can be concluded that the use of Hurst exponent-based mean reversion filters combined with trend continuation cues has significant potential to improve robustness in classification based ER forecasting in the crypto market.

6.2 Future Work

The future work proposed can be summarized as follows:

1. As significant improvement are achieved with the ensemble model present, a suggested next step may be to test this tool across a larger pool of synthetic pairs in the crypto market, as well as in other financial markets, to assess the impact on robustness across dynamical systems and markets.
2. The testing phase was conducted using unseen data, however to take into consideration the reactions of the market to active interaction, from the perspective of game theory and sensitivity to initial conditions in nonlinear systems, live tests using CEX APIs must be conducted to validate the results of this research.
3. The custom layer or the Hurst exponent based mean reversion filter was not tuned during or before training (which is a very significant standalone achievement), which means that there is very significant potential for further adding to the improvements in robustness and performance.
4. As feature engineering was purely based on historical price action, the addition of external data, as well as market condition data, may help improve ML performance on top of what has been achieved.
5. Going into further detail and building more complex statistical models of synthetic pairs in the crypto market will help further refine the level of precision at which the portfolio impact of the proposed methods are measured.
6. It is also important to study the relationship between accuracy and negative predictive value vs. precision, which may impact model training to improve performance through the use of custom loss functions and position sizing / market interaction strategies.

Bibliography

[1] TradingView, "Btc/usd vs gold chart," https://www.tradingview.com/chart/INDEX:BTCUSD/TVC:GOLD, 2025, accessed: 2025-07-22.

[2] A. Ballis and K. Drakos, "The explosion in cryptocurrencies: a black hole analogy," *Financial Innovation*, vol. 7, p. 8, 2021, published: 27 January 2021. [Online]. Available: https://link.springer.com/article/10.1186/s40854-020-00222-0

[3] Binance. (2025) Binance API Documentation. [Online]. Available: https://www.binance.com/en/binance-api

[4] Nichtich, "File:binary-classification-labeled.svg," https://commons.wikimedia.org/wiki/File:Binary-classification-labeled.svg, 2008, last updated: 3 January 2025.

[5] R. de Roover, *The Rise and Decline of the Medici Bank: 1397-1494*, 1963. [Online]. Available: https://gwern.net/doc/history/medici/1963-deroover-theriseanddeclineofthemedicibank.pdf

[6] A. P. Usher, "The origins of banking: The primitive bank of deposit, 1200–1600," *The Economic History Review*, vol. 4, no. 4, pp. 399–428, 1934, open PDF: https://cooperative-individualism.org/usher-abbott_the-origins-of-banking-1934-apr.pdf. [Online]. Available: https://www.jstor.org/stable/2589849

[7] G. M. Gallarotti, *The Anatomy of an International Monetary Regime: The Classical Gold Standard, 1880-1914*. New York: Oxford University Press, 1995. [Online]. Available: https://books.google.de/books?hl=en&lr=&id=Jx7oCwAAQBAJ&oi=fnd

[8] M. D. Bordo, "The classical gold standard: Some lessons for today," 1992. [Online]. Available: http://qed.econ.queensu.ca/pub/faculty/keayi/232links/232read/Bordo92.pdf

[9] M. de Cecco, "The international debt problem in the interwar period," European University Institute, Department of Economics, EUI Working Paper No. 84/103, April 1984. [Online]. Available: https://cadmus.eui.eu/bitstream/handle/1814/22902/1984_EUIWP_103.pdf?sequence=1

[10] O. Accominotti and D. Chambers, "The returns to currency trading: Evidence from the interwar period," Mar. 2013, working Paper. [Online]. Available: https://economics.yale.edu/sites/default/files/accominotti-130401.pdf

[11] B. Eichengreen, *Globalizing Capital: A History of the International Monetary System*, 2nd ed. Princeton and Oxford: Princeton University Press, 2008. [Online]. Available: https://delong.typepad.com/files/eichengreen-globalizing.pdf

[12] M. D. Bordo, "The bretton woods international monetary system: A historical overview," in *A Retrospective on the Bretton Woods System: Lessons for International Monetary Reform*, M. D. Bordo and B. Eichengreen,

Eds. University of Chicago Press, 1993, pp. 3–108. [Online]. Available: https://www.nber.org/system/files/chapters/c6867/c6867.pdf

[13] M. Obstfeld, "Floating exchange rates: Experience and prospects," *Brookings Papers on Economic Activity*, vol. 1985, no. 2, pp. 369–450, 1985. [Online]. Available: https://www.brookings.edu/wp-content/uploads/1985/06/1985b_bpea_obstfeld_cooper_krugman.pdf

[14] W. N. Goetzmann and P. Jorion, "A century of global stock markets," National Bureau of Economic Research, Cambridge, MA, NBER Working Paper 5901, 1997. [Online]. Available: https://www.nber.org/system/files/working_papers/w5901/w5901.pdf

[15] F. X. Diebold, "The past, present, and future of macroeconomic forecasting," *Journal of Economic Perspectives*, vol. 12, no. 2, pp. 175–192, 1998. [Online]. Available: https://www.sas.upenn.edu/~fdiebold/papers/paper19/jep2.pdf

[16] J. Yao, C. L. Tan, and H.-L. Poh, "Neural networks for technical analysis: A study on klci," *International Journal of Theoretical and Applied Finance*, vol. 2, no. 2, pp. 221–241, 1999. [Online]. Available: https://scispace.com/pdf/neural-networks-for-technical-analysis-a-study-on-klci-3qeezd92dx.pdf

[17] C.-M. Kuan and T. Liu, "Forecasting exchange rates using feedforward and recurrent neural networks," *Journal of Applied Econometrics*, vol. 10, no. 4, pp. 347–364, 1995. [Online]. Available: https://www.jstor.org/stable/2285052

[18] A. Salehpour and K. Samadzamini, "Machine learning applications in algorithmic trading: A comprehensive systematic review," *International Journal of Education and Management Engineering*, vol. 13, no. 6, pp. 41–53, 2023. [Online]. Available: https://www.mecs-press.org/ijeme/ijeme-v13-n6/IJEME-V13-N6-5.pdf

[19] F. Kamalov and I. Gurrib, "Machine learning based forecasting of significant daily returns in foreign exchange markets," *arXiv preprint arXiv:2009.10065*, 2020. [Online]. Available: https://arxiv.org/pdf/2009.10065

[20] W. A. Addy, A. O. Ajayi-Nifise, B. G. Bello, S. T. Tula, O. Odeyemi, and T. Falaiye, "Machine learning in financial markets: A critical review of algorithmic trading and risk management," *International Journal of Science and Research Archive*, vol. 11, no. 01, pp. 1853–1862, 2024. [Online]. Available: https://www.researchgate.net/profile/Titilola-Falaiye/publication/378287610_Machine_learning_in_financial_markets_A_critical_review_of_algorithmic_trading_and_risk_management/links/65d40ab728b7720cecdc912f/Machine-learning-in-financial-markets-A-critical-review-of-algorithmic-trading-and-risk-manage pdf

[21] C. Dutta, N. Ravishanker, K. Karpman, and S. Basu, "Review of statistical approaches for modeling high-frequency trading data," *Sankhya B: The Indian Journal of Statistics*, 2022. [Online]. Available: https://par.nsf.gov/servlets/purl/10378098

[22] D. K. Sharma, H. Hota, and R. Handa, "Prediction of foreign exchange rate using regression techniques," *Review of Business and Technology Research*, vol. 14, no. 1, pp. 29–33, 2017. [Online]. Available: https://mtmi.us/rbtr/sept/sept2017/06-Sharma-Hota-Handa-pp29-33.pdf

[23] S. Chan, "Exploration and prediction of factors influencing china's foreign exchange fluctuations based on multiple linear regression model," in *Proceedings of the 2025 3rd International Conference on Digital Economy and Management Science (CDEMS 2025)*, ser. Advances in Economics, Business and Management Research, vol. 336. Atlantis Press, 2025, pp. 770–773. [Online]. Available: https://www.atlantis-press.com/proceedings/cdems-25/126012795

[24] L. V. Tran, S. T. Le, and H. M. Tran, "Empirical study of cryptocurrency prices using linear regression methods," in *2022 RIVF International Conference on Computing and Communication Technologies (RIVF)*. IEEE, 2022, pp. 1–6. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10013790

[25] T. Singh, S. Mishra, D. Pandey, C. Sharma, S. Koli, and K. Joshi, "Crypto currency-bitcoin price predictor using linear regression and random forest," in *2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT)*, vol. 1, 2024, pp. 1–5.

[26] N. Uras, L. Marchesi, M. Marchesi, and R. Tonelli, "Forecasting bitcoin closing price series using linear regression and neural networks models," *PeerJ Computer Science*, vol. 6, p. e279, 2020. [Online]. Available: https://peerj.com/articles/cs-279/

[27] L. T. C. Ow, D. C. L. Ngo, and Y. Lee, "Prediction of forex trend movement using linear regression line, two-stage of multi-layer perceptron and dynamic time warping algorithms," *Journal of Information and Communication Technology*, vol. 15, no. 2, pp. 117–140, 2016. [Online]. Available: https://www.researchgate.net/publication/313738109_Prediction_of_forex_trend_movement_using_linear_regression_line_two-stage_of_multi-layer_perceptron_and_dynamic_time_warping_algorithms

[28] N. Köse, Y. E. Gür, and E. Ünal, "Deep learning and machine learning insights into the global economic drivers of the bitcoin price," *Journal of Forecasting*, vol. 44, no. 5, pp. 1666–1698, 2025. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/for.3258

[29] S. A. Gyamerah, "Are bitcoins price predictable? evidence from machine learning techniques using technical indicators," *arXiv preprint arXiv:1909.01268*, 2019. [Online]. Available: https://arxiv.org/pdf/1909.01268

[30] J. Zheng, Y. Wang, S. Li, and H. Chen, "The stock index prediction based on svr model with bat optimization algorithm," *Algorithms*, vol. 14, no. 10, p. 299, 2021. [Online]. Available: https://www.mdpi.com/1999-4893/14/10/299

[31] E. García-Gonzalo, P. J. García-Nieto, G. F. Valverde, P. R. Fernández, F. S. Lasheras, and S. L. S. Gómez, "Hybrid de-optimized gpr and narx/svr models for forecasting gold spot prices: A case study of the global commodities market," *Mathematics*, vol. 12, no. 7, p. 1039, 2024. [Online]. Available: https://www.mdpi.com/2227-7390/12/7/1039

[32] L. Khaidem, S. Saha, and S. R. Dey, "Predicting the direction of stock market prices using random forest," *arXiv preprint arXiv:1605.00003*, 2016. [Online]. Available: https://arxiv.org/pdf/1605.00003

[33] K. Smelyakov, O. Bizkovnyi, N. Sharonova, S. Smelyakov, and A. Chupryna, "Building of regression models for cryptocurrency price prediction," in *Proceedings*

of the CEUR Workshop, 2022. [Online]. Available: https://ceur-ws.org/Vol-3171/paper90.pdf

[34] S. Demirezen, "Forecasting of gold prices with arima and random forest regression and evaluating of their prediction performance," *Social Sciences Research Journal*, vol. 10, no. 03, pp. 747–758, 2021. [Online]. Available: https://socialsciencesresearchjournal.com/index.php/ssrj/article/view/329?i=1

[35] A. Natekin and A. Knoll, "Gradient boosting machines, a tutorial," *Frontiers in Neurorobotics*, vol. 7, p. 21, 2013. [Online]. Available: https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2013.00021/full

[36] S. M. Rahimpour, R. Goudarzi, V. Shahparifard, and S. N. Mirpoorian, "Algorithmic trading using technical indicators and extereme gradient boosting," in *2024 11th IEEE Swiss Conference on Data Science (SDS)*. IEEE, 2024. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10675748

[37] L. Tian, L. Feng, L. Yang, and Y. Guo, "Stock price prediction based on lstm and lightgbm hybrid model," *The Journal of Supercomputing*, vol. 78, pp. 11 768–11 793, 2022. [Online]. Available: https://link.springer.com/article/10.1007/s11227-022-04326-5

[38] L. D. Persio and N. Fraccarolo, "Energy consumption forecasts by gradient boosting regression trees," *Mathematics*, vol. 11, no. 5, p. 1068, 2023. [Online]. Available: https://www.researchgate.net/publication/368699657_Energy_Consumption_Forecasts_by_Gradient_Boosting_Regression_Trees

[39] S. F. N. Islam, A. Sholahuddin, and A. S. Abdullah, "Extreme gradient boosting (xgboost) method in making forecasting application and analysis of usd exchange rates against rupiah," *Journal of Physics Conference Series*, vol. 1722, no. 1, p. 012016, 2021. [Online]. Available: https://www.researchgate.net/publication/348359394_Extreme_gradient_boosting_XGBoost_method_in_making_forecasting_application_and_analysis_of_USD_exchange_rates_against_rupiah

[40] T. R. Li, A. S. Chamrajnagar, X. R. Fong, N. R. Rizik, and F. Fu, "Sentiment-based prediction of alternative cryptocurrency price fluctuations using gradient boosting tree model," *Frontiers in Physics*, vol. 7, p. 98, 2019. [Online]. Available: https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2019.00098/full

[41] H. Alizadegan, A. Radmehr, and M. A. Ilani, "Forecasting bitcoin prices: A comparative study of machine learning and deep learning algorithms," *Research Square*, 2024. [Online]. Available: https://assets-eu.researchsquare.com/files/rs-4390390/v1/3d260b92-c6c4-42e5-b508-da6ae5a661a7.pdf?c=1715251654

[42] J. Schmidhuber, "Deep learning in neural networks: An overview," The Swiss AI Lab IDSIA, Istituto Dalle Molle di Studi sull'Intelligenza Artificiale, University of Lugano & SUPSI, Tech. Rep., 2014. [Online]. Available: https://arxiv.org/pdf/1404.7828

[43] C. M. Bishop, "Neural networks and their applications," *Review of Scientific Instruments*, vol. 65, no. 6, pp. 1803–1832, 1994. [Online]. Available: https://web.archive.org/web/20060903153555id_/http://www.stat.purdue.edu/~zdaye/Readings/Neural_Networks_and_Their_Applications.pdf

[44] A. Fadlalla and C.-H. Lin, "An analysis of the applications of neural networks in finance," *Interfaces*, vol. 31, no. 4, pp. 112–122, 2001. [Online]. Available: https://d1wqtxts1xzle7.cloudfront.net/105597864/112-libre.pdf?1694170105=&response-content-disposition=inline%3B+filename%3DAn_Analysis_of_the_Applications_of_Neura.pdf&Expires=1753029028&Signature=VAYlm6EB511hCDDrJQq8SBMrJfRvrFE9ignf6nfANxNCh8zhhysw9e9SqVkB8~iFMv8YA6tuC7~7Q5BISA_&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

[45] K. Sako, B. N. Mpinda, and P. C. Rodrigues, "Neural networks for financial time series forecasting," *Entropy*, vol. 24, no. 5, p. 657, 2022. [Online]. Available: https://www.mdpi.com/1099-4300/24/5/657

[46] Z. Zhou and R. Wu, "Stock price prediction model based on convolutional neural networks," *Journal of Industrial Engineering & Applied Science*, vol. 2, no. 4, pp. 1–7, 2024. [Online]. Available: https://www.suaspress.org/ojs/index.php/JIEAS/article/view/v2n4a01

[47] F. Liu, Y. Li, B. Li, J. Li, and H. Xie, "Bitcoin transaction strategy construction based on deep reinforcement learning," *Applied Soft Computing*, vol. 113, p. 107952, Dec. 2021. [Online]. Available: http://dx.doi.org/10.1016/j.asoc.2021.107952

[48] S. S. Sarı and K. S. Ilgın, "Bist-100 endeks hareketlerinin brics endeksleri aracılığıyla tahmin edilmesi: Yapay sinir ağları uygulaması [estimation of ise-100 index movement through brics indexes: Application of artificial neural networks]," *Abant Sosyal Bilimler Dergisi [Journal of Abant Social Sciences]*, vol. 22, no. 1, pp. 351–366, 2022. [Online]. Available: https://dergipark.org.tr/tr/download/article-file/2344328

[49] M. A. Junior, P. Appiahene, and O. Appiah, "Forex market forecasting with two-layer stacked long short-term memory neural network (lstm) and correlation analysis," *Journal of Electrical Systems and Information Technology*, vol. 9, no. 1, p. 14, 2022. [Online]. Available: https://link.springer.com/article/10.1186/s43067-022-00054-1#citeas

[50] S. E. Charandabi and K. Kamyar, "Using a feed forward neural network algorithm to predict prices of multiple cryptocurrencies," *European Journal of Business and Management Research*, vol. 6, no. 5, pp. 15–19, 2021. [Online]. Available: https://www.ejbmr.org/index.php/ejbmr/article/view/1056

[51] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 12, pp. 6999–7019, 2022. [Online]. Available: https://arxiv.org/pdf/2004.02806

[52] Z. Hu, Y. Zhao, and M. Khushi, "A survey of forex and stock price prediction using deep learning," *Applied System Innovation*, vol. 4, no. 1, p. 9, 2021. [Online]. Available: https://www.mdpi.com/2571-5577/4/1/9

[53] X. Zhang, "Analyzing financial market trends in cryptocurrency and stock prices using cnn-lstm models," *Preprints.org*, July 2024. [Online]. Available: https://www.preprints.org/frontend/manuscript/4d0a14bd7c84790a1c10cd67966312af/download_pub

[54] J. Wang, X. Wang, J. Li, and H. Wang, "A prediction model of cnn-tlstm for usd/cny exchange rate prediction," *IEEE Access*, vol. 9, pp. 73 346–73 354, 2021. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9431182

[55] H. Song and H. Choi, "Forecasting stock market indices using the recurrent neural network based hybrid models: Cnn-lstm, gru-cnn, and ensemble models," *Applied Sciences*, vol. 13, no. 7, 2023. [Online]. Available: https://www.mdpi.com/2076-3417/13/7/4644

[56] I. E. Livieris, N. Kiriakidou, S. Stavroyiannis, and P. Pintelas, "An advanced cnn-lstm model for cryptocurrency forecasting," *Electronics*, vol. 10, no. 3, 2021. [Online]. Available: https://www.mdpi.com/2079-9292/10/3/287

[57] G. Sermpinis, A. Karathanasopoulos, R. Rosillo, and D. de la Fuente, "Neural networks in financial trading," *Annals of Operations Research*, 2019. [Online]. Available: https://eprints.gla.ac.uk/178881/1/178881.pdf

[58] MathWorks. Long short-term memory networks. [Online]. Available: https://www.mathworks.com/help/deeplearning/ug/long-short-term-memory-networks.html

[59] M. Nabipour, P. Nayyeri, H. Jabani, A. Mosavi, E. Salwana, and S. S. Band, "Deep learning for stock market prediction," *Entropy*, vol. 22, no. 8, p. 840, 2020. [Online]. Available: https://www.researchgate.net/publication/343339583_Deep_Learning_for_Stock_Market_Prediction

[60] Y. Gao, R. Wang, and E. Zhou, "Stock prediction based on optimized lstm and gru models," *Scientific Programming*, vol. 2021, pp. 1–8, 2021. [Online]. Available: https://onlinelibrary.wiley.com/doi/full/10.1155/2021/4055281

[61] K. Zarzycki and M. Ławryńczuk, "Advanced predictive control for gru and lstm networks," *Information Sciences*, vol. 616, pp. 229–254, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0020025522011987

[62] P. L. Seabe, C. R. B. Moutsinga, and E. Pindza, "Forecasting cryptocurrency prices using lstm, gru, and bi-directional lstm: A deep learning approach," *Fractal and Fractional*, vol. 7, no. 2, 2023. [Online]. Available: https://www.mdpi.com/2504-3110/7/2/203

[63] M. R. Pahlevi, K. Kusrini, and T. Hidayat, "Comparison of lstm and gru models for forex prediction," *Sinkron : Jurnal dan Penelitian Teknik Informatika*, vol. 8, no. 4, Oct. 2023. [Online]. Available: https://www.jurnal.polgan.ac.id/index.php/sinkron/article/view/12709

[64] A. Zhang, Z. C. Lipton, M. Li, and A. Smola. 10.2. gated recurrent units (gru). Dive into Deep Learning. [Online]. Available: https://d2l.ai/chapter_recurrent-modern/gru.html

[65] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[66] J. Xiao, S. Bi, and T. Deng, "Comparative analysis of lstm, gru, and transformer models for stock price prediction," *arXiv preprint arXiv:2411.05790*, 2024. [Online]. Available: https://arxiv.org/pdf/2411.05790

[67] J. Lian, "Comparative analysis of lstm, gru and transformer deep learning models for cryptocurrency zec price prediction performance," in *Proceedings of the 9th International Conference on Financial Innovation and Economic Development (ICFIED 2024)*, 2024, pp. 396–405. [Online]. Available: https://www.atlantis-press.com/article/125999624.pdf

[68] D. V. Godoy, "Deep learning visuals," https://dvgodoy.github.io/dl-visuals/Transformers/, 2024, accessed: [Insert Date Accessed, e.g., 22 July 2025].

[69] J. Gong and S. Sun, "A new approach of stock price prediction based on logistic regression model," in *2009 International Conference on New Trends in Information and Service Science*, 2009, pp. 1366–1371. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/5260596

[70] F. Zhou, Q. Zhang, D. Sornette, and L. Jiang, "Cascading logistic regression onto gradient boosted decision trees for forecasting and trading stock indices," *Applied Soft Computing*, vol. 84, p. 105747, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1568494619305289

[71] A. Salah and G. Hayette, "A meta-analysis of supervised and unsupervised machine learning algorithms and their application to active portfolio management," *Expert Systems with Applications*, vol. 271, p. 126611, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417425002337

[72] Y. Guo, "Application of generative adversarial networks in financial time series data augmentation," *SSRN Electronic Journal*, Feb 2025. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5142078

[73] L. Xie and S. Yu, "Unsupervised feature extraction with convolutional autoencoder with application to daily stock market prediction," *Concurrency and Computation: Practice and Experience*, vol. 33, no. 16, p. e6282, 2021. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/cpe.6282?casa_token=424hzGvMLrgAAAAA%3AVWfUQ3XfGD2pzdoncXhjP7xqermTQ6gQKCA-JKn0Yh24J6awQ3wpimOAg_Zpt2R9iO-NGf2Cxh-KockDFg

[74] L. Thordin, "Analysing customer behaviour in the fx market using order flow data and machine learning techniques," Master's Thesis, Lund University, Lund, Sweden, September 2014, centre for Mathematical Sciences, Mathematical Statistics. [Online]. Available: https://lup.lub.lu.se/luur/download?func=downloadFile&recordOId=5043462&fileOId=5043464

[75] Z. Issa and B. Horvath, "Non-parametric online market regime detection and regime clustering for multidimensional and path-dependent data structures," 2023. [Online]. Available: https://arxiv.org/abs/2306.15835

[76] M. J. Rajaei and Q. H. Mahmoud, "A survey on pump and dump detection in the cryptocurrency market using machine learning," *Future Internet*, vol. 15, no. 8, 2023. [Online]. Available: https://www.mdpi.com/1999-5903/15/8/267

[77] T. Huotari, J. Savolainen, and M. Collan, "Deep reinforcement learning agent for s&p 500 stock selection," *Axioms*, vol. 9, no. 4, p. 130, Nov 2020. [Online]. Available: https://www.researchgate.net/publication/346804179_Deep_Reinforcement_Learning_Agent_for_SP_500_Stock_Selection

[78] S. Kalva and N. Satuluri, "Stock market investment strategy using deep-q-learning network," in *Multi-disciplinary Trends in Artificial Intelligence*, R. Morusupalli, T. S. Dandibhotla, V. V. Atluri, D. Windridge, P. Lingras, and V. R. Komati, Eds. Cham: Springer Nature Switzerland, 2023, pp. 484–495. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-36402-0_45#citeas

[79] H. Zhang, Z. Jiang, and J. Su, "A deep deterministic policy gradient-based strategy for stocks portfolio management," 2021. [Online]. Available: https://arxiv.org/abs/2103.11455

[80] F. Rundo, "Deep lstm with reinforcement learning layer for financial trend prediction in fx high frequency trading systems," *Applied Sciences*, vol. 9, no. 20, 2019. [Online]. Available: https://www.mdpi.com/2076-3417/9/20/4460

[81] M. Qin, S. Sun, W. Zhang, H. Xia, X. Wang, and B. An, "Earnhft: Efficient hierarchical reinforcement learning for high frequency trading," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 13, Mar. 2024, pp. 14 669–14 676. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/29384

[82] K. Kumlungmak and P. Vateekul, "Multi-agent deep reinforcement learning with progressive negative reward for cryptocurrency trading," *IEEE Access*, vol. 11, pp. 66 440–66 455, 2023.

[83] M. R. SAHIB, H. ELKINA, and T. ZAKI, "From technical indicators to trading decisions: A deep learning model combining cnn and lstm," *(IJACSA) International Journal of Advanced Computer Science and Applications*, vol. 15, no. 6, pp. 847–855, 2024. [Online]. Available: https://thesai.org/Downloads/Volume15No6/Paper_85-From_Technical_Indicators_to_Trading_Decisions.pdf

[84] I. Noorani and F. Mehrdoust, "Forecasting bitcoin price by a hybrid structure based on arima, svm and lssvm models," *Annals of Financial Economics*, vol. 19, no. 04, p. 2450020, 2024. [Online]. Available: https://doi.org/10.1142/S2010495224500209

[85] M. Jiang, J. Liu, L. Zhang, and C. Liu, "An improved stacking framework for stock index prediction by leveraging tree-based ensemble models and deep learning algorithms," *Physica A: Statistical Mechanics and its Applications*, vol. 541, p. 122272, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0378437119313093

[86] D. Stempień and R. Ślepaczuk, "Hybrid models for financial forecasting: Combining econometric, machine learning, and deep learning models," *arXiv preprint arXiv:2505.19617*, May 2025. [Online]. Available: https://arxiv.org/pdf/2505.19617

[87] M. Patsiarikas, G. Papageorgiou, and C. Tjortjis, "Using machine learning on macroeconomic, technical, and sentiment indicators for stock market forecasting," *Information*, vol. 16, no. 7, 2025. [Online]. Available: https://www.mdpi.com/2078-2489/16/7/584

[88] R. M. Nabi, S. A. Saeed, and A. M. W. Abdi, "Feature engineering for stock price prediction," *International Journal of Advanced Science and Technology*, vol. 29, no. 12s, pp. 2486–2496, June 2020. [Online]. Available: https://www.researchgate.net/publication/342339410_Feature_Engineering_for_Stock_Price_Prediction

[89] W. Choi, S. Jang, S. Kim, C. Park, S. Park, and S. Song, "Return prediction by machine learning for the korean stock market," *Journal of the Korean Statistical Society*, vol. 53, no. 1, pp. 248–280, Mar 2024. [Online]. Available: https://link.springer.com/article/10.1007/s42952-023-00245-0

[90] R. J. Hyndman and G. Athanasopoulos, *Forecasting: principles and practice*, 2nd ed. Melbourne, Australia: OTexts, 2018, ch. 8.1 Stationarity and differencing. [Online]. Available: https://otexts.com/fpp2/stationarity.html

[91] A. G. Ţiţan, "The efficient market hypothesis: Review of specialized literature and empirical research," *Procedia Economics and Finance*, vol. 32, pp. 442–449, 2015, emerging Markets Queries in Finance and Business 2014, EMQFB 2014, 24-25 October 2014, Bucharest, Romania. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2212567115014161

[92] E. Hurwitz and T. Marwala, "Common mistakes when applying computational intelligence and machine learning to stock market modelling," 2012. [Online]. Available: https://arxiv.org/abs/1208.4429

[93] A. N. Atwa, A. Sedky, and M. Kholief, "Challenges of conventional feature extraction techniques for multivariate time-bars assets," *International Journal of Data Science and Analytics*, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s41060-025-00824-w

[94] A. Rostamian and J. G. O'Hara, "Event prediction within directional change framework using a cnn-lstm model," *Neural Computing and Applications*, vol. 34, pp. 17 193–17 205, 2022. [Online]. Available: https://link.springer.com/article/10.1007/s00521-022-07687-3

[95] J. Rahman, H. Rahman, N. Islam, T. Tanchangya, M. Ridwan, and M. Ali, "Regulatory landscape of blockchain assets: Analyzing the drivers of nft and cryptocurrency regulation," *BenchCouncil Transactions on Benchmarks, Standards and Evaluations*, vol. 5, no. 1, p. 100214, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772485925000274

[96] T. Nguyen Thi Thu and V. Dang Xuan, "Supervised support vector machine in predicting foreign exchange trading," *International Journal of Intelligent Systems and Applications*, vol. 10, no. 9, pp. 48–56, 2018. [Online]. Available: https://www.mecs-press.org/ijisa/ijisa-v10-n9/IJISA-V10-N9-6.pdf

[97] P. Cho, J. H. Park, and J. W. Song, "Equity research report-driven investment strategy in korea using binary classification on stock price direction," *IEEE Access*, vol. 9, pp. 46 364–46 373, 2021. [Online]. Available: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9382307

[98] R. Bruni, "Stock market index data and indicators for day trading as a binary classification problem," *Data in Brief*, vol. 10, pp. 569–575, 2017. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352340916308058/pdfft?md5=57500e80575882ffe8350dbe560e7db3&pid=1-s2.0-S2352340916308058-main.pdf

[99] J. Jiang, C. Yang, X. Wang, and B. Li, "Why regression? binary encoding classification brings confidence to stock market index price prediction," *arXiv preprint arXiv:2506.03153*, May 2025. [Online]. Available: https://arxiv.org/pdf/2506.03153

[100] D.-H. Kwon, J.-B. Kim, J.-S. Heo, C.-M. Kim, and Y.-H. Han, "Time series classification of cryptocurrency price trend based on a recurrent lstm neural network," *Journal of Information Processing Systems*, vol. 15, no. 3, pp. 694–706, 2019. [Online]. Available: https://koreascience.kr/article/JAKO201919163608818.page

[101] E. Akyildirim, A. Goncu, and A. Sensoy, "Prediction of cryptocurrency returns using machine learning," *Annals of Operations Research*, vol. 297, no. 1, pp. 3–36, Jan 2021. [Online]. Available: https://link.springer.com/article/10.1007/s10479-020-03575-y

[102] S. M. Qureshi, A. Saeed, F. Ahmad, A. R. Khattak, S. H. Almotiri, M. A. Al Ghamdi, and M. Shah Rukh, "Evaluating machine learning models for predictive accuracy in cryptocurrency price forecasting," *PeerJ Computer Science*, vol. 11, p. e2626, Feb 2025. [Online]. Available: https://peerj.com/articles/cs-2626/

[103] A. W. Lo and A. C. MacKinlay, *A Non-Random Walk Down Wall Street*, fifth printing, and first paperback printing, 2002 ed. 41 William Street, Princeton, New Jersey 08540: Princeton University Press, 1999, printed in the United States of America. Paperback ISBN 0-691-09256-7. Includes bibliographical references and index. [Online]. Available: https://assets.press.princeton.edu/chapters/s6558.pdf

[104] D. H. Bailey, J. M. Borwein, M. Lopez de Prado, and Q. J. Zhu, "The probability of backtest overfitting," *SSRN*, 2013. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2326253

[105] M. L. de Prado, *Advances in Financial Machine Learning.* Hoboken, New Jersey: John Wiley & Sons, Inc., 2018, cover image: © Erikona/Getty Images. Published simultaneously in Canada. [Online]. Available: https://agorism.dev/book/finance/ml/Marcos%20Lopez%20de%20Prado%20-%20Advances%20in%20Financial%20Machine%20Learning-Wiley%20%282018%29.pdf

[106] H. White, "A reality check for data snooping," *Econometrica*, vol. 68, no. 5, pp. 1097–1126, 2000, accessed on JSTOR. [Online]. Available: https://www.jstor.org/stable/2999444

[107] D.-G. Anghel, "A reality check on trading rule performance in the cryptocurrency market: Machine learning vs. technical analysis," *Finance Research Letters*, vol. 39, p. 101655, 2021. [Online]. Available: https://doi.org/10.1016/j.frl.2020.101655

[108] Y. Lu, H. Zhang, and Q. Guo, "Stock and market index prediction using informer network," *arXiv preprint arXiv:2305.14382*, 2023. [Online]. Available: https://arxiv.org/pdf/2305.14382

[109] A. A. Salunke, "Robust stock market forecasting using huber loss: Mitigating extreme events and outliers in lstm models," 2023, bachelor of Technology thesis. [Online]. Available: https://www.researchgate.net/profile/Aatmaj-Salunke/publication/373240182_Robust_Stock_Market_Forecasting_Using_Huber_Loss_Mitigating_Extreme_Events_and_Outliers_in_LSTM_Models/links/64e2b343177c5904130b9808/Robust-Stock-Market-Forecasting-Using-Huber-Loss-Mitigating-Extreme-Events-and-Outliers-i pdf

[110] A. Landi and T. Lemishko, "Real-world viability of cointegration-based forex pairs trading strategy with walk-forward optimization," SSRN, Tech. Rep., 2024, sSRN Working Paper, posted: 12 Feb 2025. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5068086

[111] T. Lemishko, A. Landi, and J. Caicedo-Llano, "Cointegration-based strategies in forex pairs trading," SSRN, Tech. Rep., 2024, sSRN Working Paper, posted: 15 Apr 2024; last revised: 18 Dec 2024. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4771108

[112] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed. Boulder, CO: Westview Press, 2015.

[113] A. Mashuri, N. H. Adenan, N. S. A. Karim, T. S. Wei, and Z. Zeng, "Application of chaos theory in different fields – a literature review," *Journal of Science and Mathematics Letters*, vol. 12, no. 1, pp. 92–101, 2024, review Article. Received: 1 October 2023; Accepted: 15 December 2023; Published: 5 January 2024. [Online]. Available: https://doi.org/10.37134/jsml.vol12.1.11.2024

[114] P. de Grauwe and M. Grimaldi, "A chaotic model of the exchange rate: The role of fundamentalists and chartists," *Open Economies Review*, vol. 4, no. 4, pp. 351–379, 1993. [Online]. Available: https://link.springer.com/article/10.1007/BF01011136

[115] V. Soloviev, A. Bielinskyi, O. Serdyuk, V. Solovieva, and S. Semerikov, "Lyapunov exponents as indicators of the stock market crashes," in *Proceedings of the 2nd International Workshop on Informatics & Data-Driven Education (IDDE 2020) co-located with 16th International Conference on ICT in Education, Research and Industrial Applications (ICTERI 2020)*, ser. CEUR Workshop Proceedings, vol. 2732. CEUR-WS.org, 2020, copyright © 2020 for this paper by its authors. Use permitted under CC BY 4.0. [Online]. Available: https://ceur-ws.org/Vol-2732/20200455.pdf

[116] R. ur Rehman Ghaffar, M. S. Nazir, H. Anwar, and A. Tanveer, "Bitcoin price prediction using machine learning algorithms," *Financial Innovation*, vol. 8, no. 1, pp. 1–25, 2022. [Online]. Available: https://jfin-swufe.springeropen.com/articles/10.1186/s40854-022-00394-x

[117] P. Kroha and M. Škoula, "Hurst exponent and trading signals derived from market time series," in *Proceedings of the 20th International Conference on Enterprise Information Systems (ICEIS)*. SCITEPRESS, 2019, pp. 371–378. [Online]. Available: https://www.scitepress.org/Papers/2018/66670/66670.pdf